

math

MAMMOTH

Grade 1-A Worktext



Addition
within 0-10



Subtraction
within 0-10



Two-digit numbers
and place value



Telling time



Light Blue
Series

Maria Miller

2026 edition

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2026 Edition

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Foreword

Math Mammoth Grade 1 comprises a complete math curriculum for the first grade mathematics studies. The curriculum meets the Common Core standards.

The main areas of study for Math Mammoth Grade 1 are:

1. The concepts of addition and subtraction, strategies for addition and subtraction facts, and addition and subtraction word problems;
2. Developing understanding of place value for two-digit numbers;
3. Developing understanding and some basic strategies for two-digit addition and subtraction.

Additional topics we study are telling time (whole and half hours), geometric shapes, measurement, and counting coins.

This book, the 1-A worktext, covers addition, subtraction, and word problems (chapters 1 and 2), place value with two-digit numbers (chapter 3), and telling time (chapter 4). The book 1-B covers addition and subtraction facts, shapes, fractions, measuring, adding and subtracting two-digit numbers, graphs, and counting coins.

I heartily recommend that you read the full user guide in the following pages.

I wish you success in teaching math!

Maria Miller, the author

User Guide

Note: You can also find the information that follows online, at <https://www.mathmammoth.com/userguides/>.

The Common Core Standards documentation is available at:

<https://www.mathmammoth.com/preview/standards/MM-CCS-Grade1-2026.pdf>

Basic principles in using Math Mammoth Complete Curriculum

Math Mammoth is mastery-based, which means it concentrates on a few major topics at a time, in order to study them in depth. The two books (parts A and B) are like a “framework”, but you still have a lot of liberty in planning your child’s studies. You can even use it in a *spiral* manner, if you prefer. Simply have your student study in 2-3 chapters simultaneously. In first grade, I suggest studying chapters 1-3 in order, but you can be flexible with the other chapters and schedule them earlier or later.

Math Mammoth is not a scripted curriculum. In other words, it is not spelling out in exact detail what the teacher is to do or say. Instead, Math Mammoth gives you, the teacher, various tools for teaching:

- **The two student worktexts** (parts A and B) contain all the lesson material and exercises. They include the explanations of the concepts (the teaching part) in blue boxes. The worktexts also contain some advice for the teacher in the “Introduction” of each chapter.

The teacher can read the teaching part of each lesson before the lesson, or read and study it together with the student in the lesson, or let the student read and study on his own. If you are a classroom teacher, you can copy the examples from the “blue teaching boxes” to the board and go through them on the board.

- There are hundreds of **videos** matched to the curriculum available at <https://www.mathmammoth.com/videos/>. There isn’t a video for every lesson, but there are dozens of videos for each grade level. You can simply have the author teach your child or student!
- Don’t automatically assign all the exercises. Use your judgment, trying to assign just enough for your student’s needs. You can use the skipped exercises later for review. For most students, I recommend to start out by assigning about half of the available exercises. Adjust as necessary.
- Each chapter introduction contains a **list of links to various free online games** and activities. These games can be used to supplement the math lessons, for learning math facts, or just for some fun.
- The student books contain some **mixed review lessons**, and the curriculum also provides you with additional **cumulative review lessons**.
- There is a **chapter test** for each chapter of the curriculum, and a comprehensive end-of-year test.
- The **worksheet maker** allows you to make additional worksheets for most calculation-type topics in the curriculum. This is a single html file. You will need Internet access to be able to use it.
- You can use the free online exercises at <https://www.mathmammoth.com/practice/>. This is an expanding section of the site, so check often to see what new topics we are adding to it!
- Some grade levels have **cut-outs** to make fraction manipulatives or geometric solids.
- The answer key is included when you purchase the digital version (the download). If you purchase the printed version, the answer key is available as a separate book.

How to get started

Have ready the first lesson from the student worktext. Go over the first teaching part (within the blue boxes) together with your child. Go through a few of the first exercises together, and then assign some problems for your child to do on their own.

Repeat this if the lesson has other blue teaching boxes. Naturally, you can also use the videos at <https://www.mathmammoth.com/videos/>

Many students can eventually study the lessons completely on their own — the curriculum becomes self-teaching. However, students definitely vary in how much they need someone to be there to actually teach them.

Pacing the curriculum

Each chapter introduction contains a suggested pacing guide for that chapter. You will see a summary on the right. (This summary does not include the optional lessons, nor time for the optional tests.)

Most lessons are 2 pages long, intended for one day. Some lessons are 3 pages, again intended for one day but you can take two if needed. There are also a few optional lessons, not included in the tables on the right.

It can also be helpful to calculate a general guideline as to how many pages per week the student should cover in order to go through the curriculum in one school year.

The table below lists how many pages there are for the student to finish in this particular grade level, and gives you a guideline for how many pages per day to finish, assuming a 170-day (34-week) school year. The page count in the table below *includes* the optional lessons.

Example:

Grade level	School days	Days for tests and reviews	Lesson pages	Days for the student book	Pages to study per day	Pages to study per week
1-A	83	8	149	75	2	10
1-B	87	10	151	77	2	10
Grade 1 total	170	18	300	152	2	10

The table below is for you to fill in. Allow several days for tests and additional review before tests — I suggest at least twice the number of chapters in the curriculum. Then, to get a count of “pages to study per day”, **divide the number of lesson pages by the number of days for the student book**. Lastly, multiply this number by 5 to get the approximate page count to cover in a week.

Grade level	Number of school days	Days for tests and reviews	Lesson pages	Days for the student book	Pages to study per day	Pages to study per week
1-A			149			
1-B			151			
Grade 1 total			300			

Now, something important. Whenever the curriculum has lots of similar practice problems (a large set of problems), feel free to **only assign 1/2 or 2/3 of those problems**. If your student gets it with less amount of exercises, then that is perfect! If not, you can always assign the rest of the problems for some other day. In fact, you could even use these unassigned problems the next week or next month for some additional review.

In general, 1st-2nd graders might spend 25-40 minutes a day on math. Third-fourth graders might spend 30-60 minutes a day. Fifth-sixth graders might spend 45-75 minutes a day. If your student finds math enjoyable, they can of course spend more time with it! However, it is not good to drag out the lessons on a regular basis, because that can then affect the student's attitude towards math.

Working space, the usage of additional paper, and mental math

The curriculum generally includes working space directly on the page for students to work out the problems. However, feel free to let your students use extra paper when necessary. They can use it, not only for the “long” algorithms (where you line up numbers to add, subtract, multiply, and divide), but also to draw diagrams and pictures to help organize their thoughts. Some students won't need the additional space (and may resist the thought of extra paper), while some will benefit from it. Use your discretion.

Some exercises don't have any working space, but just an empty line for the answer (e.g. $200 + \underline{\hspace{1cm}} = 1,000$). Typically, I have intended that such exercises be done using *mental math*.

However, there are some students who struggle with mental math (often this is because of not having studied and used it in the past). As always, the teacher has the final say (not me!) as to how to approach the exercises and how to use the curriculum. We do want to prevent extreme frustration (to the point of tears). The goal is always to provide SOME challenge, but not too much, and to let students experience success enough so that they can continue to enjoy learning math.

Students struggling with mental math will probably benefit from studying the basic principles of mental calculations from the earlier levels of Math Mammoth curriculum. To do so, look for lessons that list mental math strategies. They are taught in the chapters about addition, subtraction, place value, multiplication, and division. My article at https://www.mathmammoth.com/lessons/practical_tips_mental_math also gives you a summary of some of those principles.

Using tests

For each chapter, there is a **chapter test**, which can be administered right after studying the chapter. **The tests are optional**. Some families might prefer not to give tests at all. The main reason for the tests is for diagnostic purposes, and for record keeping. These tests are not aligned or matched to any standards.

In the digital version of the curriculum, the tests are provided as PDF files. You can edit them (such as to change the numbers in them) to provide a different test using PDF apps that have editing capabilities. You can even use the annotation tools (such as text boxes) available in most PDF apps. Remember to save the edited file under a different file name, or you will lose the original.

The end-of-year test is best administered as a diagnostic or assessment test, which will tell you how well the student remembers and has mastered the mathematics content of the entire grade level.

Using cumulative reviews and the worksheet maker

The student books contain mixed review lessons which review concepts from earlier chapters. The curriculum also comes with additional cumulative review lessons, which are just like the mixed review lessons in the student books, with a mix of problems covering various topics. These are found in their own folder in the digital version, and in the Tests & Cumulative Reviews book in the print version.

The cumulative reviews are optional; use them as needed. They are named indicating which chapters of the main curriculum the problems in the review come from. For example, “Cumulative Review, Chapter 4” includes problems that cover topics from chapters 1-4.

Both the mixed and cumulative reviews allow you to spot areas that the student has not grasped well or has forgotten. When you find such a topic or concept, you have several options:

1. Check if the worksheet maker lets you make worksheets for that topic.
2. Check for any online games and resources in the Introduction part of the particular chapter in which this topic or concept was taught.
3. If you have the digital version, you could reprint the lesson from the student worktext, and have the student restudy that.
4. Perhaps you only assigned 1/2 or 2/3 of the exercise sets in the student book at first, and can now use the remaining exercises.
5. Check if our online practice area at <https://www.mathmammoth.com/practice/> has something for that topic.
6. Khan Academy has free online exercises, articles, and videos for most any math topic imaginable.

Concerning challenging word problems and puzzles

While this is not absolutely necessary, I heartily recommend supplementing Math Mammoth with challenging word problems and puzzles. You could do that once a month, for example, or more often if the student enjoys it.

The goal of challenging story problems and puzzles is to **develop the student’s logical and abstract thinking and mental discipline**. I recommend starting these in fourth grade, at the latest. Then, students are able to read the problems on their own and have developed mathematical knowledge in many different areas. Of course I am not discouraging students from doing such in earlier grades, either.

Math Mammoth curriculum contains lots of word problems, and they are usually multi-step problems. Several of the lessons utilize a bar model for solving problems. Even so, the problems I have created are usually tied to a specific concept or concepts. I feel students can benefit from solving problems and puzzles that require them to think “out of the box” or are just different from the ones I have written.

I recommend you use the free Math Stars problem-solving newsletters as one of the main resources for puzzles and challenging problems:

Math Stars Problem Solving Newsletter (grades 1-8)

<https://www.homeschoolmath.net/teaching/math-stars.php>

I have also compiled a list of other resources for problem solving practice, which you can access at this link:

<https://l.mathmammoth.com/challengingproblems>

Another idea: you can find puzzles online by searching for “brain puzzles for kids,” “logic puzzles for kids” or “brain teasers for kids.”

Frequently asked questions and contacting us

If you have more questions, please first check the FAQ at <https://www.mathmammoth.com/faq-lightblue>

If the FAQ does not cover your question, you can then contact us using the contact form at the Math Mammoth.com website.

Chapter 0: Kindergarten Math Review

Introduction

This chapter is optional. You can use it to check that your child/student has mastered the important concepts of kindergarten math. The lessons do not review every concept from kindergarten math, but they do cover the ones I have deemed important:

- writing the numerals from 0 to 20
- reciting the counting sequence up to 100
- counting up to 20 individual items
- comparing numbers between 1 and 10
- the concepts of addition and subtraction; representing addition and subtraction with drawings
- understanding that a number can be decomposed into pairs in more than one way (e.g. $6 = 3 + 3$ but also, $6 = 2 + 4$)
- solving very simple addition and subtraction word problems (within 0-10)
- correctly naming basic shapes

We will practice several of these still in first grade, but the skills that are essential before going on to the other chapters in Grade 1 are being able to correctly count up to 20 items, being able to write the numerals from 0 to 20, and the basic concepts of addition and subtraction.

Please note: decomposing the numbers from 11-19 as sums of “10 + some number” will also be covered in detail later in grade 1. The reason it is introduced in kindergarten is to *prepare* children to learn place value. Before going on to chapter 1 of Math Mammoth Grade 1, it is essential is that the child is able to recite the counting sequence at least up to 20, and to write those numerals. It is not essential for the child to understand that 18 is 10 + 8; that will be covered in chapter 3.

Pacing Suggestion for Chapter 0

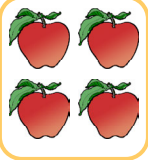
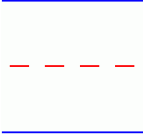
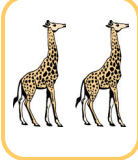
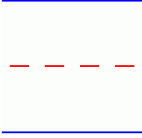
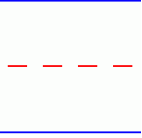
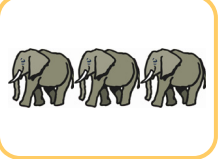
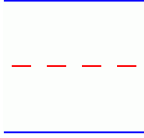
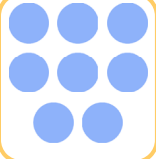
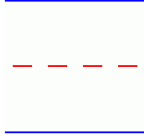
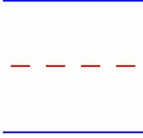
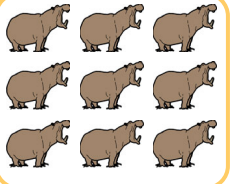
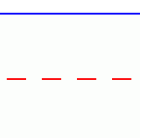
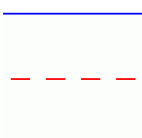
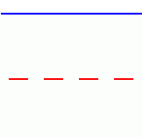
The Lessons in Chapter 0	page	span	suggested pacing	your pacing
Counting and Writing Numbers	12	2 pages	1 day	
Teen Numbers and Comparing	14	2 pages	1 day	
Two Groups and a Total	16	2 pages	1 day	
Addition	18	2 pages	1 day	
Subtraction as “Taking Away”	20	2 pages	1 day	
Shapes and Positions	22	2 pages	1 day	
TOTALS		12 pages	6 days	

Counting and Writing Numbers

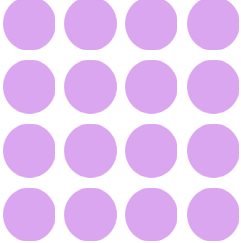
1. **To the teacher:** This exercise is to be done orally.

- a. Ask the student to count from 40 to 60. b. Ask the student to count from 87 to 100.
c. Ask the student to count from 10 to 100 by tens.

2. Count and write the number.

a. 		b. 		c. 	
d. 		e. 		f. 	
g. 		h. 		i. 	

3. Count and write the number.

 a. _____	 b. _____
 c. _____	 d. _____

4. Group (draw a large circle around) the given number of items.



a. Make a group of eleven marbles.



b. Make a group of sixteen chicks.

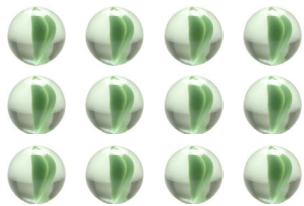


c. Make a group of seventeen stars.



d. Make a group of nineteen strawberries.

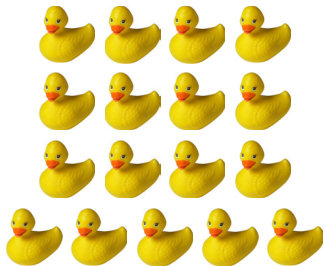
5. Count and write the number.



a. _____



b. _____



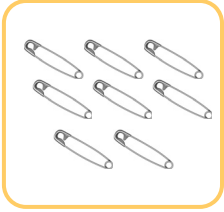
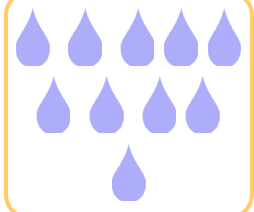
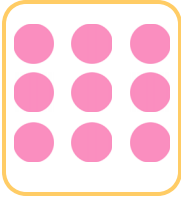
c. _____



d. _____

Teen Numbers and Comparing

1. Circle the group that has MORE things. Circle both if they have the same number of things.

a.  	b.  
c.  	d.  
e.  	f.  

2. Circle the bigger number.

a. 9 7	b. 8 10	c. 5 4
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3. Circle the smaller number.

a. 1 4	b. 5 6	c. 9 8
-----------	-----------	-----------

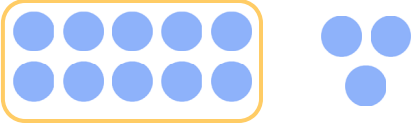
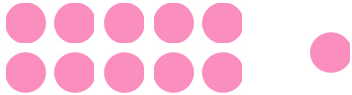
4. Write the number that is 1 more.

a. 8 _____	b. 6 _____	c. 3 _____
d. 12 _____	e. 16 _____	f. 19 _____

5. Each illustration has between 11 and 19 objects. Circle the ten objects.

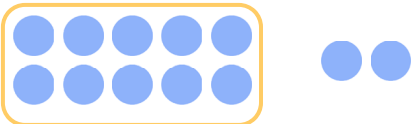
Then count the ones that are not in the ten-group.

Lastly, write the total as a sum of “10 + some number”.

 a. $10 + \underline{3}$ <u>13</u>	 b. $10 + \underline{\quad}$ <u> </u>
 c. $\underline{\quad} + \underline{\quad}$ <u> </u>	 d. $\underline{\quad} + \underline{\quad}$ <u> </u>

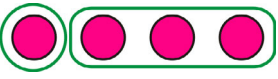
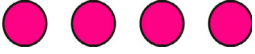
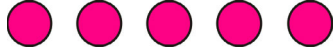
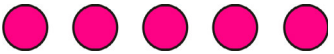
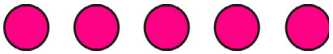
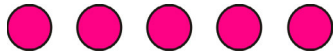
6. Write each number as a sum of 10 and some other number.

Draw dots to illustrate the equation: draw a group of 10 dots, and some more dots.

 a. $12 = \underline{\quad} + \underline{\quad}$	b. $17 = \underline{\quad} + \underline{\quad}$
c. $14 = \underline{\quad} + \underline{\quad}$	d. $19 = \underline{\quad} + \underline{\quad}$

Two Groups and a Total

1. Make two groups. Write how many are in the second group.

a. 4  1 and <u>3</u>	b. 4  2 and _____	c. 5  4 and _____
d. 5  1 and _____	e. 5  3 and _____	f. 5  0 and _____

2. Decompose 6, 7, and 8 into pairs in different ways. First, draw as many dots as the number shows. Then divide them into two groups, in different ways. Lastly write how many are in each group.

a. 6 _____ and _____	b. 6 _____ and _____	c. 6 _____ and _____
d. 7 _____ and _____	e. 7 _____ and _____	f. 7 _____ and _____
g. 8 _____ and _____	h. 8 _____ and _____	i. 8 _____ and _____

3. The number at the top is the total. Draw the missing dots on the face of the blank dice. Write on the lines how many dots are on the face of each dice.

a. 3 • _____ and _____	b. 6 • • _____ and _____	c. 5 • • _____ and _____
d. 4 • • • _____ and _____	e. 6 • • • _____ and _____	f. 5 • _____ and _____

“One and four makes five.”	1 and 4 5
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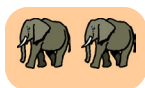
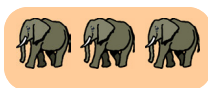
4. Write how many are in each group. Write the total in the box.

a. _____ and _____	b. _____ and _____	c. _____ and _____
---	---	---

5. Draw circles for each number. Write the total in the box.

a. 2 and 2	b. 5 and 1	c. 1 and 4
--	--	--

Addition



$$3 + 2 = 5$$

THREE plus TWO equals FIVE



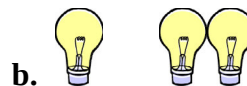
$$1 + 3 = 4$$

ONE plus THREE equals FOUR

1. Write the number of things in each group. Add. Read the addition sentences aloud using “plus” and “equals”.



$$\underline{1} + \underline{3} = \boxed{}$$



$$\underline{} + \underline{} = \boxed{}$$



$$\underline{} + \underline{} = \boxed{}$$



$$\underline{} + \underline{} = \boxed{}$$



$$\underline{} + \underline{} = \boxed{}$$



$$\underline{} + \underline{} = \boxed{}$$

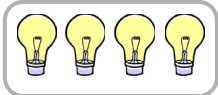
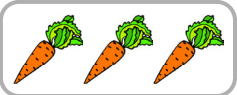


$$\underline{} + \underline{} = \boxed{}$$



$$\underline{} + \underline{} = \boxed{}$$

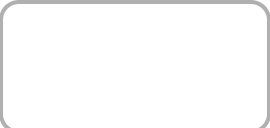
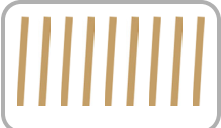
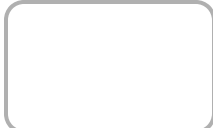
2. Add with zero.

a.   _____ + _____ = _____	b.   _____ + _____ = _____
c.   _____ + _____ = _____	d.   _____ + _____ = _____

3. Add. You can draw sticks, circles, or other pictures to help you.

a. $4 + 3 =$ _____	b. $5 + 5 =$ _____	c. $7 + 1 =$ _____
d. $9 + 0 =$ _____	e. $3 + 6 =$ _____	f. $3 + 4 =$ _____

4. Draw another group of sticks so that there are 10 in total. Write the numbers in each group.

a.   _____ and _____	b.   _____ and _____
c.   _____ and _____	d.   _____ and _____

Subtraction as "Taking Away"

From six carrots we take away two.
(Cover the two crossed-out carrots.)
Then there are four left.



$$6 - 2 = 4$$

Six minus two equals four

1. Cross out objects. How many are left?



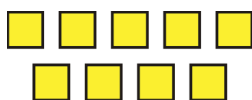
a. $6 - 3 = \underline{3}$



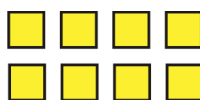
b. $8 - 6 = \underline{\quad}$



c. $7 - 3 = \underline{\quad}$



d. $9 - 3 = \underline{\quad}$



e. $8 - 4 = \underline{\quad}$



f. $10 - 5 = \underline{\quad}$



g. $6 - 3 = \underline{\quad}$

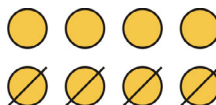


h. $6 - 5 = \underline{\quad}$

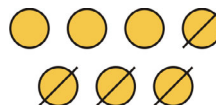


i. $5 - 5 = \underline{\quad}$

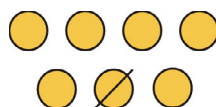
2. Write a subtraction sentence to match each picture.



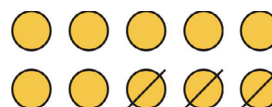
a. $\underline{\quad} - \underline{\quad} = \underline{\quad}$



b. $\underline{\quad} - \underline{\quad} = \underline{\quad}$



c. $\underline{\quad} - \underline{\quad} = \underline{\quad}$



d. $\underline{\quad} - \underline{\quad} = \underline{\quad}$

3. Subtract. You can draw sticks or circles to help you.

a. $8 - 3 = \underline{\quad}$	b. $5 - 1 = \underline{\quad}$	c. $10 - 7 = \underline{\quad}$
d. $7 - 2 = \underline{\quad}$	e. $6 - 3 = \underline{\quad}$	f. $9 - 7 = \underline{\quad}$

4. Add and subtract.

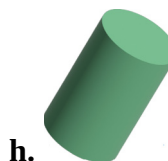
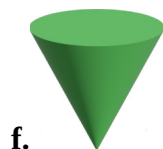
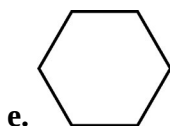
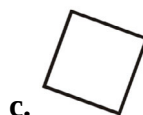
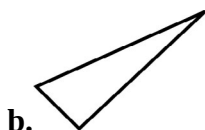
a. $4 + 1 = \underline{\quad}$ $2 + 2 = \underline{\quad}$	b. $4 + 2 = \underline{\quad}$ $1 + 1 = \underline{\quad}$	c. $3 + 1 = \underline{\quad}$ $0 + 5 = \underline{\quad}$
d. $2 - 2 = \underline{\quad}$ $5 - 3 = \underline{\quad}$	e. $3 - 2 = \underline{\quad}$ $4 - 3 = \underline{\quad}$	f. $4 - 0 = \underline{\quad}$ $3 - 1 = \underline{\quad}$

5. Solve. You can draw something to help you, or act it out.

a. Henry saw 3 cookies on one plate and 5 cookies on another. How many cookies in total did Henry see?	b. There were nine birds in a tree. Then two of them flew away. How many birds are still in the tree?
c. Lily drew five bunnies sleeping. Then she drew two more. How many bunnies did Lily draw in all?	

Shapes and Positions

1. Name each shape.



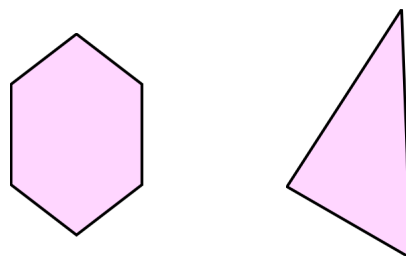
2. Draw here a triangle, circle, and rectangle. (An approximate sketch is sufficient.)

a. Draw a triangle.

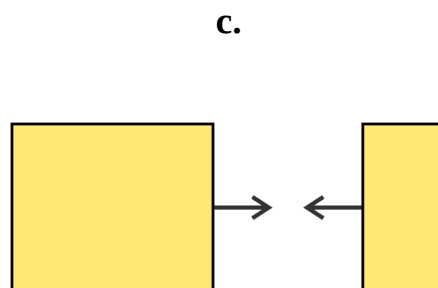
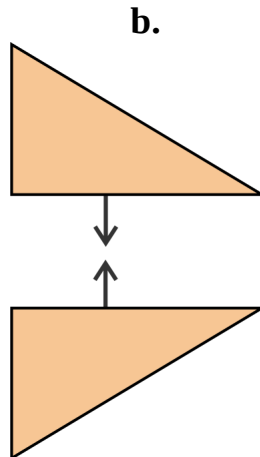
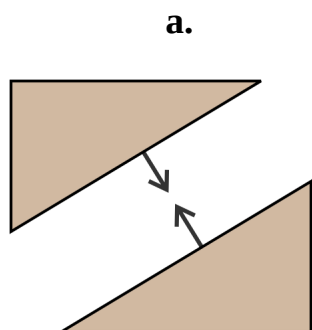
b. Draw a circle.

c. Draw a rectangle.

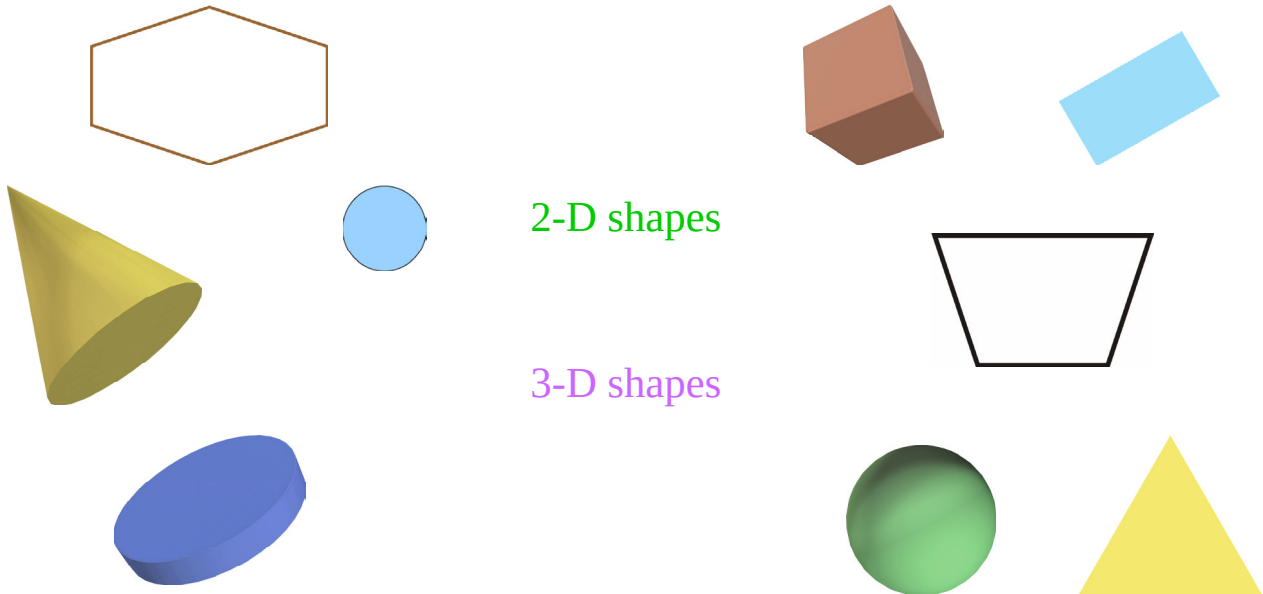
3. What is the same and what is different about these two shapes?
(Can be done orally.)



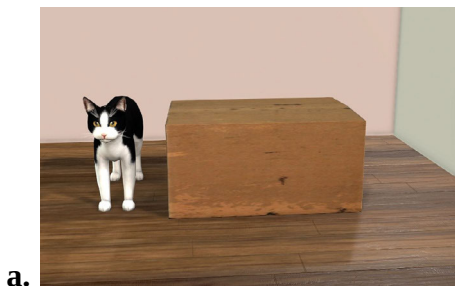
4. If you put together these shapes, what new shape will you get?
(You may cut out the shapes.)



5. Are the shapes flat/two-dimensional (2-D) or solids/three-dimensional (3-D)?
Draw a line from each shape to the correct answer.



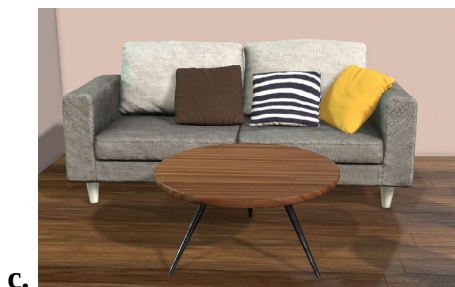
6. Where is the cat? Describe the relative positions of objects (orally).



The cat is _____
the box.



The cat is _____
the box.



The table is _____
the sofa.



The ball is _____
the table.

Chapter 1: Addition Within 0-10

Introduction

The first chapter of *Math Mammoth Grade 1* focuses on various concepts related to addition, and on the basic addition facts within 0-10.

We start out by reviewing the concept of addition, which has already been taught in kindergarten, in the lesson *Basic Addition*. The following lesson (*Counting on to Add*) explains how you can add 1, 2, or 3 to a number by counting on 1, 2, or 3 steps *from* that number. This is a simple strategy to us adults, but children may need it pointed out.

The next two lessons build on the thought of counting on, as students learn to illustrate sums with jumps on a number line. A number line is an important way to model addition, as it helps to build number sense and will tie in with measurement, later on.

Then, we review comparing numbers and using the symbols $>$, $<$, and $=$ to record the comparisons.

Next, we introduce “missing number” problems (e.g. $1 + \underline{\quad} = 5$) over the course of two lessons. These problems are very important, as they connect with fact families and with subtraction.

The child might confuse the missing number problem $1 + \underline{\quad} = 5$ with $1 + 5 = \underline{\quad}$. To help them see the difference, word these problems like this: “One and *how many more* make five?” You can also model these problems by drawings. In our example of $1 + \underline{\quad} = 5$, you would first draw one stick, and then tell the student, “We need a total of five sticks. Draw more until there are five of them.” The number of sticks that the child needs to draw in order to make five is the missing number. So, you can say, “First there was one stick, then you needed to add (draw) some more to make 5. How many more did you draw?”

Then we come to the lessons titled *Sums to...* These lessons practice the number bonds (number combinations) that add up to a particular number. In the case of 5, they are 0 and 5, 1 and 4, and 2 and 3. The goal is to help the child to memorize the addition facts within 0-10. We will deal with these addition facts again in chapter 5, in context of fact families, and along with the corresponding subtraction facts.

My approach to memorizing the basic addition facts within 0-10 is many-fold:

1. Structured drills, such as used in the lessons *Sums to 5*, *Sums to 6*, and so on. In each of these lessons, the child learns the number combinations that add up to the specific number (aka number bonds). This understanding is the basis for the drills. Thus, the drills are not totally random, because they are based on the pattern or the structure in the facts. This will connect the facts to a context, and help the child to better understand the facts on a conceptual level, instead of merely memorizing them at random.
2. Using addition facts in games and in everyday life is very helpful — and especially in games, because all children like to play games.
3. Random drilling may also be used, sparingly, as one tool among others.
4. Memory helpers can be silly mnemonics or writing math facts on a poster and hanging it on the wall. Not all children need these, but feel free to use them if you like.

These same addition facts are used a lot in the following chapter about subtraction, and they are studied and practiced again in chapter 5. They are also used constantly in all later math work. I recommend that children become fluent with addition and subtraction facts within 0-10 by the end of first grade.

Before closing the chapter, we focus on the thought of equality and the usage of the equals sign ($=$). Many children develop a misconception of the equals sign being an “operator,” as if it means that you need to add/subtract/multiply/divide, or “operate” on the numbers in the problem. A child with this misconception will treat the equation $9 = \underline{\quad} + 4$ as the addition problem $9 + 4$. The lessons on equality and comparisons focus on preventing that erroneous view, and on building the proper understanding of the symbol: that of denoting

equality of what is on either side of the symbol. For example, students will see and work with equations such as $5 = 4 + 1$ and $3 + 5 = 4 + 4$.

In this chapter, students also encounter addition tables, number patterns, word problems, and adding three or four numbers. So, while on the surface it may look like that all we do is add small numbers, actually a lot happens! Please also see the following pages for games that I recommend while studying this chapter. Games are important at this level, as they help children to practice the addition facts and also make math fun.

Lastly, keep in mind the free videos matched to the curriculum at <https://www.mathmammoth.com/videos/>.

Good Mathematical Practices

- Learning the meaning of the equals sign, and learning to use it correctly, is part of attending to precision in mathematics. Too many people are “loose” in their usage of this sign, writing things such as:
 $20 + 40 = 60 \times 3 = 180$. This is an attempt to record two different calculations in one line, but it is imprecise, because in mathematics, anything on either side of the equals sign is supposed to be equal. This claims that $20 + 40$ is equal to 60×3 , which is equal to 180.
- As children study the sums of 5, sums of 6, sums of 7, and so on, they will see patterns in these addition facts, and will learn these facts within a structure, not as random facts. Mathematics has been called “the science of patterns,” and thus, recognizing patterns and structures is the pure essence of mathematics.

Pacing Suggestion for Chapter 1

Please add one day to the pacing for the test if you will use it.

The Lessons in Chapter 1	page	span	suggested pacing	your pacing
Basic Addition	30	2 pages	1 day	
Counting On to Add	32	2 pages	1 day	
Adding on the Number Line 1	34	3 pages	1 day	
Adding on the Number Line 2	37	2 pages	1 day	
Comparing Numbers	39	2 pages	1 day	
Missing Numbers 1	41	2 pages	1 day	
Missing Numbers 2	43	2 pages	1 day	
Sums to 5	45	2 pages	1 day	
Sums to 6	47	2 pages	1 day	
Word Problems	49	2 pages	1 day	
Sums to 7	51	2 pages	1 day	
Sums to 8	53	2 pages	1 day	
Adding Three Numbers	55	2 pages	1 day	
Adding Many Numbers	57	2 pages	1 day	
Sums to 9	59	2 pages	1 day	
Equality	61	2 pages	1 day	
Comparisons	63	2 pages	1 day	
Addition Practice	65	2 pages	1 day	
Sums to 10	67	2 pages	1 day	
More Practice	69	2 pages	1 day	
Chapter 1 Review	71	4 pages	2 days	
Chapter 1 Test (optional)				
TOTALS		45 pages	22 days	

Math Talks

I invite you to take a look at these math-inspired real-life photographs. Use them to have short conversations with your student(s) about how we can see math in real life, with no pressure to get any “right answer.”

<https://www.mathmammoth.com/MathTalks>

Each photo has a connection to some simple math concept, such as counting, addition, or shapes. Many times, you can find several math concepts in it, such as both addition and subtraction, or both squares and multiplication. But there is no obligation to find all the possibilities. The idea is just to have a relaxed time, sharing one’s thoughts and listening to others’ ideas.

To help the child get started, you can ask, “What do you notice? What do you wonder?” Another great way to use the photos is to ask the child or students to make a story about the photo that uses numbers or math in some way.

I suggest you have a “math talk” a few times a week. It should only take a few minutes. Soon you will start noticing opportunities for such talks in your life, also! Just point out to your child or students objects in life that relate to math in some way, or ask them to make a math story of some scene. If your child has access to a camera, you can also ask them to take “math talk” photos themselves.

Games and Activities

Some Went Hiding

You need: The same number of small objects as the sum you are studying. For example, to study the sums with 5, you need 5 objects (marbles, blocks, etc).

Game play: The first player shows the objects but quickly hides some of them behind his back without showing how many. Then he shows the remaining objects to the next player, who has to say how many “went hiding.” If the player gives the right answer, it is then his turn to hide some and ask the next player to answer. If he gives a wrong answer, he forfeits his turn.

Variation: Instead of getting a turn to hide objects, the player who answers correctly may gain points or other rewards for the right answer.

10 Out (or 6 Out, 7 Out, 8 Out, etc.)

You need: A deck of number cards with numbers 1-10, or regular playing cards without the face cards.

Preparation: Choose a target sum, such as 10. Deal seven cards to each player. Place the rest face down in a pile in the middle of the table.

Game play: During your turn, first take one card from the pile. Then try to find pairs of cards in your hand that add up to 10, and discard any such pairs. Discard the card 10 also if you have it. If you cannot find any such pairs, ask for any one card you want (such as 6) from the player to your right (as in “Go Fish”). That player, if he has it, must give it, and you will then discard the pair that makes 10. Then it is the next player’s turn. The player who first discards all the cards from his hand is the winner.

Variations:

- * Deal more than seven cards.
- * Deal fewer cards if there are a lot of players or the players are very young.
- * Allow players to discard *three* cards that add up to 10.
- * Instead of ten, players discard cards that add up to 5, 6, 7, 8, or 9.

Number Bonds in the Pond

You need: A standard deck (or several) of playing cards or number cards

Preparation: Choose a target sum for the game. If the target sum is 5, make a deck of cards consisting of numbers 1 through 4. If the target sum is 6, make a deck of numbers 1-5. And so on. (The deck always consists of numbers that are from 1 through $X - 1$ where X is the target sum.) Place a target number card face up between the players, and spread out the rest of the cards face down, like a pond, between the players.

Game play: During your turn, if you don't have any cards in your hand, take two cards from the pond. If you do, take one card from the pond. Now check if any two cards in your hand add up to the target number. If so, put those cards in your personal pile. If not, it is the next player's turn. The game ends when there are no more cards in the pond. The winner is the person with most cards in their personal pile.

Variation: Allow three cards/numbers to be added to reach the target number.

Notes: Depending on the number of players, you may need several decks of cards to make the pond. When first playing this game with your child/students, start with 5 as the target number, and advance to target numbers of 6, 7, and onward. Playing this game several times will help the child to memorize the number bonds (the sums) associated with a particular target number.

Addition Crabs

You need: Two dice, eleven markers for each of the two players, and the game board:

<https://www.mathmammoth.com/download/Addition-Crabs.pdf>

Game Play: On your turn, roll the two dice and add the amounts the dice show. Then place a marker on the crab with that sum. If you cannot go, you skip your turn. The player that gets all their markers on the crabs first is the winner.

Addition Challenge

You need: A standard deck of playing cards from which you remove the face cards and perhaps also some of the other higher-numbered cards, such as tens, nines, and eights. Alternatively, a set of dominoes works well for children who do not yet know the sums beyond 12.

Game Play: In each round, each player is dealt two cards face up, and calculates the sum. The player with the highest sum gets all the cards from the other players. After enough rounds have been played to use all of the cards, the player with the most cards wins. If two or more players have the same sum, those players get an additional two cards and use those to resolve the tie.

Any **board game** where you move the piece by rolling two dice also works well to practice addition.

Games and Activities at Math Mammoth Practice Zone

Single-Digit Addition Practice

Practice basic addition. For this chapter, choose generic practice and also missing addend problems.

<https://www.mathmammoth.com/practice/addition-single-digit>

Addition Hidden Picture Game

Choose a number range (such as 1 to 10) and uncover a hidden picture while solving addition problems.

<https://www.mathmammoth.com/practice/mystery-picture>

Number Bonds

Practice number bonds, either with pictures or with numbers.

<https://www.mathmammoth.com/practice/number-bonds>

“7 Up” Addition Facts

You will see seven cards dealt face up. Simply choose any two cards that make 10 (or your chosen sum) to discard. When there are no cards that make that sum, click the deck to deal more cards.

<https://www.mathmammoth.com/practice/seven-up>

Fruity Math

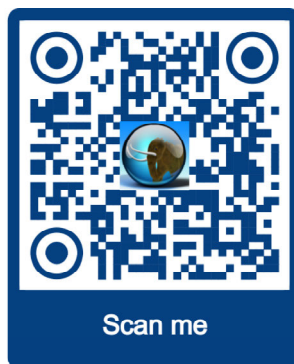
Add two single-digit numbers (such as $4 + 5$). Click the fruit with the correct answer and try to get as many points as you can within two minutes.

<https://www.mathmammoth.com/practice/fruity-math>

Further Resources on the Internet

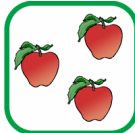
These resources match the topics in this chapter, and offer online practice, online games (occasionally, printable games), and interactive illustrations of math concepts. We heartily recommend you take a look. Many people love using these resources to supplement the bookwork, to illustrate a concept better, and for some fun. Enjoy!

<https://links.mathmammoth.com/gr1ch1>



Basic Addition

1. In the second box, draw enough things to show the second number. Then add.

a.   $2 + 1 = \underline{\quad}$	b.   $3 + 2 = \underline{\quad}$	c.   $1 + 2 = \underline{\quad}$
d.   $4 + 1 = \underline{\quad}$	e.   $2 + 3 = \underline{\quad}$	f.   $0 + 4 = \underline{\quad}$

2. Add. Compare the two additions in each box. What do you notice?
(You can draw marbles to help.)

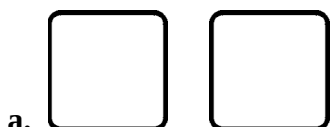
  a. $2 + 3 = \underline{\quad}$   $3 + 2 = \underline{\quad}$	b. $1 + 2 = \underline{\quad}$ $2 + 1 = \underline{\quad}$	c. $3 + 1 = \underline{\quad}$ $1 + 3 = \underline{\quad}$
d. $1 + 4 = \underline{\quad}$ $4 + 1 = \underline{\quad}$	e. $0 + 2 = \underline{\quad}$ $2 + 0 = \underline{\quad}$	f. $5 + 0 = \underline{\quad}$ $0 + 5 = \underline{\quad}$

3. Solve. You can draw or act out the story to help you.

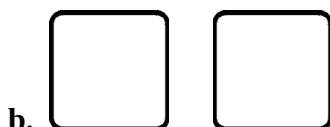
a. Joanna picked four flowers on the yard. Then she found one more and picked that one too. How many flowers does she have now?

b. Eric was playing with one toy car. Then his brother gave him four more. How many cars does he have now?

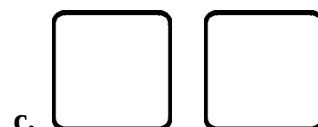
4. Draw dots in each box for the numbers. Then add.



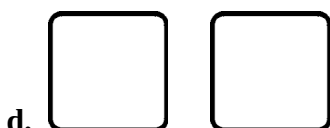
$$2 + 2 = \underline{\quad}$$



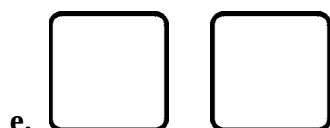
$$1 + 4 = \underline{\quad}$$



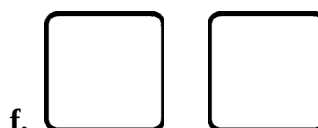
$$0 + 5 = \underline{\quad}$$



$$5 + 1 = \underline{\quad}$$



$$2 + 3 = \underline{\quad}$$



$$5 + 2 = \underline{\quad}$$

5. Add. If you want to, you can draw circles or sticks to help you.

a. $1 + 2 = \underline{\quad}$

b. $3 + 0 = \underline{\quad}$

c. $2 + 2 = \underline{\quad}$

d. $2 + 3 = \underline{\quad}$

e. $1 + 4 = \underline{\quad}$

f. $0 + 5 = \underline{\quad}$

g. $3 + 2 = \underline{\quad}$

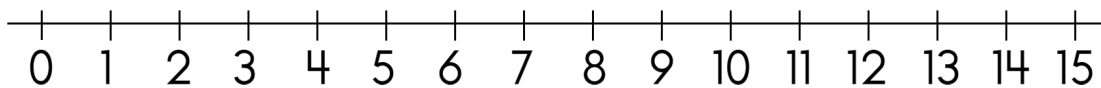
h. $2 + 1 = \underline{\quad}$

i. $4 + 1 = \underline{\quad}$

Counting On to Add

To add 1, you can just count on from that number to the next number.

$$9 + 1 = ?$$



The next number after 9 is 10. So, $9 + 1$ equals 10.

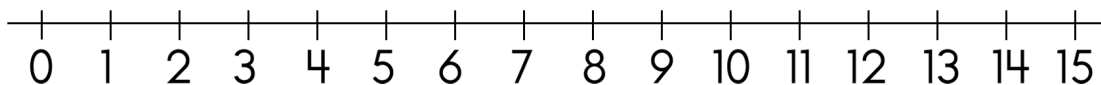
(Just think how slow it would be to count first 1, 2, 3, 4, 5, 6, 7, 8, 9, and THEN count one more. You don't have to do that! You can start at 9.)

To add 2, count on two steps from the number.

$$12 + 2 = ?$$

We count: 12, 13, 14. (See the number line above.) So, $12 + 2 = 14$.

1. Count on to add. You can use the number line.



a. $11 + 1 =$ _____

b. $8 + 2 =$ _____

c. $7 + 2 =$ _____

d. $14 + 1 =$ _____

e. $9 + 2 =$ _____

f. $13 + 1 =$ _____

g. $8 + 1 =$ _____

h. $12 + 2 =$ _____

i. $10 + 1 =$ _____

2. Add. Compare the problems in each box.

a. $10 + 1 =$ _____

b. $12 + 1 =$ _____

c. $9 + 1 =$ _____

$10 + 2 =$ _____

$12 + 2 =$ _____

$9 + 2 =$ _____

To add 3, count on three steps from the number.

$$8 + 3 = ?$$

We count: 8, 9, 10, 11. (A number line can help, also.) So, $8 + 3 = 11$.

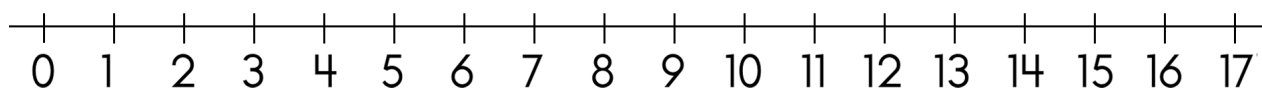
3. Add.

a. $11 + 3 =$ _____

b. $13 + 3 =$ _____

c. $7 + 3 =$ _____

4. Add 1, 2, and 3 to each number. Notice the patterns!



a.

$10 + 1 =$ _____

$10 + 2 =$ _____

$10 + 3 =$ _____

b.

$13 + 1 =$ _____

$13 + 2 =$ _____

$13 + 3 =$ _____

c.

$15 + 1 =$ _____

$15 + 2 =$ _____

$15 + 3 =$ _____

5. Find the additions that are correct.

a. $9 + 3 = 12$

b. $4 + 3 = 8$

c. $9 + 2 = 10$

d. $11 + 1 = 13$

e. $10 + 2 = 12$

Puzzle Corner

Count on to find the answers.

$20 + 3 =$ _____

$56 + 2 =$ _____

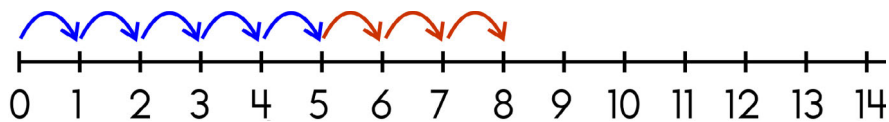
$81 + 3 =$ _____

$33 + 4 =$ _____

$62 + 2 =$ _____

$74 + 3 =$ _____

Adding on the Number Line 1



First jump 5 steps... **Then jump 3 more.** **You land on 8 .**

So, $5 + 3 = 8$.

1. Draw little jumps (steps) to show the addition. Start at zero. You can also make “jumps” using your finger! (Alternatively, use tape to make a number line on the floor, and actually jump!)

a. $2 + 3 =$ _____

b. $5 + 2 =$ _____

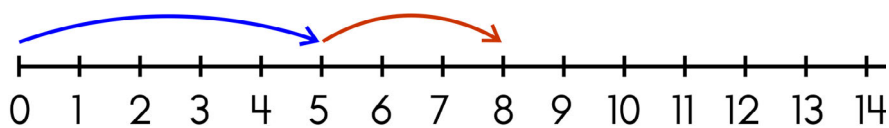
c. $3 + 1 =$ _____

d. $4 + 3 =$ _____

e. $6 + 3 =$ _____

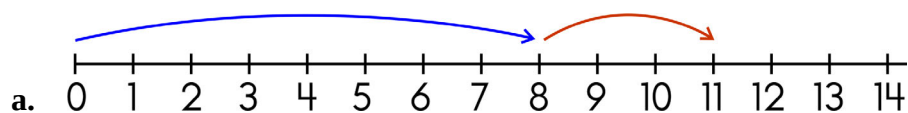
f. $3 + 4 =$ _____

We can also show $5 + 3 = 8$ using only two arrows:

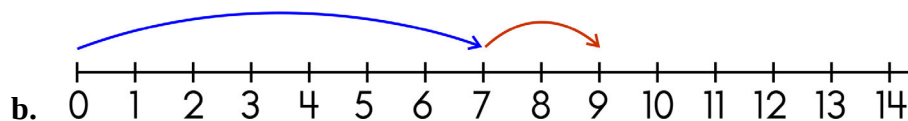


One arrow is 5 steps long, and lands at 5. The other is 3 steps long. Make little jumps with your finger to verify that.

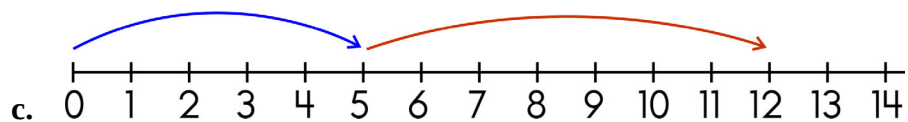
2. Write the addition. (You can make little jumps with your finger to see how many steps long an arrow is.)



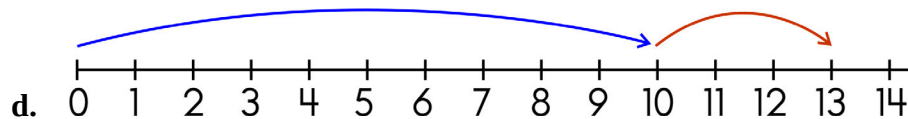
$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$



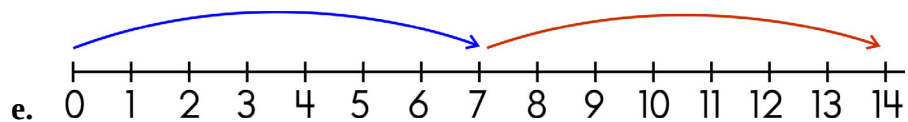
$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$



$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$



$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

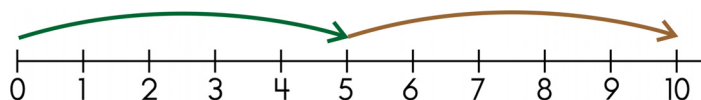


$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

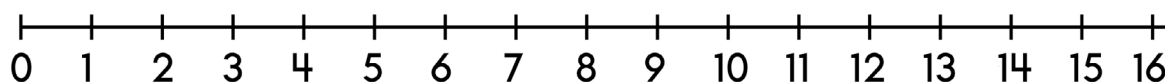
3. Lisa says the illustration shows $5 + 10$.

Do you agree?

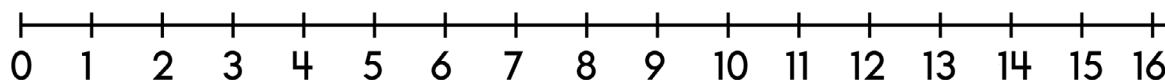
Explain your thinking.



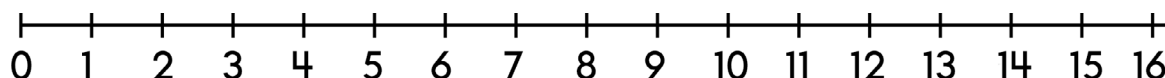
4. Draw arrows (jumps) to show the addition.



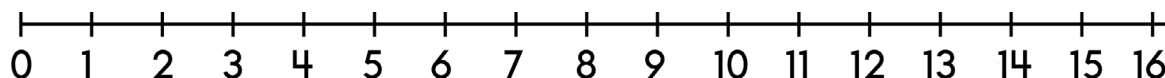
a. $11 + 4 =$ _____



b. $10 + 2 =$ _____



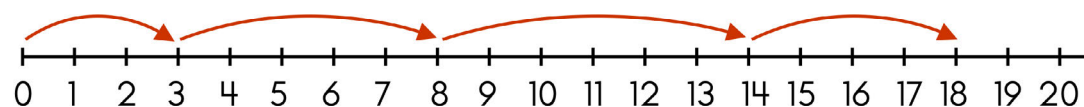
c. $8 + 8 =$ _____



d. $9 + 6 =$ _____

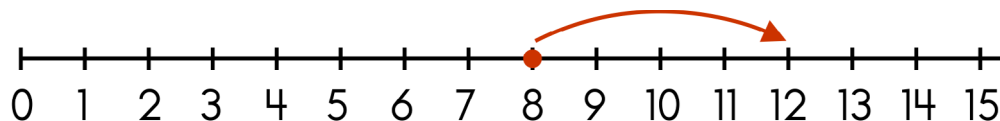
Puzzle Corner

What addition sentence is illustrated by the arrows?



Adding on the Number Line 2

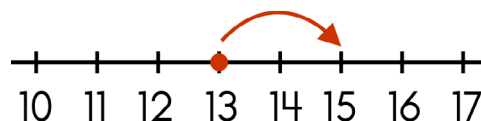
What addition is illustrated here? (Think about it.)



The dot at 8 is the starting point. Then, the “jump” is 4 steps or 4 units long.

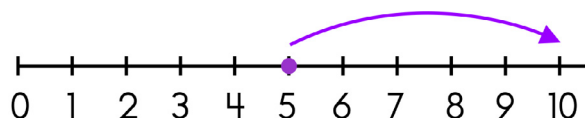
This is a different way to illustrate $8 + 4 = 12$. It matches with the idea of counting on: you start at 8 and count on four numbers: 8 ... 9, 10, 11, 12.

To show $13 + 2$ in this manner, we draw a dot at 13, and an arrow that is 2 steps long:

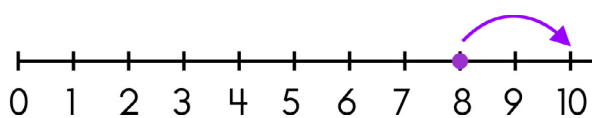


The arrow ends at 15. That means $13 + 2 = 15$.

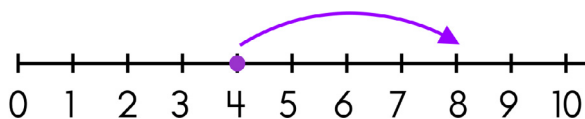
1. Write the addition sentence that is illustrated on the number line.



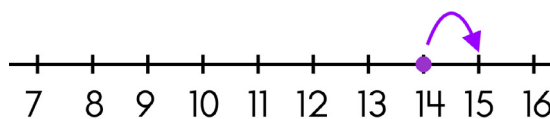
a. _____ + _____ = _____



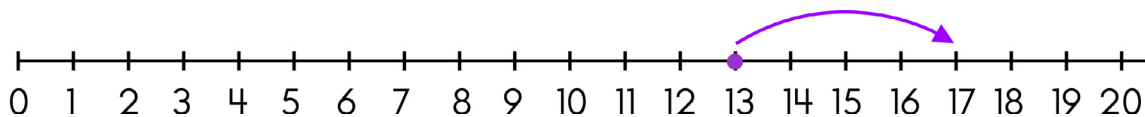
b.



c.

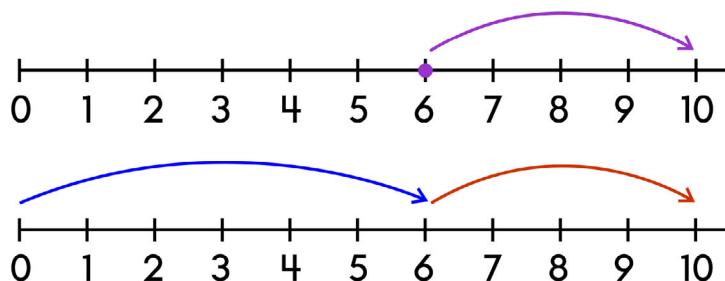


d.

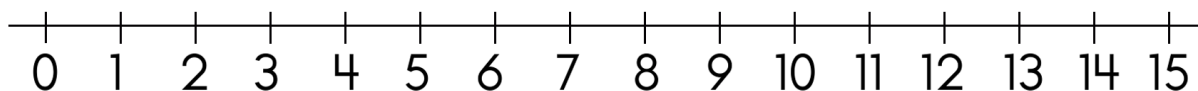


e.

2. Both of these illustrate (show) the same addition! Which?



3. Add. You can count on or draw a jump on the number line.



a. $8 + 2 =$ _____	b. $4 + 3 =$ _____	c. $11 + 2 =$ _____
d. $10 + 3 =$ _____	e. $9 + 5 =$ _____	f. $13 + 3 =$ _____

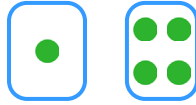
4. Complete. Notice the patterns!

a. $6 + 0 =$ _____ $6 + 1 =$ _____ $6 + 2 =$ _____ $6 + 3 =$ _____	b. $9 + 0 =$ _____ $9 + 1 =$ _____ $9 + 2 =$ _____ $9 + 3 =$ _____	c. $11 + 0 =$ _____ $11 + 1 =$ _____ $11 + 2 =$ _____ $11 + 3 =$ _____
---	---	---

5. You can add the numbers in either order! Which way is easier?

a. $7 + 2 =$ _____ $2 + 7 =$ _____	b. $2 + 5 =$ _____ $5 + 2 =$ _____	c. $8 + 2 =$ _____ $2 + 8 =$ _____
---------------------------------------	---------------------------------------	---------------------------------------

Comparing Numbers



$$1 < 4$$

One is less than four.

< means "IS LESS THAN".



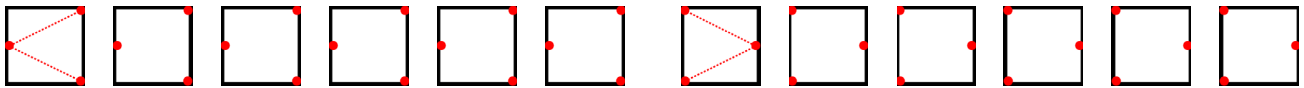
$$5 > 3$$

Five is greater than three.

> means "IS GREATER THAN".

The symbols < and > are like "hungry alligator's mouths."
The mouth always opens towards the **bigger** number.

1. Practice writing the symbols < and > .



2. Circle the bigger number. Read the comparison sentences aloud.



a. $1 < 3$

"1 is less than 3."



b. $2 < 5$

"2 is less than 5."



c. $6 > 3$

"6 is greater than 3."

3. Circle the bigger number. Read aloud using "less than" or "greater than".

a. $6 > 0$

b. $7 < 8$

c. $4 < 7$

d. $6 > 3$

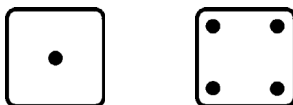
e. $1 < 7$

f. $6 > 5$

g. $3 < 10$

h. $0 < 5$

4. Write < or > in the box.



a. 1 4



b. 2 5



c. 6 3

5. Write $<$ or $>$ between the numbers. You can draw dots to help you.

a. 2 5	b. 7 4	c. 6 5	d. 0 4
---	---	---	---

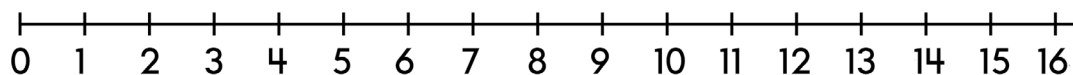
6. Write $<$ or $>$ between the two numbers.

- | | | | |
|--|--|---|--|
| a. 1 4 | b. 4 5 | c. 2 4 | d. 5 3 |
| e. 6 7 | f. 8 6 | g. 10 8 | h. 5 9 |
| i. 3 7 | j. 1 0 | k. 4 6 | l. 0 8 |

7. Circle the number that, if written on the empty line, would make the comparison true.

a. 2 5 7 6 $<$ _____	b. 4 8 2 4 $>$ _____	c. 8 5 1 5 $<$ _____
d. 1 9 2 _____ $>$ 2	e. 9 6 8 _____ $<$ 7	f. 3 1 4 _____ $<$ 3

8. Write $<$ or $>$ between the two numbers. Use the number line to help.



- | | | | |
|--|--|--|--|
| a. 12 10 | b. 11 14 | c. 16 13 | d. 11 12 |
|--|--|--|--|

Missing Numbers 1

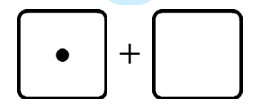
Something is missing from this addition. The TOTAL is not missing. The total is 5.

The first dice has one dot. And the second?

That's what is missing! Draw 4 dots in the second dice.

Read: "1 plus what number makes 5?"
or, "1 and how many more makes 5?"

5

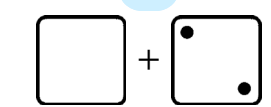


1 + _____

There should be a total of 4 dots. The second dice has two. There are none in the first dice, so you need to draw them.

Read: "What number and 2 makes 4?"
or, "2 and how many more makes 4?"

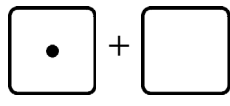
4



_____ + 2

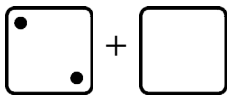
1. Complete the additions. Draw the missing dots. The total is on top.

3



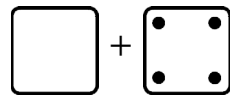
a. 1 + _____

4



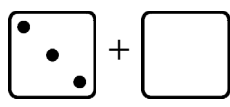
b. 2 + _____

5



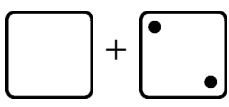
c. _____ + 4

5



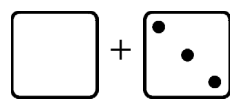
d. 3 + _____

5



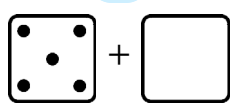
e. _____ + 2

4



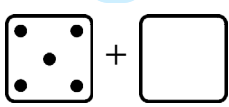
f. _____ + 3

7



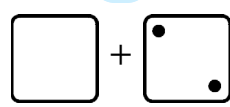
g. 5 + _____

6



h. 5 + _____

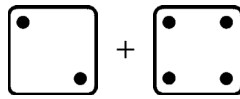
6



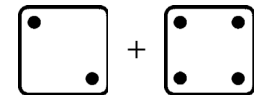
i. _____ + 2

Since $2 + 4$ equals 6, the missing number is *four*.

The “6” can be on either side of the equals sign.



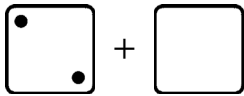
$$2 + \underline{\quad} = 6$$



$$6 = 2 + \underline{\quad}$$

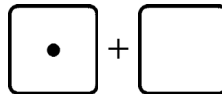
2. Draw more dots to show the missing number. Write the missing number.

a.



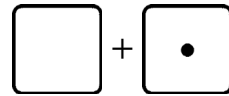
$$2 + \underline{\quad} = 4$$

b.



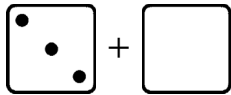
$$1 + \underline{\quad} = 4$$

c.



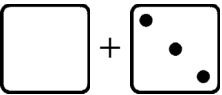
$$\underline{\quad} + 1 = 5$$

d.



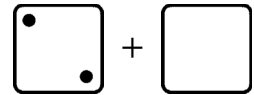
$$3 + \underline{\quad} = 5$$

e.



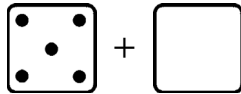
$$\underline{\quad} + 3 = 4$$

f.



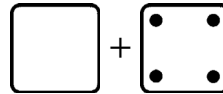
$$3 = 2 + \underline{\quad}$$

g.



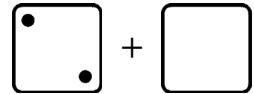
$$6 = 5 + \underline{\quad}$$

h.



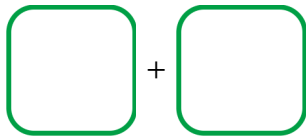
$$6 = \underline{\quad} + 4$$

i.

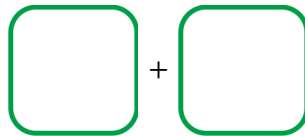


$$5 = 2 + \underline{\quad}$$

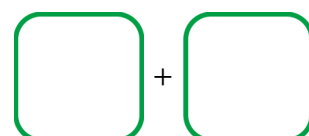
3. Draw dots to illustrate each addition problem. Find what number is missing.



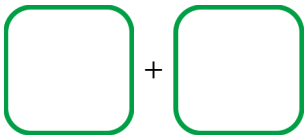
$$a. 4 + \underline{\quad} = 5$$



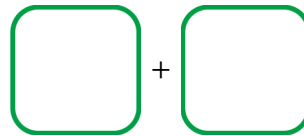
$$b. 1 + \underline{\quad} = 2$$



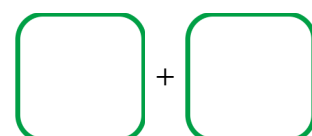
$$c. \underline{\quad} + 3 = 5$$



$$d. 3 = 2 + \underline{\quad}$$



$$e. 4 = 1 + \underline{\quad}$$



$$f. 6 = \underline{\quad} + 2$$

Missing Numbers 2

There are no dots on the face of either dice.

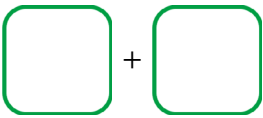
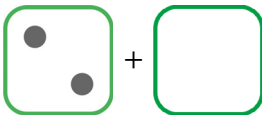
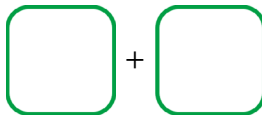
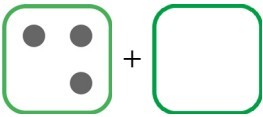
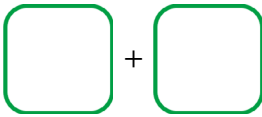
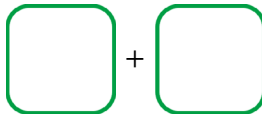
The first dice is missing its dots. The second is *supposed* to have none, since there is a zero below it.

Draw 4 dots on the first dice, because $\underline{4} + 0 = \text{4}$.

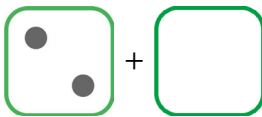
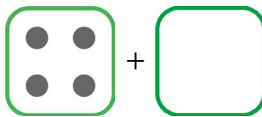
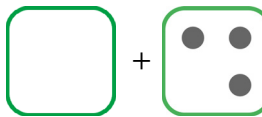
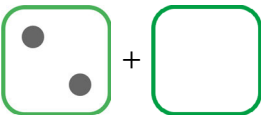
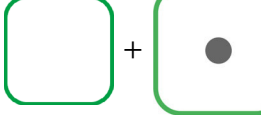
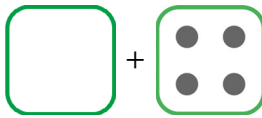
$$\square + \square$$

$$\underline{\quad} + 0 = \text{4}$$

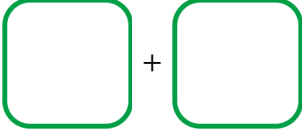
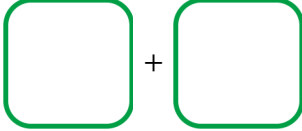
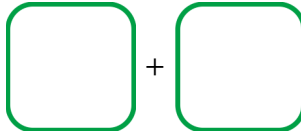
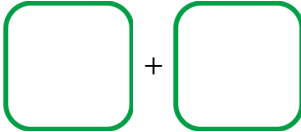
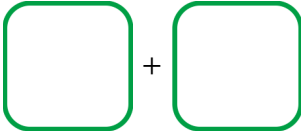
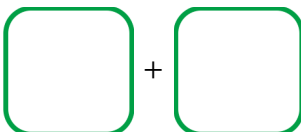
1. Draw dots for the missing numbers. Note: some boxes are *supposed* to have zero dots.

 a. $0 + \underline{\quad} = 4$	 b. $2 + \underline{\quad} = 2$	 c. $\underline{\quad} + 0 = 3$
 d. $3 + \underline{\quad} = 3$	 e. $\underline{\quad} + 0 = 5$	 f. $0 + \underline{\quad} = 1$

2. Draw dots in the empty box for the missing number. Read the problems aloud:
“2 plus how many makes 4?”

 a. $2 + \underline{\quad} = 4$	 b. $4 + \underline{\quad} = 4$	 c. $\underline{\quad} + 3 = 4$
 d. $5 = 2 + \underline{\quad}$	 e. $3 = \underline{\quad} + 1$	 f. $5 = \underline{\quad} + 4$

3. Draw dots to illustrate each addition problem. Write the missing number.

 a. 4 + _____ = 5	 b. 1 + _____ = 2	 c. _____ + 4 = 4
 d. 6 = 5 + _____	 e. 6 = _____ + 2	 f. 7 = 5 + _____

4. Fill in the missing numbers. You can draw dots or sticks to help you.

a. 2 +  = 5	b.  + 2 = 4	c.  + 1 = 3
d. 6 =  + 3	e. 6 = 3 + 	f. 6 = 0 + 

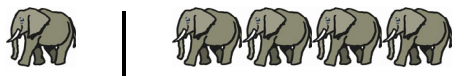
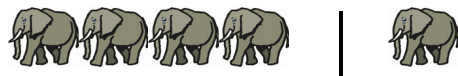
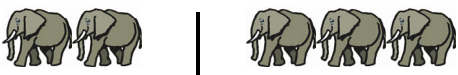
5. Fill in the missing numbers. The marbles illustrate the total. Notice the patterns!

a.  0 + _____ = 3 1 + _____ = 3 2 + _____ = 3 3 + _____ = 3	b.  0 + _____ = 4 1 + _____ = 4 2 + _____ = 4 3 + _____ = 4 4 + _____ = 4	c.  0 + _____ = 5 1 + _____ = 5 2 + _____ = 5 3 + _____ = 5 4 + _____ = 5 5 + _____ = 5
---	--	---

Sums to 5

1. Here are the different ways to group five elephants into two groups. The “|” symbol separates the two groups. Fill in the missing numbers.

Note: Additions are called **sums**. For example, $2 + 3$ is a sum: it is the sum of 2 and 3.

5	
 $\underline{0} + \underline{5}$	 $\underline{\quad} + \underline{\quad}$
 $\underline{\quad} + \underline{\quad}$	 $\underline{\quad} + \underline{\quad}$
 $\underline{\quad} + \underline{\quad}$	 $\underline{\quad} + \underline{\quad}$

2. Play “5 Out” *and/or* “Some Went Hiding” with 5 objects (see the introduction).

3. Add.

a. $4 + 1 = \underline{\quad}$

$2 + 2 = \underline{\quad}$

$3 + 2 = \underline{\quad}$

$1 + 2 = \underline{\quad}$

b. $2 + 3 = \underline{\quad}$

$1 + 3 = \underline{\quad}$

$1 + 4 = \underline{\quad}$

$2 + 1 = \underline{\quad}$

c. $1 + 1 = \underline{\quad}$

$0 + 5 = \underline{\quad}$

$1 + 4 = \underline{\quad}$

$3 + 2 = \underline{\quad}$

4. **Drill.** Say the answers aloud as your teacher points to problems. (Don't write the answers.)

$1 + \square = 5$

$4 + \square = 5$

$\square + 2 = 5$

$\square + 3 = 5$

$2 + \square = 5$

$3 + \square = 5$

$\square + 0 = 5$

$\square + 1 = 5$

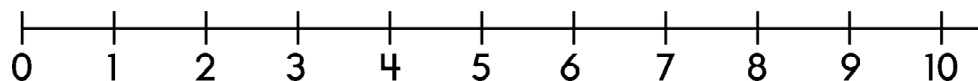
$0 + \square = 5$

$5 + \square = 5$

$\square + 4 = 5$

$\square + 5 = 5$

5. Add. Compare the problems in each group.



a. $4 + 0 = \underline{\hspace{2cm}}$

$4 + 1 = \underline{\hspace{2cm}}$

$4 + 2 = \underline{\hspace{2cm}}$

b. $6 + 0 = \underline{\hspace{2cm}}$

$6 + 1 = \underline{\hspace{2cm}}$

$6 + 2 = \underline{\hspace{2cm}}$

c. $3 + 0 = \underline{\hspace{2cm}}$

$3 + 1 = \underline{\hspace{2cm}}$

$3 + 2 = \underline{\hspace{2cm}}$

d. $7 + 0 = \underline{\hspace{2cm}}$

$7 + 1 = \underline{\hspace{2cm}}$

$7 + 2 = \underline{\hspace{2cm}}$

e. $5 + 0 = \underline{\hspace{2cm}}$

$5 + 1 = \underline{\hspace{2cm}}$

$5 + 2 = \underline{\hspace{2cm}}$

f. $8 + 0 = \underline{\hspace{2cm}}$

$8 + 1 = \underline{\hspace{2cm}}$

$8 + 2 = \underline{\hspace{2cm}}$

6. Draw more things to illustrate each missing number. Complete the addition sentence.



$\underline{2} + \underline{\hspace{2cm}} = 5$



$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = 3$



$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = 4$



$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = 5$



$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = 6$



$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = 6$

Sums to 6

1. Here are the ways to group six chicks into two groups. Fill in the missing numbers.

6	
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	

2. Play “6 Out” *and/or* “Some Went Hiding” with 6 objects (see the introduction).

3. **Drill.** Say the answers aloud as your teacher points to problems. (Don’t write the answers.)

$1 + \square = 6$

$4 + \square = 6$

$\square + 2 = 6$

$\square + 3 = 6$

$2 + \square = 6$

$3 + \square = 6$

$\square + 0 = 6$

$\square + 1 = 6$

$6 + \square = 6$

$5 + \square = 6$

$\square + 4 = 6$

$\square + 5 = 6$

4. Fill in the missing numbers. You can draw dots to help you.

a.  $2 + \boxed{} = 6$	b.  $2 + \boxed{} = 5$	c.  $4 + \boxed{} = 6$
d.  $3 + \boxed{} = 6$	e.  $1 + \boxed{} = 6$	f.  $5 + \boxed{} = 6$
g.  $1 + \boxed{} = 5$	h. $0 + \boxed{} = 6$	i.  $3 + \boxed{} = 5$

5. Jack and Jill share 5 cucumbers and 6 lemons in different ways. Find how many Jill gets. You can cover the cucumbers or lemons with your hand to help.

a. 5






Jack gets:	Left for Jill:
2	
1	
5	
3	
0	
4	

b. 6



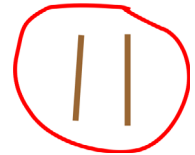



Jack gets:	Left for Jill:
1	
4	
5	
0	
2	
3	

Word Problems

Example. Brian lives with his mom, dad, and his two brothers. One day, Brian and his mom were inside, and the rest of the family were outside. How many family members were outside?

To help you solve a word problem, you can draw a picture of the situation, or act it out with objects. For example, here you could draw sticks to represent the various people.



Two are inside... how many are outside?

Do you know how many people are in that family? Yes! You can figure that out!

1. Finish solving the example above. How many family members were outside?
2. Solve the problems. Draw pictures to help you, or use objects to act out the situation. Think: Are you asked the total? Or do you already know the total?

a. Lisa has three goldfish and Lauren has six. How many goldfish do they have together?

b. Jill has seven dolls, and she wants to have hats for all of them. She has found three hats so far. How many does she still need?

c. A fish bowl has four fish swimming in it. Lisa added four more. How many fish are now in it?

d. Paul has nine toy cars. Six of them are in the living room. The rest of them Paul cannot find. How many cars are missing?

3. Solve.

a. Six friends were playing hide-and-seek. Four of them went hiding. How many did not go hiding?

b. Brenda ate two cookies, and later she ate four more. How many cookies did she eat?

4. Which number is greater? Or are they equal? Write $<$, $>$ or $=$.

a. 7 7

b. 7 8

c. 6 4

d. 10 10

e. 8 4

f. 2 2

g. 0 0

h. 8 7

i. 4 4

j. 1 5

k. 6 8

l. 2 0

5. Add.

a.	b.	c.	d.
$3 + 1 = \underline{\quad}$	$2 + 2 = \underline{\quad}$	$3 + 1 = \underline{\quad}$	$3 + 3 = \underline{\quad}$
$4 + 2 = \underline{\quad}$	$1 + 4 = \underline{\quad}$	$1 + 3 = \underline{\quad}$	$1 + 5 = \underline{\quad}$

Puzzle Corner

What numbers can go into these puzzles?
(There are many possibilities.)

	+		= 6
+		+	
	+		= 5
= 5		= 6	

	+		= 7
+		+	
	+		= 6
= 7		= 6	

Sums to 7

1. Here are the different ways to group seven marbles into two groups. Fill in the missing numbers.

7	
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____

2. Play “7 Out” *and/or* “Some Went Hiding” with 7 objects (see the introduction).

3. **Drill.** Say the answers aloud as your teacher points to problems. (Don’t write the answers.)

$5 + \square = 7$	$2 + \square = 7$	$6 + \square = 7$	$\square + 3 = 7$	$\square + 7 = 7$
$3 + \square = 7$	$1 + \square = 7$	$0 + \square = 7$	$\square + 2 = 7$	$\square + 1 = 7$
$7 + \square = 7$	$4 + \square = 7$	$4 + \square = 7$	$\square + 6 = 7$	$\square + 5 = 7$

4. Solve.

a. Mom has two eggs at home.
The cake recipe calls for five eggs.
How many more eggs will she need?

b. Jack saw three chicks under Mommy Hen. Then, five more chicks ran to her. How many chicks does the hen have?

5. This is a new way to write addition! The answer goes *under* the line.

$$\begin{array}{r} \text{a.} \quad 4 \\ + 3 \\ \hline 7 \end{array}$$

$$\begin{array}{r} \text{b.} \quad 1 \\ + 5 \\ \hline \end{array}$$

$$\begin{array}{r} \text{c.} \quad 5 \\ + 0 \\ \hline \end{array}$$

$$\begin{array}{r} \text{d.} \quad 4 \\ + 1 \\ \hline \end{array}$$

$$\begin{array}{r} \text{e.} \quad 4 \\ + 0 \\ \hline \end{array}$$

$$\begin{array}{r} \text{f.} \quad 2 \\ + 5 \\ \hline \end{array}$$

$$\begin{array}{r} \text{g.} \quad 0 \\ + 3 \\ \hline \end{array}$$

$$\begin{array}{r} \text{h.} \quad 1 \\ + 3 \\ \hline \end{array}$$

$$\begin{array}{r} \text{i.} \quad 3 \\ + 3 \\ \hline \end{array}$$

$$\begin{array}{r} \text{j.} \quad 2 \\ + 2 \\ \hline \end{array}$$

6. Fill in the missing numbers. You may draw dots to help. Notice the patterns!

a.



$$0 + \underline{\quad} = 7$$

$$1 + \underline{\quad} = 7$$

$$2 + \underline{\quad} = 7$$

$$3 + \underline{\quad} = 7$$

b.

$$0 + \underline{\quad} = 6$$

$$1 + \underline{\quad} = 6$$

$$2 + \underline{\quad} = 6$$

$$3 + \underline{\quad} = 6$$

c.

$$0 + \underline{\quad} = 5$$

$$1 + \underline{\quad} = 5$$

$$2 + \underline{\quad} = 5$$

$$3 + \underline{\quad} = 5$$

Sums to 8

1. Write the sums to show the different ways to group eight marbles into two groups.

8	
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	

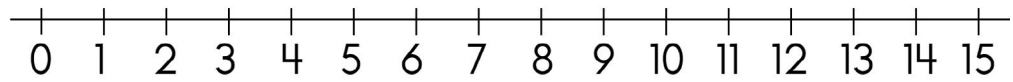
2. Play “8 Out” *and/or* “Some Went Hiding” with 8 objects (see the introduction).

3. **Drill.** Say the answers aloud as your teacher points to problems. (Don’t write the answers.)

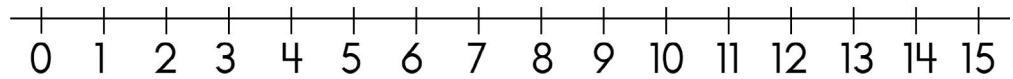
 + 5 = 8	 + 4 = 8	2 + = 8	3 + = 8	8 + = 8
 + 0 = 8	 + 6 = 8	5 + = 8	7 + = 8	6 + = 8
 + 2 = 8	 + 3 = 8	1 + = 8	4 + = 8	 + 1 = 8

4. Fill in the answers, and illustrate each sum (addition) on the number line.

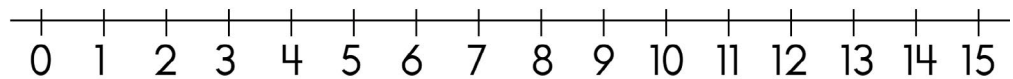
a. $6 + 3 =$



b. $9 + 4 =$



c. $11 + 3 =$



5. Add.

a.
$$\begin{array}{r} 4 \\ + 2 \\ \hline \end{array}$$

b.
$$\begin{array}{r} 6 \\ + 2 \\ \hline \end{array}$$

c.
$$\begin{array}{r} 3 \\ + 3 \\ \hline \end{array}$$

d.
$$\begin{array}{r} 7 \\ + 1 \\ \hline \end{array}$$

e.
$$\begin{array}{r} 5 \\ + 2 \\ \hline \end{array}$$

f.
$$\begin{array}{r} 1 \\ + 2 \\ \hline \end{array}$$

g.
$$\begin{array}{r} 6 \\ + 1 \\ \hline \end{array}$$

h.
$$\begin{array}{r} 4 \\ + 3 \\ \hline \end{array}$$

i.
$$\begin{array}{r} 5 \\ + 1 \\ \hline \end{array}$$

j.
$$\begin{array}{r} 3 \\ + 2 \\ \hline \end{array}$$

6. Draw the missing marbles. Write the sums.



a. $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 6$



b. $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 7$



c. $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 7$



d. $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 8$

Adding Three Numbers

Mom put one sandwich in one container, three in another, and four in yet another. How many sandwiches are there in all?

(You can draw something (sticks, dots) to represent the sandwiches, to help you solve this.)

There are **1 + 3 + 4** sandwiches in all. When we add three numbers, we can add them in any order we wish:

We can add the first two numbers first:

$$\begin{array}{c} 1 + 3 + 4 \\ \swarrow \quad \searrow \\ 4 + 4 = \end{array}$$

Or, we can add the last two numbers first:

$$\begin{array}{c} 1 + 3 + 4 \\ \quad \swarrow \quad \searrow \\ 1 + 7 = \end{array}$$

Or, we can add the first and the last number first:

$$\begin{array}{c} 1 + 3 + 4 \\ \swarrow \quad \searrow \\ 3 + 5 = \end{array}$$

It doesn't matter in which order you add three numbers.

1. Add in different orders. Which way is easier for you?

a. Add the last two numbers first.

$$2 + 5 + 2 =$$

Or, add the first and last number first.

$$2 + 5 + 2 =$$

b. Add the first two numbers first.

$$3 + 1 + 5 =$$

Or, add the first and last number first.

$$3 + 1 + 5 =$$

c. Add the last two numbers first.

$$2 + 5 + 3 =$$

Or, add the first two numbers first.

$$2 + 5 + 3 =$$

d. Add the first two numbers first.

$$1 + 6 + 2 =$$

Or, add the last two numbers first.

$$1 + 6 + 2 =$$

2. Add.

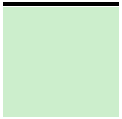
a. $1 + 2 + 3 =$

b. $4 + 0 + 3 =$

c. $2 + 5 + 1 =$

3. Add. Again, you can add in any order.

a.

$$\begin{array}{r} 5 \\ 1 \\ + 4 \\ \hline \end{array}$$


b.

$$\begin{array}{r} 3 \\ 1 \\ + 4 \\ \hline \end{array}$$


c.

$$\begin{array}{r} 2 \\ 2 \\ + 4 \\ \hline \end{array}$$


d.

$$\begin{array}{r} 7 \\ 1 \\ + 1 \\ \hline \end{array}$$


e.

$$\begin{array}{r} 4 \\ 0 \\ + 3 \\ \hline \end{array}$$


f.

$$\begin{array}{r} 2 \\ 6 \\ + 2 \\ \hline \end{array}$$


g.

$$\begin{array}{r} 1 \\ 1 \\ + 8 \\ \hline \end{array}$$


h.

$$\begin{array}{r} 3 \\ 2 \\ + 4 \\ \hline \end{array}$$


i.

$$\begin{array}{r} 7 \\ 2 \\ + 1 \\ \hline \end{array}$$


j.

$$\begin{array}{r} 4 \\ 1 \\ + 4 \\ \hline \end{array}$$


4. Solve. You can draw pictures to help.

- a. Molly picked two pretty flowers. Then she found some more, and picked three more flowers. Then she picked yet two more. How many flowers did Molly pick?

- b. Jack has 10 rabbits. One morning when he came to see them, he only saw 6 rabbits. How many were missing?

- c. Jenny and Penny each have four goldfish. Betty has two goldfish. How many do the three girls have together?

Adding Many Numbers

To add four or more numbers, you can (again) add in any order you wish.

For example, each of these sums are equal:

$$1 + 2 + 3 + 4$$

$$1 + 4 + 3 + 2$$

$$4 + 2 + 3 + 1$$

And so on.

To find the sum, we could add like this:

$$\begin{array}{r} 1 + 2 + 3 + 4 \\ \backslash / \\ 3 + 3 + 4 \\ \backslash / \\ 6 + 4 = 10 \end{array}$$

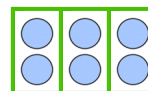
But we could add in some other order, also.

1. Add. You can color the numbers you want to add first!

a. $1 + 8 + 1 =$	b. $2 + 2 + 5 =$	c. $4 + 1 + 4 =$
d. $4 + 0 + 3 + 2 =$	e. $5 + 0 + 1 + 2 =$	
f. $2 + 1 + 5 + 2 =$	g. $3 + 1 + 2 + 1 =$	
h. $7 + 1 + 1 + 1 =$	i. $1 + 1 + 2 + 1 =$	

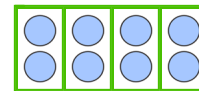
2. Match the addition problems to the right pictures and solve them.

a. $3 + 3 + 3 =$ _____



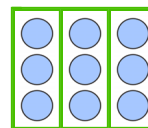
b. $2 + 2 + 2 + 2 =$ _____

c. $4 + 4 + 4 =$ _____

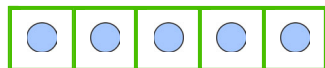
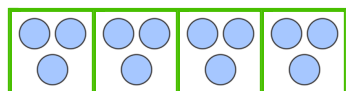
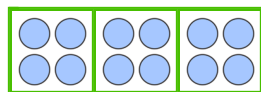


d. $2 + 2 + 2 =$ _____

e. $3 + 3 + 3 + 3 =$ _____

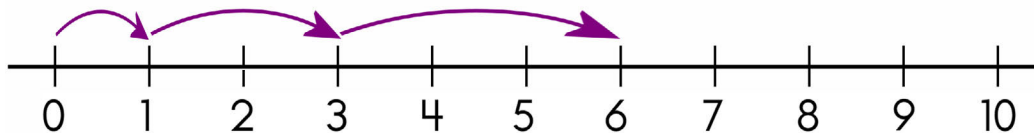


f. $1 + 1 + 1 + 1 + 1 =$ _____

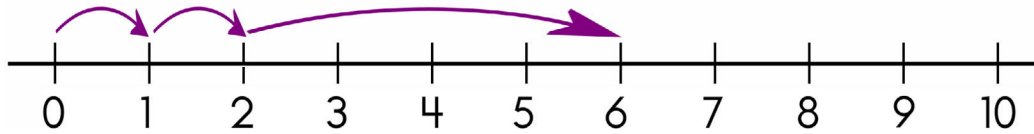


3. Write the additions that match the number line jumps.

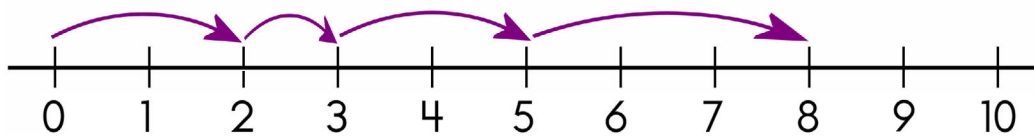
Remember to check how many steps long each arrow is.



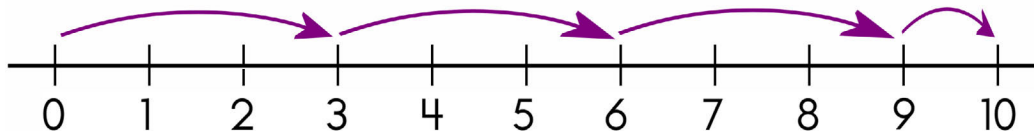
a. _____ + _____ + _____ = _____



b. _____ + _____ + _____ = _____



c. _____ + _____ + _____ + _____ = _____



d. _____ + _____ + _____ + _____ = _____

4. Are these additions right? If not, correct them.

a. $1 + 2 + 3 = 6$

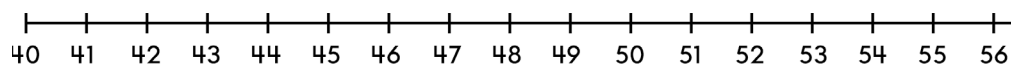
b. $2 + 2 + 3 = 8$

c. $1 + 7 + 2 = 9$

d. $2 + 5 + 2 = 9$

Puzzle Corner

Illustrate the sum $43 + 6 + 5$ on the number line.



Sums to 9

1. Write the sums to show the different ways to group nine marbles into two groups.

9	
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____

2. Play “9 Out” *and/or* “Some Went Hiding” with 9 objects (see the introduction).

3. **Drill.** Say the answers aloud as your teacher points to problems. (Don’t write the answers.)

$\square + 8 = 9$

$\square + 4 = 9$

$2 + \square = 9$

$3 + \square = 9$

$\square + 7 = 9$

$\square + 2 = 9$

$\square + 6 = 9$

$9 + \square = 9$

$6 + \square = 9$

$\square + 1 = 9$

$\square + 7 = 9$

$\square + 3 = 9$

$0 + \square = 9$

$4 + \square = 9$

$\square + 5 = 9$

4. Fill in the missing numbers. You may draw dots to help. Notice the patterns!

a. $2 + \underline{\quad} = 9$

$3 + \underline{\quad} = 9$

$4 + \underline{\quad} = 9$

$5 + \underline{\quad} = 9$

b. $2 + \underline{\quad} = 7$

$3 + \underline{\quad} = 7$

$4 + \underline{\quad} = 7$

$5 + \underline{\quad} = 7$

c. $2 + \underline{\quad} = 8$

$3 + \underline{\quad} = 8$

$4 + \underline{\quad} = 8$

$5 + \underline{\quad} = 8$

5. Write an addition to match the situation.

Luis put seashells in boxes. He put four shells in one box, and five in another.

Altogether, the two boxes have _____ shells.

$\underline{\quad} + \underline{\quad} = \underline{\quad}$

6. Solve. Think: “Is it asking the total? Or, do I already know the total?” Draw to help.

a. There are two red chairs in the living room, six red chairs in the kitchen, and none in the other rooms. How many red chairs are in the house?

b. A full crayon box has eight crayons. You see four crayons in the box. How many crayons are missing?

7. Add.

a.
$$\begin{array}{r} 2 \\ + 5 \\ \hline \end{array}$$

b.
$$\begin{array}{r} 1 \\ + 6 \\ \hline \end{array}$$

c.
$$\begin{array}{r} 4 \\ + 4 \\ \hline \end{array}$$

d.
$$\begin{array}{r} 7 \\ + 1 \\ \hline \end{array}$$

e.
$$\begin{array}{r} 7 \\ + 2 \\ \hline \end{array}$$

f.
$$\begin{array}{r} 3 \\ + 5 \\ \hline \end{array}$$

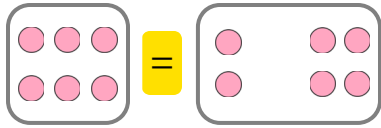
g.
$$\begin{array}{r} 4 \\ + 2 \\ \hline \end{array}$$

h.
$$\begin{array}{r} 3 \\ + 4 \\ \hline \end{array}$$

i.
$$\begin{array}{r} 1 \\ + 5 \\ \hline \end{array}$$

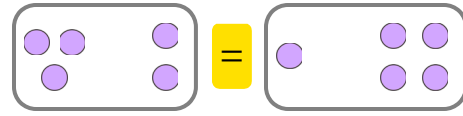
j.
$$\begin{array}{r} 4 \\ + 5 \\ \hline \end{array}$$

Equality



$$6 = 2 + 4$$

Six equals two plus four.



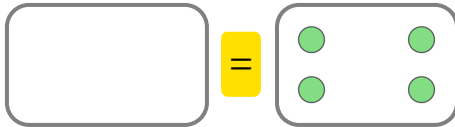
$$3 + 2 = 1 + 4$$

Notice how both sides equal 5!

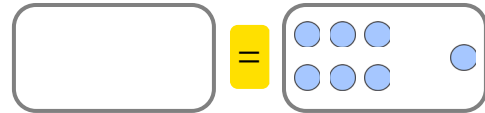
The symbol = means “EQUALS”. The things on the two sides of that sign have to be equal in value.

The symbol = does *not* mean “calculate the answer.”

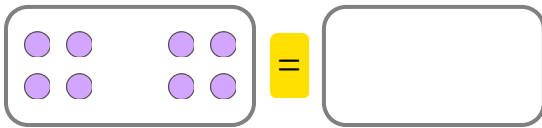
1. Draw more dots to make both sides have the same number of dots. Fill in the missing numbers. To help you, you can draw a vertical line through the equals signs, creating two sides. Both sides have to have the same number of dots.



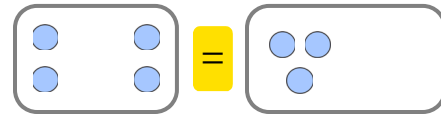
a. $\underline{\hspace{2cm}} = 2 + 2$



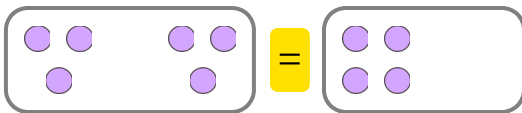
b. $\underline{\hspace{2cm}} = 6 + 1$



c. $4 + 4 = \underline{\hspace{2cm}}$



d. $2 + 2 = 3 + \underline{\hspace{2cm}}$

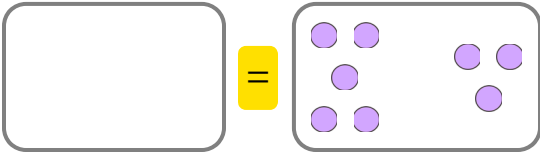
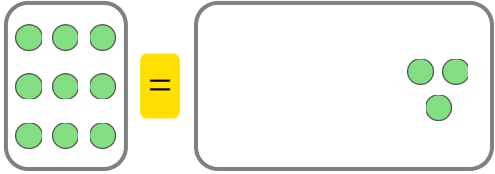
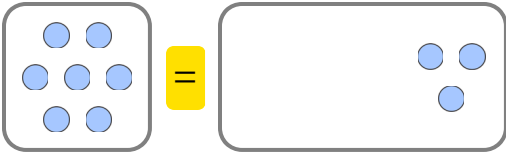
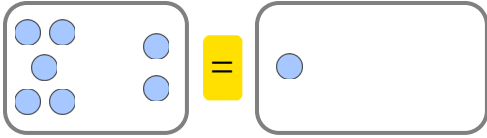


e. $3 + 3 = 4 + \underline{\hspace{2cm}}$



f. $1 + \underline{\hspace{2cm}} = 3 + 3$

2. Draw more dots to make both sides have the same number of dots. Fill in the missing numbers.

 a. <u> </u> = 5 + 3	 b. 9 = <u> </u> + 3
 c. 7 = <u> </u> + 3	 d. 5 + 2 = 1 + <u> </u>

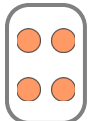
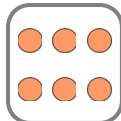
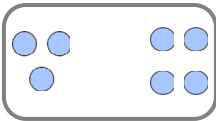
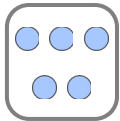
3. Fill in the missing numbers. You can draw dots to help you.

a. <u> </u> = 6 + 1	b. <u> </u> = 4 + 4	c. <u> </u> = 9 + 1
d. 4 = <u> </u> + 1	e. 8 = <u> </u> + 4	f. 6 = <u> </u> + 2
g. 5 = 4 + <u> </u>	h. 7 = 5 + <u> </u>	i. 6 = 3 + <u> </u>

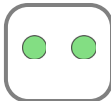
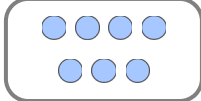
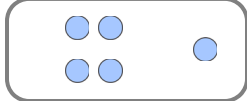
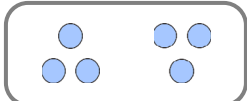
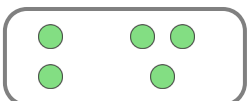
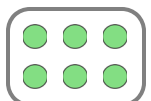
4. Fill in the missing numbers. You can draw dots to help you.

a. <u> </u> + 2 = 3 + 3	b. <u> </u> + 1 = 5 + 2	c. <u> </u> + 2 = 3 + 1
d. 2 + 2 = <u> </u> + 3	e. 5 + 3 = 4 + <u> </u>	f. 6 + 1 = 3 + <u> </u>

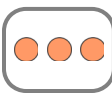
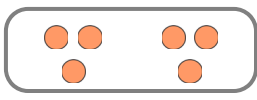
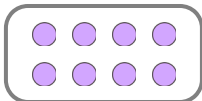
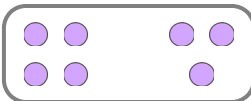
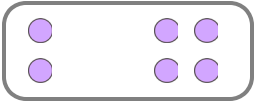
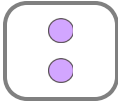
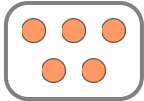
Comparisons

 $<$  $4 < 6$ Four is <u>less than</u> six.	 $>$  $3 + 4 > 5$ Three plus four is <u>greater than</u> five. (Because 7 is greater than 5.)	$<$ means “IS LESS THAN” $>$ means “IS GREATER THAN”
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1. Fill in the missing numbers.

 $<$  a. $\underline{\quad} < \underline{\quad} + \underline{\quad}$	 $>$  b. $\underline{\quad} > \underline{\quad} + \underline{\quad}$
 $>$  c. $\underline{\quad} + \underline{\quad} > \underline{\quad}$	 $<$  d. $\underline{\quad} + \underline{\quad} < \underline{\quad}$

2. Compare, writing $<$ or $>$ in the box. Also, fill in the missing numbers.

  a. $\underline{\quad} \square \underline{\quad} + \underline{\quad}$	  b. $\underline{\quad} \square \underline{\quad} + \underline{\quad}$
  c. $\underline{\quad} + \underline{\quad} \square \underline{\quad}$	  d. $\underline{\quad} + \underline{\quad} \square \underline{\quad}$

Here we compare $3 + 2$ and 4 .
First add $3 + 2$. It is 5 .
(We write 5 below the $3 + 2$.)

Then we compare 5 and 4 .
Five is greater so we draw $>$.

$$\begin{array}{ccc} 3 + 2 & ? & 4 \\ \downarrow & & \downarrow \\ 5 & > & 4 \end{array}$$

3. First add. Write the answer below. Then compare and write $<$, $>$ or $=$.

a. $5 + 2$? 4 $\downarrow$ $\downarrow$ 4	b. $4 + 4$? 7 $\downarrow$ $\downarrow$ 7	c. 2 ? $1 + 1$ $\downarrow$ $\downarrow$ 	d. 7 ? $3 + 6$ $\downarrow$ $\downarrow$
e. $4 + 1$? 3 $\downarrow$ $\downarrow$ 	f. $3 + 5$? 8 $\downarrow$ $\downarrow$ 	g. 9 ? $6 + 2$ $\downarrow$ $\downarrow$ 	h. 6 ? $2 + 8$ $\downarrow$ $\downarrow$

4. Add in your head. Compare the sum to the other number. Then write $<$, $>$, or $=$.

a. $1 + 4$ 3	b. $2 + 2$ 5	c. 2 $0 + 0$	d. 3 $0 + 3$
e. $4 + 4$ 9	f. $8 + 1$ 9	g. 0 $6 + 2$	h. 8 $3 + 4$

Puzzle Corner

Compare.

a. $3 + 1 + 4$ $2 + 3 + 5$

b. $11 + 1 + 2$ $2 + 10 + 1$

c. $1 + 15 + 2$ $2 + 15 + 1$

d. $1 + 5 + 1 + 2$ $2 + 3 + 3 + 3$

Addition Practice

1. Remember: you can add the numbers in either order! Which way is easier?

a.	b.	c.	d.
$6 + 1 = \underline{\quad}$	$7 + 2 = \underline{\quad}$	$1 + 8 = \underline{\quad}$	$2 + 5 = \underline{\quad}$
$1 + 6 = \underline{\quad}$	$2 + 7 = \underline{\quad}$	$8 + 1 = \underline{\quad}$	$5 + 2 = \underline{\quad}$

2. Add three numbers. Try to find the easiest order!

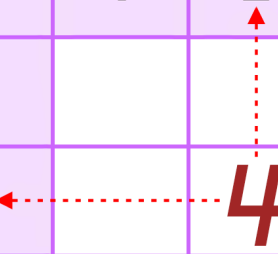
a. $1 + 8 + 1 = \underline{\quad}$	b. $4 + 1 + 2 = \underline{\quad}$	c. $3 + 1 + 3 = \underline{\quad}$
$2 + 2 + 2 = \underline{\quad}$	$0 + 7 + 2 = \underline{\quad}$	$1 + 1 + 1 = \underline{\quad}$

3. **Double** means two times the same thing! Draw dots or sticks. Write the total in the box.

 a. Double 2 4	b. Double 3	c. Double 4
d. Double 5	e. Double 0	f. Double 1

4. Fill in the addition tables. Add the number above and the number to the left.

+	1	2	3
1			
2			
3			



+	1	2	3
4			
5			
6			

5. Add.

a. $4 + 4 = \underline{\hspace{2cm}}$ $6 + 2 = \underline{\hspace{2cm}}$	b. $4 + 3 = \underline{\hspace{2cm}}$ $7 + 2 = \underline{\hspace{2cm}}$	c. $2 + 4 = \underline{\hspace{2cm}}$ $3 + 6 = \underline{\hspace{2cm}}$
d. $1 + 7 = \underline{\hspace{2cm}}$ $3 + 5 = \underline{\hspace{2cm}}$	e. $5 + 4 = \underline{\hspace{2cm}}$ $5 + 2 = \underline{\hspace{2cm}}$	f. $8 + 1 = \underline{\hspace{2cm}}$ $1 + 6 = \underline{\hspace{2cm}}$

6. Compare and write $<$, $>$ or $=$. (Some of these are equal!)

a. $4 + 1 \square 5$	b. $7 \square 4 + 4$	c. $6 \square 2 + 3$
d. $2 + 5 \square 7$	e. $5 \square 5 + 0$	f. $10 \square 5 + 5$
g. $2 + 2 \square 3$	h. $9 \square 1 + 8$	i. $2 \square 2 + 2$

7. Let's make charts!

a. Add 1 $5 + 1 = \underline{\hspace{2cm}}$ $6 + 1 = \underline{\hspace{2cm}}$ $7 + 1 = \underline{\hspace{2cm}}$ $8 + 1 = \underline{\hspace{2cm}}$ $9 + 1 = \underline{\hspace{2cm}}$	b. Add 2 $2 + 2 = \underline{\hspace{2cm}}$ $3 + 2 = \underline{\hspace{2cm}}$ $4 + 2 = \underline{\hspace{2cm}}$ $5 + 2 = \underline{\hspace{2cm}}$ $6 + 2 = \underline{\hspace{2cm}}$	c. Add 3 $2 + 3 = \underline{\hspace{2cm}}$ $3 + 3 = \underline{\hspace{2cm}}$ $4 + 3 = \underline{\hspace{2cm}}$ $5 + 3 = \underline{\hspace{2cm}}$ $6 + 3 = \underline{\hspace{2cm}}$
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Sums to 10

1. Write the sums to show the different ways to group ten marbles into two groups.

 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	 _____ + _____
 _____ + _____	

2. Play “10 Out” *and/or* “Some Went Hiding” with 10 objects (see the introduction).

3. **Drill.** Say the answers aloud as your teacher points to problems. (Don't write the answers.)

$\square + 6 = 10$ $\square + 4 = 10$ $1 + \square = 10$ $6 + \square = 10$ $3 + \square = 10$

$\square + 3 = 10$ $\square + 5 = 10$ $7 + \square = 10$ $9 + \square = 10$ $4 + \square = 10$

$\square + 8 = 10$ $\square + 9 = 10$ $2 + \square = 10$ $5 + \square = 10$ $8 + \square = 10$

4. Solve. Think: Are you asked the total? Or do you already know the total?

You can draw or act out the situation, to help you.

a. There were three birds in the tree. Seven more flew in. How many birds are now in the tree?	b. Tina got seven books from the library. She has read three of them. How many books has she not read?
c. Jessica has ten dolls. She sees four of them in her room. How many are somewhere else?	d. Larry has three toy cars and his brother also has three. How many do they have together?

5. Connect two numbers together if they make ten.

4	1	6	4	4	3
5	2	3	8	2	2
3	7	1	9	2	5
3	6	5	7	5	7
9	0	3	2	3	8

More Practice

1. Fill in the missing numbers. Notice the patterns!

a.

$$2 + \underline{\quad} = 10$$

$$3 + \underline{\quad} = 10$$

$$4 + \underline{\quad} = 10$$

$$5 + \underline{\quad} = 10$$

b.

$$2 + \underline{\quad} = 9$$

$$3 + \underline{\quad} = 9$$

$$4 + \underline{\quad} = 9$$

$$5 + \underline{\quad} = 9$$

c.

$$2 + \underline{\quad} = 8$$

$$3 + \underline{\quad} = 8$$

$$4 + \underline{\quad} = 8$$

$$5 + \underline{\quad} = 8$$

2. Compare, writing $<$, $>$ or $=$ in the box. (Note: first you need to add.)

a. $1 + 9$ 9

c. $4 + 4$ 9

e. 6 $5 + 2$

g. 9 $5 + 4$

b. $5 + 5$ 10

d. $3 + 5$ 7

f. 10 $6 + 3$

h. 7 $7 + 1$

3. Draw a line to the correct answer.

$$7 + 1$$

$$2 + 6$$

$$3 + 4$$

$$5 + 2$$

$$4 + 4$$

$$1 + 6$$

$$5 + 3$$

7

8

9

$$7 + 3$$

$$3 + 6$$

$$4 + 6$$

$$1 + 8$$

$$5 + 4$$

$$3 + 7$$

$$2 + 8$$

10

4. Solve. (Remember, you can draw or act out the situation.)

a. Bill has ten toy cars but he can find only six. How many are missing?	b. Jack saw two birds on the lawn and five on the fence. How many birds did he see in all?
c. Together, Jessica and Jenny have ten books. Jenny has five of them. How many does Jessica have?	d. The store has ten dolls. Two of them are on the bottom shelf. The rest are on the top shelf. How many dolls are on the top shelf?

5. Fill in as much of the addition table as you can, and do not worry about the rest.
Color the square blue if the answer is **8**.

+	1	2	3	4	5	6	7
0							
1							
2							
3							
4							
5							

Chapter 1 Review

1. Write the different sums of 5 and of 6 .

$$5 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$5 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$5 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$6 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$6 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$6 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$6 = \underline{\quad\quad} + \underline{\quad\quad}$$

2. Draw a line to the correct answer.

$$4 + 1$$

$$2 + 3$$

$$3 + 3$$

$$4 + 2$$

5

$$5 + 1$$

$$0 + 6$$

$$1 + 4$$

$$2 + 4$$

6

3. Write the missing numbers.

a.

$$\underline{\quad\quad} + 2 = 6$$

$$\underline{\quad\quad} + 1 = 6$$

$$2 + \underline{\quad\quad} = 5$$

$$1 + \underline{\quad\quad} = 5$$

$$6 + \underline{\quad\quad} = 6$$

b.

$$6 = 0 + \underline{\quad\quad}$$

$$5 = 0 + \underline{\quad\quad}$$

$$6 = 3 + \underline{\quad\quad}$$

$$6 = 4 + \underline{\quad\quad}$$

$$5 = 4 + \underline{\quad\quad}$$

4. Compare, writing $<$, $>$, or $=$ in the box.

a. $2 + 2$ 5

b. $2 + 4$ 5

c. $4 + 4$ 5

d. $0 + 5$ 5

e. 6 $2 + 4$

f. 6 $2 + 5$

5. Write the different sums of 7 and of 8.

$$7 = \underline{\quad} + \underline{\quad}$$

$$7 = \underline{\quad} + \underline{\quad}$$

$$7 = \underline{\quad} + \underline{\quad}$$

$$7 = \underline{\quad} + \underline{\quad}$$

$$7 = \underline{\quad} + \underline{\quad}$$

$$7 = \underline{\quad} + \underline{\quad}$$

$$8 = \underline{\quad} + \underline{\quad}$$

$$8 = \underline{\quad} + \underline{\quad}$$

$$8 = \underline{\quad} + \underline{\quad}$$

$$8 = \underline{\quad} + \underline{\quad}$$

$$8 = \underline{\quad} + \underline{\quad}$$

$$8 = \underline{\quad} + \underline{\quad}$$

6. Draw a line to the correct answer.

7

$$4 + 3$$

$$2 + 6$$

$$3 + 5$$

$$4 + 4$$

$$5 + 2$$

$$1 + 6$$

$$5 + 3$$

$$7 + 1$$

$$6 + 2$$

8

7. Add.

$$\begin{array}{r} \text{a.} \quad 4 \\ + 2 \\ \hline \end{array}$$

$$\begin{array}{r} \text{b.} \quad 5 \\ + 3 \\ \hline \end{array}$$

$$\begin{array}{r} \text{c.} \quad 1 \\ + 7 \\ \hline \end{array}$$

$$\begin{array}{r} \text{d.} \quad 1 \\ + 4 \\ \hline \end{array}$$

$$\begin{array}{r} \text{e.} \quad 2 \\ + 5 \\ \hline \end{array}$$

$$\begin{array}{r} \text{f.} \quad 3 \\ + 3 \\ \hline \end{array}$$

8. Are these correct? Circle true or false.

a. $7 + 3 = 10$ true or false	b. $7 = 1 + 5$ true or false
c. $9 = 5 + 5$ true or false	d. $2 + 2 = 1 + 3$ true or false
e. $2 + 4 = 7$ true or false	f. $3 + 3 = 7 + 2$ true or false

9. Write the different sums of 9 and of 10.

$$9 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$9 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$9 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$9 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$9 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$9 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$10 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$10 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$10 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$10 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$10 = \underline{\quad\quad} + \underline{\quad\quad}$$

$$10 = \underline{\quad\quad} + \underline{\quad\quad}$$

10. Draw a line to the correct answer.

$$2 + 7$$

$$3 + 6$$

$$4 + 6$$

$$5 + 5$$

9

$$9 + 1$$

10

$$1 + 8$$

$$5 + 4$$

$$3 + 7$$

$$2 + 8$$

11. Add.

a. $4 + 1 + 1 = \underline{\quad\quad}$

b. $8 + 2 + 0 = \underline{\quad\quad}$

c. $1 + 3 + 6 = \underline{\quad\quad}$

d. $2 + 2 + 4 = \underline{\quad\quad}$

e. $5 + 2 + 0 + 0 = \underline{\quad\quad}$

f. $1 + 2 + 2 + 1 = \underline{\quad\quad}$

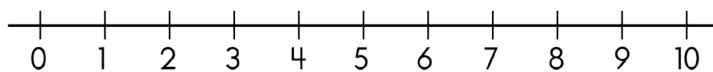
g. $2 + 3 + 1 + 3 = \underline{\quad\quad}$

12. Solve.

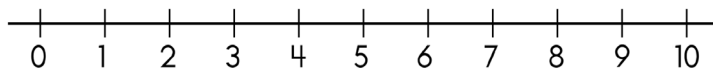
Emily put three chairs in a row. Behind them she put another three chairs, and then behind them three more chairs. (You can draw a picture.)
How many chairs did she use?

13. What numbers can make 10? Choose numbers that make 10, and draw arrows to illustrate the sums on the number lines.

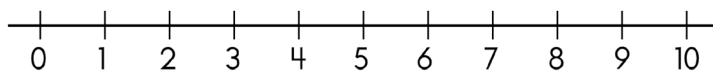
a. $10 = \underline{\quad\quad} + \underline{\quad\quad}$



b. $10 = \underline{\quad\quad} + \underline{\quad\quad}$



c. $10 = \underline{\quad\quad} + \underline{\quad\quad}$



14. Fill in as much as you can of the addition table.

+	2	4	3	6	7	5	8
1							
3							
4							
2							

Puzzle Corner

Complete these number puzzles. Is there more than one solution?

	+		= 9
+		+	
	+		= 9
= 10		= 8	

	+		= 9
+		+	
	+		= 8
= 9		= 8	

Chapter 2: Subtraction Within 0-10

Introduction

The second chapter of *Math Mammoth Grade 1* covers the concept of subtraction, the relationship between addition and subtraction, and the various meanings of subtraction.

In the first lesson, *Subtraction Is Taking Away*, the child learns the basic meaning of subtraction as taking away objects, and learns to write subtractions from an illustration where some objects are crossed out. The child can figure out the answers by simply counting how many objects are left.

If the child does not yet know the word “minus,” it is a good idea to introduce it first orally. Use blocks or other concrete objects. For example, show the child eight blocks and take away three blocks. Then use both kinds of wordings: “Eight blocks, take away three blocks, leaves five blocks. Eight blocks *minus* three blocks *equals* five blocks.” Then let the child do the same. Play with concrete objects until the child can use the words “minus” and “equals” in his or her own speech.

In the next lesson, the child counts down to subtract, which ties in subtraction with the number line. This is a transitional strategy to solve subtraction problems, because later students will learn more efficient ways to subtract, but it is important conceptually. For now, the student can solve $9 - 3$ by counting down three steps from nine: eight, seven, six. So the answer is six.

The following lesson, *Subtraction and Addition in the Same Picture*, begins the study of the relationship between addition and subtraction. This concept will span several lessons. This first lesson presents two sets of objects, such as blue and white balls, and the student writes both an addition sentence and a subtraction sentence from this illustration.

The lesson *When Can You Subtract?* concentrates on the idea that some subtractions, such as $4 - 5$, are meaningless when you think of taking away. The child also makes subtraction patterns in this lesson.

Then we continue studying the connection between addition and subtraction in the lesson *Two Subtractions from One Addition*. As an example, the child writes both $8 - 3 = 5$ and $8 - 5 = 3$ from the addition $3 + 5 = 8$. This idea ties in with fact families, a concept that is coming up soon.

In the lesson *Two Parts—One Total*, we study word problems that do not involve the idea of taking away but have two parts making up a total. For example, if there are 10 flowers of which some are white and some are red, and seven of them are white, how many are red? We know the “parts” (the red and white flowers) add up to 10, so we can write a missing addend sentence $7 + \underline{\quad} = 10$. This can be solved by subtracting $10 - 7$ or by knowing the addition fact $7 + 3 = 10$.

Then we study fact families. This means writing two additions and two subtractions using the same three numbers. Fact families will be used extensively in the next chapter.

In the lesson *How Many More?* students find how many more or how many fewer objects one person has than the other by drawing the objects. This lesson can easily be done with manipulatives if desired.

In the very next lesson, “*How Many More? Problems and Differences*,” we continue the theme, this time writing a missing addend addition for problems that ask “how many more.” For example, Veronica has 4 marbles and Ann has 6. We write the missing addend sentence $4 + \underline{\quad} = 6$ to find how many more Ann has. In the next lesson the child then learns to write subtraction sentences for such problems.

Feel free to mix the lessons from this chapter with the lessons from Chapter 3 (place value) if your student or child gets bored or has attention issues with the topic of subtraction.

Good Mathematical Practices

- In the previous chapter, students mainly encountered word problems of these types of situations: “add to” and two parts forming a whole. These situations are still present in the problems in this chapter, but we will now also include “take from” type of situations and comparisons (how many more/fewer).

This means the word problems in this chapter may involve either addition or subtraction. Note that often, the same problem can be solved either by subtraction or by a missing-number addition. Consider for example this problem: “Julie has eight shirts, but she can only find two. How many are missing?” It can be solved by subtraction $8 - 2$ or by thinking of the addition $2 + \underline{\quad} = 8$.

Including a variety of word problem situations is important, so that children learn to think and not merely blindly apply the operation in the title of the lesson. Practice makes perfect, including in problem solving, and our goal is to start early.

If your student makes a mistake (whether in word problems or elsewhere), don't put down the mistake. Mistakes are actually valuable as they enable the neurons in the brain to grow new connections. Thinking about mistakes and trying to correct them literally causes your brain to grow! So, just encourage your student(s) to try again when they get something wrong. You can also ask them to explain their thinking, and gently guide them towards the correct “path.”

Pacing Suggestion for Chapter 2

Please add one day to the pacing for the test if you will use it.

The Lessons in Chapter 2	page	span	suggested pacing	your pacing
Basic Subtraction	80	2 pages	1 day	
Subtracting on the Number Line	82	2 pages	1 day	
Count Down to Subtract	84	2 pages	1 day	
Subtraction and Addition in the Same Picture	86	3 pages	1 day	
When Can You Subtract?	89	2 pages	1 day	
Equations	91	2 pages	1 day	
Two Subtractions from One Addition	93	3 pages	1 day	
Two Parts—One Total	96	2 pages	1 day	
Fact Families	98	2 pages	1 day	
More Fact Families	100	2 pages	1 day	
Subtraction Practice	102	2 pages	1 day	
How Many More? Part 1	104	2 pages	1 day	
How Many More? Part 2	106	2 pages	1 day	
How Many More? Part 3	108	2 pages	1 day	
Difference	110	2 pages	1 day	
“How Many More” Problems and Subtraction	112	2 pages	1 day	
Solving Subtraction Problems with Addition	114	2 pages	1 day	
Chapter 2 Mixed Review	116	2 pages	1 day	
Chapter 2 Review	118	1 page	1 day	
Chapter 2 Test (optional)				
TOTALS		39 pages	19 days	

Math Talks

I invite you to take a look at these math-inspired real-life photographs. Use them to have short conversations with your student(s) about how we can see math in real life, with no pressure to get any “right answer.”

<https://www.mathmammoth.com/MathTalks>

Each photo has a connection to some simple math concept, such as counting, addition, or shapes. Many times, you can find several math concepts in it, such as both addition and subtraction, or both squares and multiplication. But there is no obligation to find all the possibilities. The idea is just to have a relaxed time, sharing one’s thoughts and listening to others’ ideas.

To help the child get started, you can ask, “What do you notice? What do you wonder?” Another great way to use the photos is to ask the child or students to make a story about the photo that uses numbers or math in some way.

I suggest you have a “math talk” a few times a week. It should only take a few minutes. Soon you will start noticing opportunities for such talks in your life, also! Just point out to your child or students objects in life that relate to math in some way, or ask them to make a math story of some scene. If your child has access to a camera, you can also ask them to take “math talk” photos themselves.

Games and Activities

Number Line Activity

You need: A six-sided dice and two play markers. Also, draw a number line from 0 to 15 on a blank paper. Mark number 10 in a special way. For example, you can make it bolder or bigger, or circle it.

Game Play: Place both markers at zero on the number line. The goal is to reach number 10. On your turn, roll the dice, and say an addition sentence starting with the position where your marker is, adding to it what the dice shows. For example, if your marker is at 2, and you roll a 5, you say “ $2 + 5 = 7$ ”, and then move your marker to seven. Once your marker is past 10, you say a subtraction sentence instead. For example, if your marker is at 12, and you roll a 4, you will say, “ $12 - 4 = 8$ ”, and move your marker to 4. The winner is the person who gets their game piece to land at 10.

It is important to say the addition/subtraction sentence before moving the piece. Otherwise, this activity amounts to counting steps on a number line and is reduced to a counting activity.

Note: If a child cannot do a particular subtraction or addition, they can use the number line to figure out the answer, before moving their piece.

Rabbit Race

You need: A set of number cards from 1-10 OR a regular dice OR a 10-sided dice. A marker for each player. Download the game board from here:
<https://www.mathmammoth.com/download/Rabbit-Race.pdf>

Game Play: First, choose a number to subtract from, such as 10. This number can be any from 6 to 10. If you use 6, 7, 8, or 9 as this number, adjust your deck of number cards accordingly or use the regular dice (not the 10-sided dice). We don’t want to end up with negative numbers!

On your turn, take a card from the deck or roll the dice. Subtract the number you get from 10. Then move your marker that many steps. Follow the instructions on the game board for any special squares. The winner is the player who first reaches the END.

Let's Make a Difference!

You need: A large deck of number cards with numbers 1-10.

Preparation: Deal six cards to each player. Place the rest face down in a pile in the middle of the table.

Game play: On your turn, check if you can make a subtraction sentence using any three cards in your hand. If so, say that subtraction aloud, and put those three cards in your personal pile. If not, take *one* card from the middle pile, and check again. If you still cannot make a subtraction, your turn ends. At the end of your turn, draw enough cards from the middle pile so that you once again have at least six cards in your hand (if you already have six or more, don't draw more at this point). The turn then passes to the next player.

Play in this manner until there are no more cards in the middle pile. At that point, the player with most cards in their personal pile wins.

Note: Essentially, players can check if they have three cards that make an *addition* sentence. However, it is best if the child notices that on his/her own.

Variation: For an easier game, remove 10s, 9s, and perhaps even 8s from the deck.

Subtraction Challenge

You need: A standard deck of playing cards from which you remove the face cards.

Game Play: In each round, each player is dealt two cards face up, and has to calculate their difference (subtract). The player with the highest difference (highest answer) gets all the cards played in that round, including from the other players, and puts them in his personal pile. If two or more players have the same answer, then those players get an additional two cards, and use those to resolve the tie.

After enough rounds have been played to use all of the cards, the player with the most cards wins.

Games and Activities at Math Mammoth Practice Zone

Connection between Addition and Subtraction

<https://www.mathmammoth.com/practice/add-sub-connection>

Fact Families

<https://www.mathmammoth.com/practice/fact-families>

Subtraction Hidden Picture Game

Choose a number range (such as 1 to 10) and uncover a hidden picture while solving subtraction problems.

<https://www.mathmammoth.com/practice/mystery-picture-subtraction>

Number Bonds

Practice number bonds, either with pictures or with numbers.

<https://www.mathmammoth.com/practice/number-bonds>

Fruity Math

Choose addition or subtraction, level 1. Click the fruit with the correct answer and try to get as many points as you can within two minutes.

<https://www.mathmammoth.com/practice/fruity-math>

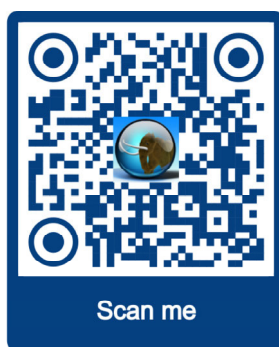
Further Resources on the Internet

We have compiled a list of Internet resources that match the topics in this chapter, including pages that offer:

- **online practice** for concepts;
- online **games**, or occasionally, printable games;
- **animations** and interactive **illustrations** of math concepts;
- **articles** that teach a math concept.

We heartily recommend you take a look! Many of our customers love using these resources to supplement the bookwork. You can use these resources as you see fit for extra practice, to illustrate a concept better and even just for some fun. Enjoy!

<https://l.mathmammoth.com/gr1ch2>



Basic Subtraction

From five bananas
we take away three.
(Cover three bananas.)



$$5 - 3 = 2$$

There are two left.

Five minus three equals two.

1. Write a subtraction sentence to match each picture.



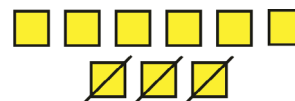
a. $5 - 1 = \underline{\quad}$



b. $\underline{\quad} - \underline{\quad} = \underline{\quad}$



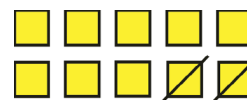
c. $\underline{\quad} - \underline{\quad} = \underline{\quad}$



d. $\underline{\quad} - \underline{\quad} = \underline{\quad}$

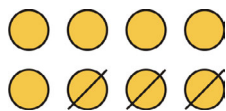


e. $\underline{\quad} - \underline{\quad} = \underline{\quad}$



f. $\underline{\quad} - \underline{\quad} = \underline{\quad}$

2. Solve. Draw dots and cross out some of them to match the subtraction problem.



a. $8 - 3 = \underline{\quad}$

b. $5 - 1 = \underline{\quad}$

c. $10 - 7 = \underline{\quad}$

d. $9 - 5 = \underline{\quad}$

e. $10 - 4 = \underline{\quad}$

f. $8 - 6 = \underline{\quad}$

3. Solve. You can draw dots to help you.

a. $7 - 2 = \underline{\hspace{2cm}}$	b. $10 - 1 = \underline{\hspace{2cm}}$	c. $9 - 7 = \underline{\hspace{2cm}}$
d. $6 - 3 = \underline{\hspace{2cm}}$	e. $7 - 3 = \underline{\hspace{2cm}}$	f. $10 - 4 = \underline{\hspace{2cm}}$

4. Solve the problems.

a. There were seven birds in a tree. Three flew away. How many are left?	b. Mom has four silver plates in the cupboard and five on the table. How many does she have in all?
c. Five girls were jumping rope. Then four more came. How many girls are now jumping rope?	d. Tina had six bunches of flowers. She sold six of them. How many were left?

5. Compare. Write $<$, $>$, or $=$.

a. $2 + 6$ 9

$3 + 6$ 9

$4 + 6$ 9

b. $4 - 2$ 5

$3 - 1$ 3

$4 - 4$ 0

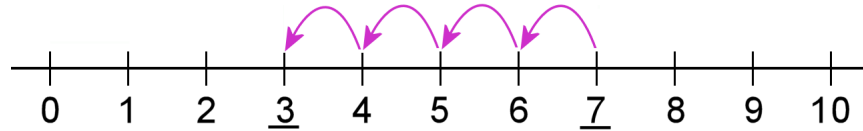
c. 10 $10 + 4$

10 $10 + 0$

10 $10 + 7$

Subtracting on the Number Line

Count 4 steps backwards from 7. You end up at 3.

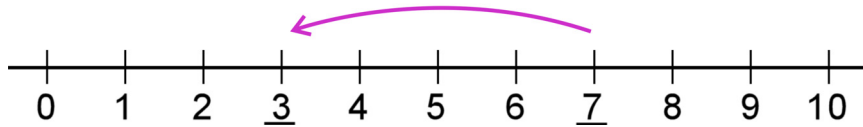


$$7 - 4 = 3$$

“Seven” “Six, five, four, three.”

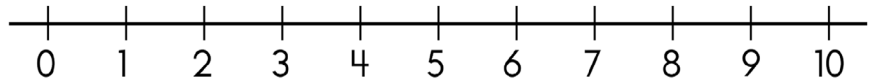
Start. Count down FOUR steps.

You can also use a single arrow:

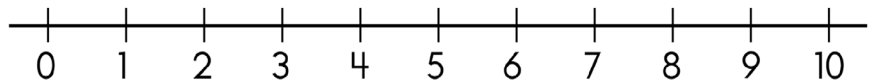


1. Draw a single arrow or several little steps to illustrate each subtraction, and solve.

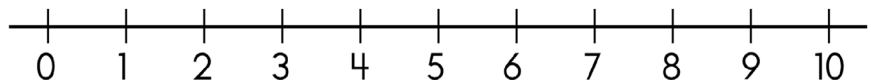
a. $7 - 2 =$ _____



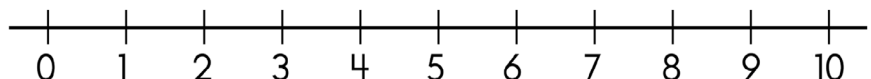
b. $8 - 4 =$ _____



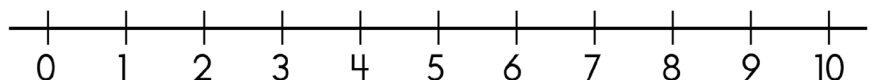
c. $6 - 5 =$ _____



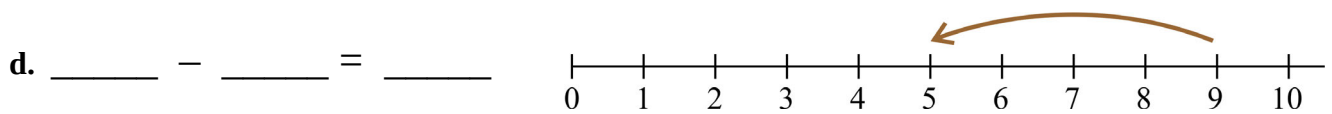
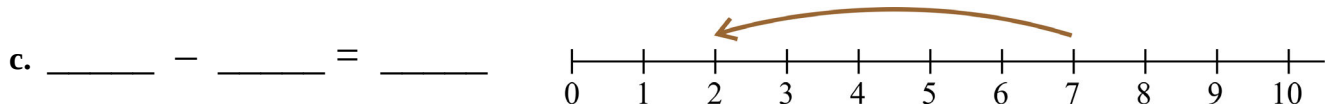
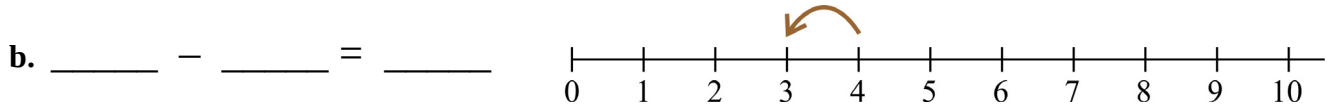
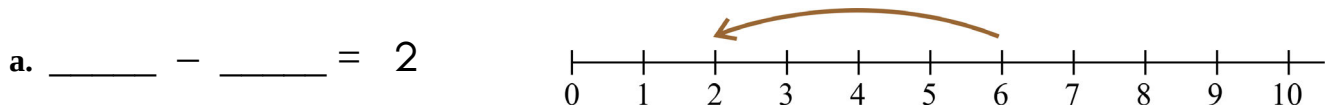
d. $9 - 3 =$ _____



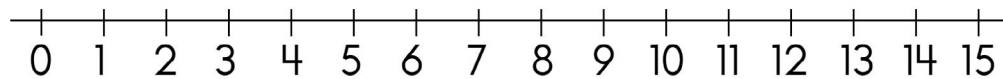
e. $10 - 3 =$ _____



2. Write the subtraction sentence that the arrow illustrates.



3. Solve. You can use the number line to help. (Start at a number, and make little jumps.)



a. $10 - 5 = \underline{\quad}$	b. $7 - 6 = \underline{\quad}$	c. $10 - 4 = \underline{\quad}$
d. $4 - 4 = \underline{\quad}$	e. $8 - 4 = \underline{\quad}$	f. $11 - 2 = \underline{\quad}$
g. $10 - 1 = \underline{\quad}$	h. $12 - 4 = \underline{\quad}$	i. $15 - 5 = \underline{\quad}$

4. Fill in the patterns.

a.	b.	c.
$10 - 1 = \underline{\quad}$	$8 - 1 = \underline{\quad}$	$13 - 1 = \underline{\quad}$
$10 - 2 = \underline{\quad}$	$8 - 2 = \underline{\quad}$	$13 - 2 = \underline{\quad}$
$10 - 3 = \underline{\quad}$	$8 - 3 = \underline{\quad}$	$13 - 3 = \underline{\quad}$
$10 - 4 = \underline{\quad}$	$8 - 4 = \underline{\quad}$	$13 - 4 = \underline{\quad}$

Count Down to Subtract

To subtract without any pictures or objects, you can count down in your head. Count down as many steps as what you need to subtract.

8 **− 3** = 5
 “Eight” “Seven, six, five.”
 Start. Count down THREE steps.

10 **− 2** = 8
 “Ten” “Nine, eight.”
 Start. Count down TWO steps.

1. Write the two previous numbers.

- a. _____, _____, 7 b. _____, _____, 4 c. _____, _____, 10
 d. _____, _____, 6 e. _____, _____, 2 f. _____, _____, 8

2. Subtract one. The answer is just the previous number!

a. $6 - 1 = \underline{\hspace{2cm}}$	b. $10 - 1 = \underline{\hspace{2cm}}$	c. $9 - 1 = \underline{\hspace{2cm}}$
$8 - 1 = \underline{\hspace{2cm}}$	$7 - 1 = \underline{\hspace{2cm}}$	$3 - 1 = \underline{\hspace{2cm}}$

3. Subtract 2 or 3. You can count down. Compare the problems.

a.	b.	c.
$6 - 2 = \underline{\hspace{2cm}}$	$9 - 2 = \underline{\hspace{2cm}}$	$7 - 2 = \underline{\hspace{2cm}}$
$6 - 3 = \underline{\hspace{2cm}}$	$9 - 3 = \underline{\hspace{2cm}}$	$7 - 3 = \underline{\hspace{2cm}}$
d.	e.	f.
$10 - 2 = \underline{\hspace{2cm}}$	$5 - 2 = \underline{\hspace{2cm}}$	$8 - 2 = \underline{\hspace{2cm}}$
$10 - 3 = \underline{\hspace{2cm}}$	$5 - 3 = \underline{\hspace{2cm}}$	$8 - 3 = \underline{\hspace{2cm}}$

4. Write a subtraction or an addition to match each situation. (Write + or – in the box.)

- a. Fanny has eight puzzles. She put three in the closet. Now she has _____ puzzles left to play with.

$$\underline{\hspace{1cm}} \boxed{} \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

- b. Faith was playing with four toy horses. Then her sister Mary brought two more. Now they are playing with _____ horses.

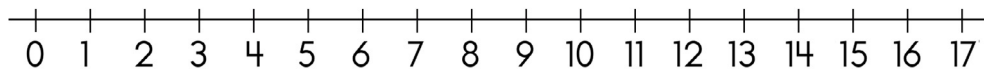
$$\underline{\hspace{1cm}} \boxed{} \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

5. Solve.

- a. Eight ducklings were walking behind Mommy Duck. Two of them stumbled and fell. How many kept walking?

- b. Ashley has three strawberries in her bowl, and so do her two brothers, Brandon and Cecil. How many strawberries in total are in those three bowls?

6. Subtract. The number line will help.



a. $14 - 2 = \underline{\hspace{2cm}}$

$14 - 4 = \underline{\hspace{2cm}}$

b. $16 - 1 = \underline{\hspace{2cm}}$

$16 - 3 = \underline{\hspace{2cm}}$

c. $11 - 3 = \underline{\hspace{2cm}}$

$11 - 4 = \underline{\hspace{2cm}}$

d. $17 - 2 = \underline{\hspace{2cm}}$

$17 - 3 = \underline{\hspace{2cm}}$

e. $12 - 1 = \underline{\hspace{2cm}}$

$12 - 2 = \underline{\hspace{2cm}}$

f. $13 - 2 = \underline{\hspace{2cm}}$

$13 - 3 = \underline{\hspace{2cm}}$

Subtraction and Addition in the Same Picture

Example 1.

Let's add the colored and the white circles:



$$4 + 6 = 10$$

Now *cover* the four colored circles.

We can write this subtraction:

$$10 - 4 = 6$$

Example 2.

There are seven circles in total:



$$3 + 4 = 7$$

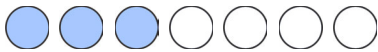
Cover the three colored circles.

We can write this subtraction:

$$7 - 3 = 4$$

1. Write an addition sentence. Then cover the *colored* things, and write a subtraction sentence.

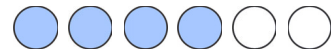
a.



$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$7 - \underline{\quad\quad} = \underline{\quad\quad}$$

b.



$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$6 - \underline{\quad\quad} = \underline{\quad\quad}$$

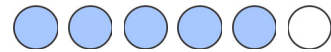
c.



$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$\underline{\quad\quad} - \underline{\quad\quad} = \underline{\quad\quad}$$

d.



$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$\underline{\quad\quad} - \underline{\quad\quad} = \underline{\quad\quad}$$

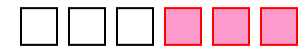
e.



$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$\underline{\quad\quad} - \underline{\quad\quad} = \underline{\quad\quad}$$

f.



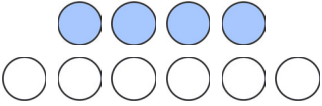
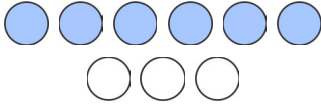
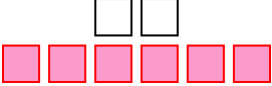
$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$\underline{\quad\quad} - \underline{\quad\quad} = \underline{\quad\quad}$$

2. Draw some colored circles and some white ones to fit the addition. Then cover the **COLORED** circles and make a subtraction sentence.

a. $7 + 1 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	b. $6 + 3 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$
c. $2 + 3 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	d. $2 + 5 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$
e. $0 + 7 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	f. $8 + 3 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$

3. Cover the colored objects, and write a subtraction sentence to fit the picture.

a.  $\underline{\quad} - \underline{\quad} = \underline{\quad}$	b.  $\underline{\quad} - \underline{\quad} = \underline{\quad}$
c.  $\underline{\quad} - \underline{\quad} = \underline{\quad}$	d.  $\underline{\quad} - \underline{\quad} = \underline{\quad}$
e.  $\underline{\quad} - \underline{\quad} = \underline{\quad}$	f.  $\underline{\quad} - \underline{\quad} = \underline{\quad}$

4. In each problem, draw some colored circles and some white circles to fit the addition sentence. Then cover the **COLORED** circles and make a subtraction sentence.

a. $9 + 1 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$	b. $7 + 2 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$
c. $10 + 4 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$	d. $10 + 2 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

5. Draw circles to fit the subtraction sentence. Write an addition sentence, too.

a. $9 - 4 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$	b. $10 - 5 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$
c. $5 - 5 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$	d. $6 - 2 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$
e. $7 - 4 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$	f. $9 - 8 = \underline{\hspace{2cm}}$ $\underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

When Can You Subtract?

Sometimes we cannot really do the subtraction problem. For example:

$$3 - 0 = \underline{\quad}$$

$$3 - 1 = \underline{\quad}$$

$$3 - 2 = \underline{\quad}$$

$$3 - 3 = \underline{\quad}$$



$$3 - 4 = ?$$

$$3 - 5 = ?$$

$$3 - 6 = ?$$

$$3 - 7 = ?$$

These we can do!

We cannot take away four or five or six triangles when there are only three!

1. Write the subtraction problems you *can* do when there are...

...four triangles



$$4 - 0 = \underline{\quad}$$

$$4 - 1 = \underline{\quad}$$

$$4 - 2 = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

...six circles



$$6 - 0 = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

...five sticks



$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

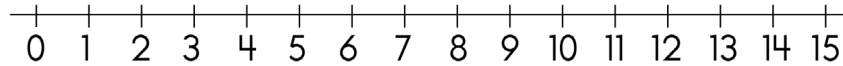
$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

You cannot do a subtraction problem in whole numbers (0, 1, 2, 3, ...) when the second number is _____ than the first number.

2. Count down to subtract.



a.	b.	c.	d.
$9 - 1 = \underline{\quad}$	$3 - 1 = \underline{\quad}$	$10 - 1 = \underline{\quad}$	$12 - 1 = \underline{\quad}$
$9 - 2 = \underline{\quad}$	$3 - 2 = \underline{\quad}$	$10 - 2 = \underline{\quad}$	$12 - 2 = \underline{\quad}$
$9 - 3 = \underline{\quad}$	$3 - 3 = \underline{\quad}$	$10 - 3 = \underline{\quad}$	$12 - 3 = \underline{\quad}$

3. Solve the subtractions that you *can* do. Don't answer the questions where you cannot take away that many.

- a. $4 - 0 =$ b. $7 - 7 =$ c. $5 - 6 =$ d. $3 - 6 =$
 e. $4 - 4 =$ f. $3 - 10 =$ g. $10 - 1 =$ h. $7 - 9 =$

4. Continue the patterns as long as you can!

a.	b.	c.
$6 - 0 = \underline{\quad}$	$10 - 5 = \underline{\quad}$	$8 - 4 = \underline{\quad}$
$6 - 1 = \underline{\quad}$	$9 - 5 = \underline{\quad}$	$7 - 4 = \underline{\quad}$
$6 - 2 = \underline{\quad}$	$8 - 5 = \underline{\quad}$	$6 - 4 = \underline{\quad}$
$6 - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - 5 = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$
$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$
$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$
$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$
$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$

Equations

An **equation** has an **equals sign**, and something (numbers, symbols) on both sides of it. Here are some equations:

$5 = 5$

$2 = 5 - 3$

$3 + 4 = 7$

$1 + 2 = 2 + 1$

This equation is **false**: $5 = 6$ Five does not equal six!

This equation is **true**: $1 = 4 - 3$ One is equal to four minus three.

1. Are these equations true or false? Circle the correct word.

a. $9 - 2 = 7$ true or false

b. $2 = 6$ true or false

c. $7 - 1 = 8$ true or false

d. $10 - 8 = 1$ true or false

e. $4 = 2 + 2$ true or false

f. $2 = 6 - 3$ true or false

g. $12 = 12$ true or false

h. $10 = 5 + 5$ true or false

2. Write one false equation and one true equation.

False equation:

True equation:

3. Choose two “math notes” or stickers that are equal in value. Then write an equation, using them and an equals sign. See the example. (Note: there are always two ways you can write the equation: what is on the right and left sides of the equals sign can be flipped.)

a. $5 + 1$ 8 6

Equation: $6 = 5 + 1$

b. 5 $2 + 2$ $1 + 3$

Equation: $\quad = \quad$

c. $4 + 3$ $5 + 2$ 6

Equation: $\quad = \quad$

d. 10 $4 + 6$ $4 + 5$

Equation: $\quad = \quad$

An equation has **two sides**: the left side and the right side. The sides are separated by the equals sign.

left side of
the equation

right side of
the equation

$$8 - 4 = 2 + 2$$

We can **flip** those sides, and the equation stays true.
(Does that make sense? If one thing is equal to another thing, then that other thing is also equal to that first thing!)

Now the sides have been flipped!

$$2 + 2 = 8 - 4$$

4. Flip the sides of the given equation, and write the new equation you get.

a. $6 + 1 = 7$
=

b. $10 - 7 = 3$
=

c. $3 - 1 = 1 + 1$
=

5. Write an equation, using two “math notes” or stickers that are equal in value.

a. $6 - 1$ 4 $4 + 1$
Equation: =

b. $6 - 3$ 2 $1 + 2$
Equation: =

c. $5 + 4$ $6 - 1$ 2 9

d. $4 + 4$ 2 $8 - 3$ 5

e. $9 - 1$ 6 $4 + 4$ 7

f. 7 $3 + 3$ $8 - 3$ 6

Build an equation using two of these and an equals sign.

Puzzle Corner

$17 + 2$ $20 - 2$ 15 $20 - 1$ $14 + 2$

Two Subtractions from One Addition

From one addition fact, you can make TWO subtraction facts:

$$3 + 4 = 7$$



$$7 - 4 = 3$$

or



$$7 - 3 = 4$$

Take away the carrots in the bottom row.

Take away the carrots in the top row.

Sometimes the two subtraction facts are the same:

$$2 + 2 = 4$$



$$4 - 2 = 2$$

or



$$4 - 2 = 2$$

Take away the carrots in the bottom row.

Take away the carrots in the top row.

1. Write one addition and two subtraction equations. First subtract the things in the bottom row, then the ones in the top row.



a. $1 + 3 = \underline{\quad 4 \quad}$

$$4 - 3 = \underline{\quad \quad}$$

or $4 - 1 = \underline{\quad \quad}$



b. $2 + 3 = 5$

$$5 - \underline{\quad \quad} = \underline{\quad \quad}$$

or $5 - \underline{\quad \quad} = \underline{\quad \quad}$



c. $\underline{\quad \quad} + \underline{\quad \quad} = \underline{\quad \quad}$

$$\underline{\quad \quad} - \underline{\quad \quad} = \underline{\quad \quad}$$

or $\underline{\quad \quad} - \underline{\quad \quad} = \underline{\quad \quad}$

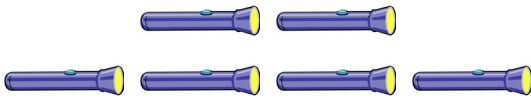


d. $\underline{\quad \quad} + \underline{\quad \quad} = \underline{\quad \quad}$

$$\underline{\quad \quad} - \underline{\quad \quad} = \underline{\quad \quad}$$

or $\underline{\quad \quad} - \underline{\quad \quad} = \underline{\quad \quad}$

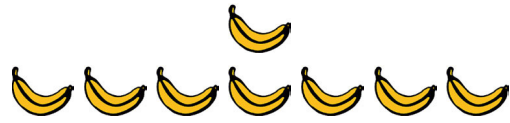
2. Write one addition and two subtraction equations.



a. _____ + _____ = _____

_____ - _____ = _____

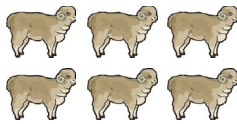
or _____ - _____ = _____



b. _____ + _____ = _____

_____ - _____ = _____

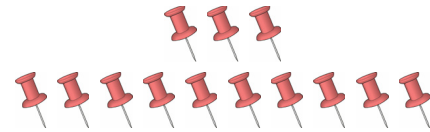
or _____ - _____ = _____



c. _____ + _____ = _____

_____ - _____ = _____

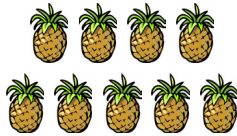
or _____ - _____ = _____



d. _____ + _____ = _____

_____ - _____ = _____

or _____ - _____ = _____



e. _____ + _____ = _____

_____ - _____ = _____

or _____ - _____ = _____



f. _____ + _____ = _____

_____ - _____ = _____

or _____ - _____ = _____

3. Are these equations true or false? Circle the correct word.

a. $10 - 2 = 7$ true or false

b. $2 = 4 - 2$ true or false

c. $7 + 1 = 8$ true or false

d. $10 - 1 = 10$ true or false

e. $8 = 2 + 5$ true or false

f. $7 = 8 - 1$ true or false

4. Complete the addition fact and the subtraction facts. Draw sticks in two groups.

 a. $6 + 1 = 7$ $7 - 1 = 6$ $7 - 6 = 1$	  b. $4 + 3 = \underline{\quad}$ $7 - \underline{\quad} = \underline{\quad}$ $7 - \underline{\quad} = \underline{\quad}$
c. $7 + 1 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	d. $5 + 4 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$
e. $2 + 6 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	f. $3 + 5 = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$

5. Write a subtraction or an addition to match the situation. (Write + or – in the box.)

Laura first ate seven cherries. Then she ate two more. She ate _____ cherries all in all.

_____ _____ = _____

Build an equation using two of these and an equals sign.

$9 + 2$

$11 + 1$

13

$14 - 2$

$10 + 4$

Puzzle Corner

Two Parts — One Total

Here you see 10 marbles. Some of them are blue and 7 are yellow.
How many are blue?

You can write an addition.
You can *also* write a subtraction,
even though nothing is taken away.



$$\underline{\quad\quad} + 7 = 10$$

$$10 - 7 = \underline{\quad\quad}$$

Cover part of the total (the yellow marbles),
and you will see the other part (the blue marbles).

1. Solve. For each problem, write an addition sentence AND a subtraction sentence.
You can draw things to help you.

- a. There are 5 blue marbles and some white marbles in a bag. In total, there are 9 marbles.
How many are white?

 marbles are white.

$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$\underline{\quad\quad} - \underline{\quad\quad} = \underline{\quad\quad}$$

- b. Mother put some blue and some red flowers in a vase. Jen counted five red ones and a total of ten. How many of the flowers are blue?

 of the flowers are blue.

$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$\underline{\quad\quad} - \underline{\quad\quad} = \underline{\quad\quad}$$

- c. There are nine children on a team, and four of them are boys. How many are girls?

 are girls.

$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$\underline{\quad\quad} - \underline{\quad\quad} = \underline{\quad\quad}$$

2. Solve. Write an addition sentence AND a subtraction sentence for each problem.
Again, you can draw something to help you.

- a. Jack has ten socks in his basket. Eight of them are white, and the rest are black.
How many are black?

_____ are black.

$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

$$\underline{\quad\quad} - \underline{\quad\quad} = \underline{\quad\quad}$$

- b. Mary saw eight chairs on the lawn, and two had blown over.
How many were still standing upright?

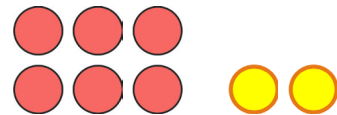
_____ are still standing upright.

$$\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$$

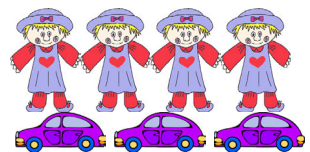
$$\underline{\quad\quad} - \underline{\quad\quad} = \underline{\quad\quad}$$

3. For each picture, make a word problem that is solved by subtraction.

a.



b.



Fact Families

Two addition facts and two subtraction facts form a **fact family** if they use the same three numbers.

For example, from 5, 3, and 2 we get the fact family on the right:

5



$$2 + 3 = 5 \quad 5 - 3 = 2$$

$$3 + 2 = 5 \quad 5 - 2 = 3$$

1. Write the fact families that match the pictures.

a. 6



$$1 + 5 = 6$$

$$5 + 1 = 6$$

$$6 - \underline{\quad} = \underline{\quad}$$

$$6 - \underline{\quad} = \underline{\quad}$$

b. 8



$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

c. 9



$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

d. 10



$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

2. Draw circles and write four *different* fact families for which the sum is 7.

a. 7 / _____ + _____ = _____ _____ + _____ = _____ _____ - _____ = _____ _____ - _____ = _____	b. 7 / _____ + _____ = _____ _____ + _____ = _____ _____ - _____ = _____ _____ - _____ = _____
c. 7 / _____ + _____ = _____ _____ + _____ = _____ _____ - _____ = _____ _____ - _____ = _____	d. 7 / _____ + _____ = _____ _____ + _____ = _____ _____ - _____ = _____ _____ - _____ = _____

3. Ann and Joe solved some problems that had missing numbers. Their answers are in the triangles. Play math teacher. Check their work and correct any mistakes that they made.

Ann: a. $8 - 4 =$ 4 b. 6 $- 4 = 1$ c. $5 -$ 2 $= 2$	Joe: d. 3 $- 4 = 1$ e. $10 - 8 =$ 1 f. $8 -$ 7 $= 1$
--	--

More Fact Families

1. Write the four facts (two additions, two subtractions) for the given numbers.

a. Numbers: 5, 3, 2

$$2 + 3 = 5$$

$$\underline{\quad} + \underline{\quad} = 5$$

$$5 - 3 = \underline{\quad}$$

$$5 - \underline{\quad} = \underline{\quad}$$

b. Numbers: 9, 4, 5

$$4 + 5 = 9$$

$$\underline{\quad} + \underline{\quad} = 9$$

$$9 - 4 = \underline{\quad}$$

$$9 - \underline{\quad} = \underline{\quad}$$

c. Numbers: 4, 0, 4

$$4 + 0 = 4$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$4 - \underline{\quad} = \underline{\quad}$$

$$4 - \underline{\quad} = \underline{\quad}$$

d. Numbers: 10, 3, 7

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

e. Numbers: 9, 3, 6

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

f. Numbers: 6, 0, 6

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

Sometimes the two addition facts are the same. When that happens, the two subtraction facts will also be the same.

For example, with 8, 4, and 4, we only get one addition fact and one subtraction fact.

8



$$4 + 4 = 8 \quad 8 - 4 = 4$$

$$(4 + 4 = 8) \quad (8 - 4 = 4)$$

2. Write the fact families.

a. Numbers: 10, 5, 5

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

b. Numbers: 9, 1, 8

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

c. Numbers: 6, 3, 3

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

d. Numbers: 7, 1, 6

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

Puzzle Corner

Find the number that the shape represents.
All numbers are 10 or less.

a. $\bigcirc - 4 - 4 = 1$

$\bigcirc$ is $\underline{\quad}$

b. $9 - \square - \square = 3$

$\square$ is $\underline{\quad}$

c. $\triangle + \triangle + 2 = 10$

$\triangle$ is $\underline{\quad}$

Subtraction Practice

1. Subtract. Write the answer below the line.

a.
$$\begin{array}{r} 10 \\ - 3 \\ \hline \end{array}$$

b.
$$\begin{array}{r} 8 \\ - 7 \\ \hline \end{array}$$

c.
$$\begin{array}{r} 6 \\ - 5 \\ \hline \end{array}$$

d.
$$\begin{array}{r} 8 \\ - 6 \\ \hline \end{array}$$

e.
$$\begin{array}{r} 8 \\ - 0 \\ \hline \end{array}$$

f.
$$\begin{array}{r} 7 \\ - 7 \\ \hline \end{array}$$

g.
$$\begin{array}{r} 7 \\ - 6 \\ \hline \end{array}$$

h.
$$\begin{array}{r} 6 \\ - 4 \\ \hline \end{array}$$

i.
$$\begin{array}{r} 6 \\ - 1 \\ \hline \end{array}$$

j.
$$\begin{array}{r} 9 \\ - 4 \\ \hline \end{array}$$

2. Fill in the missing numbers.

a. $4 - \underline{\quad} = 2$	b. $6 - \underline{\quad} = 2$	c. $10 - \underline{\quad} = 5$
d. $\underline{\quad} - 4 = 1$	e. $\underline{\quad} - 3 = 3$	f. $\underline{\quad} - 5 = 0$

3. Solve.

a. Josh has 10 toy cars. He took five of his cars to a friend's house. How many cars did he leave at home?	b. Sandy and Lily were swinging at the playground. Then four more children came. How many children are now at the playground?
c. A bowl in the kitchen had 8 bananas. Harry ate 2. How many bananas are left now?	d. Mateo has nine shirts. Right now, two are in the wash and the rest are clean. How many are clean?

4. Continue the patterns as long as you can!

a.	b.	c.
$7 - 0 = \underline{\quad}$	$10 - 4 = \underline{\quad}$	$8 - 2 = \underline{\quad}$
$7 - 1 = \underline{\quad}$	$9 - 4 = \underline{\quad}$	$7 - 2 = \underline{\quad}$
$7 - 2 = \underline{\quad}$	$8 - 4 = \underline{\quad}$	$6 - 2 = \underline{\quad}$
$7 - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$
$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$
$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$
$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$
$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$
$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$	$\underline{\quad} - \underline{\quad} = \underline{\quad}$

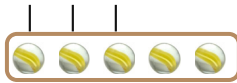
5. If the answer is four, color the box red. If the answer is five, color the box orange. If the answer is ten, color the box yellow. And, if you can't subtract, color the box gray.

$10 - 6$	$6 - 7$	$5 - 6$	$5 - 10$	$7 - 3$
$1 - 6$	$10 - 5$	$9 - 10$	$8 - 3$	$1 - 3$
$7 - 9$	$1 - 2$	$10 - 0$	$3 - 5$	$6 - 9$
$4 - 8$	$6 - 1$	$0 - 9$	$9 - 4$	$7 - 8$
$8 - 4$	$0 - 4$	$2 - 6$	$6 - 9$	$9 - 5$

How Many More? Part 1



Henry



Cindy

Cindy has more marbles.

How many more?

Three marbles are matched from each box.

Two marbles did not get matched, so Cindy has 2 more marbles than Henry.

(Henry has 2 fewer marbles than Cindy.)



Carter



Mateo

Mateo has fewer marbles.

How many fewer?

Two marbles are matched from each box.

Four marbles did not get matched, so Mateo has 4 fewer marbles than Carter.

(Carter has 4 more marbles than Mateo.)

1. Fill in how many more or how many fewer marbles one child has than another.



Jane



Jim

a. Jane has _____ more than Jim.

Jim has _____ fewer than Jane.



Mark



Mary

b. Mark has _____ more than Mary.

Mary has _____ fewer than Mark.



Ann



Liz

c. Liz has _____ more than Ann.

Ann has _____ fewer than Liz.



Faye



Sam

d. Sam has _____ more than Faye.

Faye has _____ fewer than Sam.



Peter



Frank

e. Frank has _____ more than Peter.

Peter has _____ fewer than Frank.



Susan



Bill

f. Susan has _____ more than Bill.

Bill has _____ fewer than Susan.

2. Now it is your turn to draw. Draw marbles for the child that has none.

Jane Jim a. Jane has 3 more than Jim.	Mark Sara b. Sara has 4 more than Mark.
Eric Amir c. Eric has 2 fewer than Amir.	Jack Jane d. Jane has 5 fewer than Jack.
Bill Greg e. Greg has 1 more than Bill.	Lola Liz f. Lola has 5 more than Liz.

3. Solve the problems. You can draw or act out the situations.

a. Ed has five cards, and Jack has seven. How many more cards does Jack have than Ed?	b. Manuel is 8 years old and Jack is 5. How many years older is Manuel?
c. Ava has 9 dolls, and Tina has 4. How many fewer dolls does Tina have?	

How Many More? Part 2

Compare the two problems below. They are similar, yet different!

Example 1. Julie has three more apples than Lucy. Julie has five apples. How many apples does Lucy have?

Draw the apples for both girls.

Hint: First think *who* has more apples, Julie or Lucy?

Julie:

Lucy:

Example 2. Julie has three more apples than Lucia. Lucia has five apples. How many apples does Julie have?

Draw the apples for both girls.

Hint: First think *who* has more apples, Julie or Lucia?

Julie:

Lucia:

1. Draw marbles for the child that has none.

Ed

Sally

a. Sally has two fewer than Ed.

Ann

Maria

b. Ann has four fewer than Maria.

2. Solve.

a. Ben has five more apples than Sue.
Ben has nine apples.
How many does Sue have?

b. Jill has five fewer flowers than Mary.
Jill has four flowers.
How many does Mary have?

3. It is still your turn to draw. You can decide how many marbles the children have.

 a. Jane has 5 more than Jim.	 b. Mary has 2 more than Mark.
 c. Eric has 6 fewer than Bill.	 d. Greg has 2 fewer than Bill.

4. Solve.

a. Emilio picked up two more toys than Malik. Emilio picked up 5 toys. How many toys did Malik pick up?	b. Pat drew 5 stars. Joel drew three more than Pat. How many stars did Joel draw?
c. Annie is 10 years old and Beth is 8. How many years younger is Beth?	d. Kayla ate six carrot sticks. Nia ate two fewer carrot sticks than Kayla. How many carrot sticks did Nia eat?

How Many More? Part 3

How many more toy cars does Jane have than John?

Draw more cars for John so that the children have the same amount.

I drew _____ more cars.

You can write a **missing number addition** for a “*how many more*” problem.

John



Jane



$$2 + \underline{\quad} = 5$$

Read: “2 and *how many more* makes five?”

This solves the question: “How many more is 5 than 2?”

Both of these problems can be read as “2 and *how many more* makes five?”

After all, two numbers can be added in either order.

$$2 + \underline{\quad} = 5$$

$$\underline{\quad} + 2 = 5$$

1. Draw more. Read the addition sentences.



a. $1 + \underline{\quad} = 4$

1 and how many more makes 4?



b. $3 + \underline{\quad} = 6$

3 and how many more makes 6?



c. $3 + \underline{\quad} = 7$

3 and how...

2. Solve. In (d), (e), and (f), you can draw dots if you'd like.



a. $\underline{\quad} + 5 = 7$

How many more is 7 than 5?



b. $\underline{\quad} + 2 = 6$

How many more is 6 than 2?



c. $\underline{\quad} + 1 = 8$

How many more is 8 than 1?

d. $7 + \underline{\quad} = 9$

e. $4 + \underline{\quad} = 7$

f. $3 + \underline{\quad} = 6$

3. Who has more marbles? How many more? Write an addition. You can also draw.

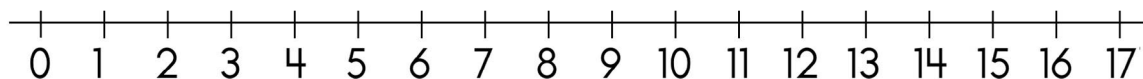
a. Jill has 10 and Sue has 7. $7 + \underline{\quad\quad} = 10$ <u> Jill </u> has <u> 3 </u> more marbles.	b. Kai has 4 and Al has 7. $\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$ <u> </u> has <u> </u> more.
c. Veronica has 4 and Ann has 6. $\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$ <u> </u> has <u> </u> more.	d. Hannah has 9 and Michelle has 2. $\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$ <u> </u> has <u> </u> more.
e. Britney has 11 and Jeanne has 10. $\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$ <u> </u> has <u> </u> more.	f. Ben has 5 and David has 10. $\underline{\quad\quad} + \underline{\quad\quad} = \underline{\quad\quad}$ <u> </u> has <u> </u> more.

4. Solve. For help, you can draw or act out the situation.

a. Ashley's shopping cart has two pineapples, and Aaron's cart has five. How many more pineapples does Aaron have? Aaron has <u> </u> more pineapples than Ashley.	 $\underline{\quad\quad} + \underline{\quad\quad} = 5$
b. A little apple tree has three apples. A bigger tree has 10. How many more apples does the bigger tree have? The bigger tree has <u> </u> more apples than the little one.	

Difference

The difference between two numbers means how far the numbers are from each other.



How far is 4 from 6? Count steps on the number line. It is two steps.

How far is 8 from 13? How much **difference** is there between 8 and 13?
Count steps on the number line.

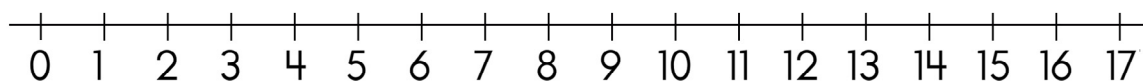
(Put your finger at 8. As you move it to 9, say "1." As you move it to 10, say "2", etc.)

How many more problems are DIFFERENCE problems.

- How many more is 7 than 3?
- 3 and how many more make 7?
- $3 + \underline{\quad} = 7$
- How far is 3 from 7?
- How much difference is there between 3 and 7?

The answer to all these is 4.

1. What is the difference between the numbers? Take steps on the number line.



a. from 6 to 10	b. from 5 to 8	c. from 7 to 11	d. from 5 to 5	e. from 1 to 10
_____ steps	_____ steps	_____ steps	_____ steps	_____ steps

2. Find the difference between the numbers. "Travel" on the number line!

From	8	4	1	3	6	10	8	9
To	10	10	9	1	5	5	12	15
Difference								

3. Find the difference between the numbers. Then write an addition. Be careful.

a. from 3 to 5 _____ steps $3 + \underline{2} = 5$	b. from 1 to 5 _____ steps $1 + \underline{\quad} = 5$	c. from 2 to 7 _____ steps $2 + \underline{\quad} = 7$
--	--	--

4. Find the difference between the numbers. Then write an addition. Be careful.

a. from 6 to 9: _____ steps $6 + \underline{\quad} = 9$	b. from 4 to 8: _____ steps _____ + _____ = _____
c. from 8 to 11: _____ steps _____ + _____ = _____	d. from 2 to 9: _____ steps _____ + _____ = _____

5. Solve. Think: is it asking for the total? Or is it asking, “How many more?”

- a. There are two cassette tapes on the table, and eight on the shelf.

How many tapes are there together?

How many more tapes are on the shelf
than on the table?

- b. There are five birds in the apple tree, and five birds in the oak tree.
Four more birds flew into the oak tree.

How many birds are in the oak tree now?

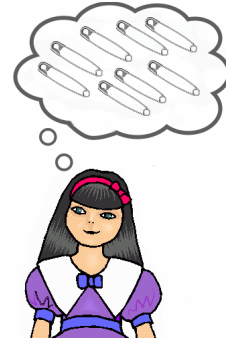
How many more birds are in the oak tree
than in the apple tree?

“How Many More” Problems and Subtraction

Nathalie needs ten safety-pins. She only has two.
How many more does she need?

$$\begin{array}{c} \text{pin} \\ \text{pin} \end{array} + \boxed{?} = 10$$

$$2 + \boxed{} = 10$$



You can use subtraction to solve a problem that asks “how many more”, like the problem $2 + \underline{\hspace{1cm}} = 10$.

Simply subtract the numbers: $10 - 2 = \underline{\hspace{1cm}}$.

It gives you the answer to $2 + \underline{\hspace{1cm}} = 10$. Why?

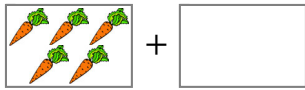
Because both problems are in the same fact family.

Use the Fact Family:

$$2 + 8 = 10, \quad 10 - 2 = 8$$

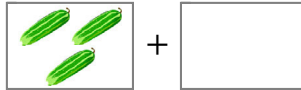
$$8 + 2 = 10, \quad 10 - 8 = 2$$

1. Fill in. The “how many more” problem has the same answer as the subtraction problem!



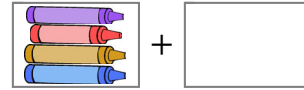
a. $5 + \underline{\hspace{1cm}} = 7$

$$7 - 5 = \underline{\hspace{1cm}}$$



b. $3 + \underline{\hspace{1cm}} = 8$

$$8 - 3 = \underline{\hspace{1cm}}$$



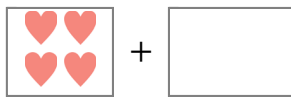
c. $4 + \underline{\hspace{1cm}} = 9$

$$9 - 4 = \underline{\hspace{1cm}}$$



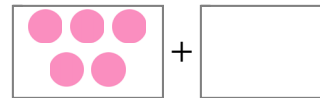
d. $5 + \underline{\hspace{1cm}} = 10$

$$10 - 5 = \underline{\hspace{1cm}}$$



e. $4 + \underline{\hspace{1cm}} = 7$

$$7 - 4 = \underline{\hspace{1cm}}$$



f. $5 + \underline{\hspace{1cm}} = 8$

$$8 - 5 = \underline{\hspace{1cm}}$$

g. $3 + \underline{\hspace{1cm}} = 10$

$$10 - 3 = \underline{\hspace{1cm}}$$

h. $2 + \underline{\hspace{1cm}} = 9$

$$9 - 2 = \underline{\hspace{1cm}}$$

i. $1 + \underline{\hspace{1cm}} = 7$

$$7 - 1 = \underline{\hspace{1cm}}$$

2. Solve the easier problem *first*.

a. $2 + \underline{\quad} = 8$ $8 - 2 = \underline{\quad}$	b. $1 + \underline{\quad} = 9$ $9 - 1 = \underline{\quad}$	c. $\underline{\quad} + 3 = 10$ $10 - 3 = \underline{\quad}$	d. $\underline{\quad} + 3 = 9$ $9 - 3 = \underline{\quad}$
The "how many more" problem has the same answer as the subtraction problem!			

3. Under each "how many more" problem, write a subtraction problem, using the same numbers. Solving the subtraction problem *first* might be easier.

a. $1 + \underline{\quad} = 7$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	b. $2 + \underline{\quad} = 9$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	c. $1 + \underline{\quad} = 10$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$
d. $\underline{\quad} + 3 = 8$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	e. $\underline{\quad} + 2 = 10$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	f. $\underline{\quad} + 3 = 9$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$

4. Solve. Think: Do you already know the total? Or is the problem asking for the total?
(You can also draw a picture to help.)

a. Mary ate two carrots. The rabbit ate six carrots. All in all, how many carrots did Mary and the rabbit eat?	b. Baby made a stack of seven blocks. Then three of the blocks got knocked down. How many blocks are in the stack now?
c. There were five lambs in the pen. Two more lambs went into the pen. How many lambs are now in the pen?	d. Lisa needs 8 dolls to be riders for her 8 toy horses. She can only find 4. How many dolls is she missing?

Solving Subtraction Problems with Addition

$$11 - 8 = ?$$

We can solve this subtraction problem by thinking of the “how many more” addition $8 + \underline{\quad} = 11$. This is the same idea as thinking of the **difference** between the numbers, or how far apart are 8 and 11 from each other.

Remember, when there are two kinds of things (two parts making a total), we can write two additions and two subtractions using the same numbers.

That is why an addition can be used to solve a subtraction.



$$8 + 3 = 11$$

$$3 + 8 = 11$$

$$11 - 8 = 3$$

$$11 - 3 = 8$$

1. Write an addition problem for each subtraction problem, using the same numbers. Solve. You can solve either the addition or the subtraction first, whichever is easier.

a. $8 - 6 = \underline{\quad}$

$\underline{\quad} + \underline{\quad} = \underline{\quad}$

b. $10 - 9 = \underline{\quad}$

$\underline{\quad} + \underline{\quad} = \underline{\quad}$

c. $9 - 7 = \underline{\quad}$

$\underline{\quad} + \underline{\quad} = \underline{\quad}$

d. $10 - 8 = \underline{\quad}$

$\underline{\quad} + \underline{\quad} = \underline{\quad}$

e. $9 - 8 = \underline{\quad}$

$\underline{\quad} + \underline{\quad} = \underline{\quad}$

f. $7 - 6 = \underline{\quad}$

$\underline{\quad} + \underline{\quad} = \underline{\quad}$

2. Write a subtraction for each “how many more” addition problem. Solve.

a. $6 + \underline{\quad} = 10$

$\underline{\quad} - \underline{\quad} = \underline{\quad}$

b. $3 + \underline{\quad} = 7$

$\underline{\quad} - \underline{\quad} = \underline{\quad}$

c. $9 + \underline{\quad} = 11$

$\underline{\quad} - \underline{\quad} = \underline{\quad}$

3. Solve. Think: Do you already know the total? Or is the problem asking for the total?
Write an addition or subtraction for each problem. You can also draw a picture to help!

a. Mom needs six cucumbers. She already has three. How many more does she need?	b. There were seven ducks on the pond. Three flew away. How many were left?
c. Jane has three teddy bears. Her friend Alicia came over and brought four teddy bears. How many teddy bears do the girls have to play with now?	d. A book has ten pages. Jerry has read six pages. How many pages does he have left to read?

4. Play math teacher again. The children's answers are inside the triangles. Check their work and correct any mistakes they made.

Ann: a. $7 - \triangle 1 = 6$ b. $8 - \triangle 3 = 4$	Shaun: c. $\triangle 2 - 4 = 2$ d. $9 - \triangle 6 = 3$	Tony: e. $\triangle 9 - 4 = 5$ f. $9 - \triangle 8 = 2$
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5. Write a matching subtraction or addition for each problem. Solve.

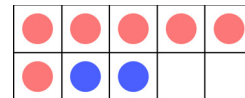
a. $7 + \underline{\quad} = 10$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	b. $2 + \underline{\quad} = 5$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$	c. $7 + \underline{\quad} = 8$ $\underline{\quad} - \underline{\quad} = \underline{\quad}$
d. $7 - 5 = \underline{\quad}$ $\underline{\quad} + \underline{\quad} = \underline{\quad}$	e. $9 - 4 = \underline{\quad}$ $\underline{\quad} + \underline{\quad} = \underline{\quad}$	f. $10 - 9 = \underline{\quad}$ $\underline{\quad} + \underline{\quad} = \underline{\quad}$

Chapter 2 Mixed Review

This is a ten-frame. It shows 5 red and 5 blue dots — a total of 10.



This ten-frame shows 6 red and 2 blue dots. $6 + 2 = 8$.



1. Connect each ten-frame to a sum it shows.



$8 + 2 = \underline{\hspace{2cm}}$

$5 + 5 = \underline{\hspace{2cm}}$



$7 + 3 = \underline{\hspace{2cm}}$

$6 + 3 = \underline{\hspace{2cm}}$



$6 + 4 = \underline{\hspace{2cm}}$

$9 + 1 = \underline{\hspace{2cm}}$



$5 + 4 = \underline{\hspace{2cm}}$

$7 + 1 = \underline{\hspace{2cm}}$



2. Add.

a.	b.	c.	d.
$3 + 3 = \underline{\hspace{2cm}}$	$5 + 2 = \underline{\hspace{2cm}}$	$2 + 6 = \underline{\hspace{2cm}}$	$3 + 5 = \underline{\hspace{2cm}}$
$3 + 4 = \underline{\hspace{2cm}}$	$4 + 2 = \underline{\hspace{2cm}}$	$4 + 3 = \underline{\hspace{2cm}}$	$4 + 2 = \underline{\hspace{2cm}}$

3. Compare. Write $<$, $>$ or $=$.

a. $4 + 3$ 5	c. $3 + 4$ 6	e. 4 $4 + 2$
b. $8 + 2$ 10	d. $9 + 2$ 9	f. 6 $3 + 3$

4. Brenda has 2 toy cars, Jason has 8, and Joe has 10.

a. How many more cars does Joe have than Jason?

b. How many cars do Jason and Brenda have together?

5. Add.

a. $3 + 1 + 4 = \underline{\hspace{2cm}}$

$3 + 2 + 2 = \underline{\hspace{2cm}}$

b. $5 + 0 + 5 = \underline{\hspace{2cm}}$

$4 + 2 + 1 = \underline{\hspace{2cm}}$

c. $1 + 1 + 3 = \underline{\hspace{2cm}}$

$5 + 1 + 3 = \underline{\hspace{2cm}}$

6. Fill in the missing numbers. Notice the patterns!

a.

$1 + \underline{\hspace{2cm}} = 8$

$2 + \underline{\hspace{2cm}} = 8$

$3 + \underline{\hspace{2cm}} = 8$

$4 + \underline{\hspace{2cm}} = 8$

b.

$1 + \underline{\hspace{2cm}} = 7$

$2 + \underline{\hspace{2cm}} = 7$

$3 + \underline{\hspace{2cm}} = 7$

$4 + \underline{\hspace{2cm}} = 7$

c.

$1 + \underline{\hspace{2cm}} = 6$

$2 + \underline{\hspace{2cm}} = 6$

$3 + \underline{\hspace{2cm}} = 6$

$4 + \underline{\hspace{2cm}} = 6$

7. Which numbers make ten? Fill in the missing numbers.

a. $10 = \boxed{\hspace{1cm}} + 6$

$10 = \boxed{\hspace{1cm}} + 5$

$10 = \boxed{\hspace{1cm}} + 1$

b. $10 = 7 + \boxed{\hspace{1cm}}$

$10 = 2 + \boxed{\hspace{1cm}}$

$10 = 4 + \boxed{\hspace{1cm}}$

c. $10 = \boxed{\hspace{1cm}} + 3$

$10 = \boxed{\hspace{1cm}} + 8$

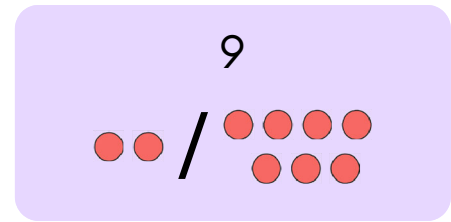
$10 = \boxed{\hspace{1cm}} + 9$

Chapter 2 Review

1. Write a fact family to match the picture.

$$\underline{\quad} + \underline{\quad} = \underline{\quad} \quad \underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad} \quad \underline{\quad} - \underline{\quad} = \underline{\quad}$$



2. Write a matching subtraction or addition for each problem. Solve.

a. $6 + \underline{\quad} = 10$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

b. $5 + \underline{\quad} = 9$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

c. $7 - 6 = \underline{\quad}$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

3. Solve.

- a. There are a total of 8 boys and girls playing in the yard. Two are girls. How many are boys?

- b. Kay has four marbles. Susan has two more marbles than Kay. Draw Kay's and Susan's marbles.

- c. Six sparrows and two robins are feeding on seeds. How many more sparrows than robins are there?

4. Fill in the missing numbers.

a. $3 + \underline{\quad} = 4$

$$1 + \underline{\quad} = 9$$

$$3 + \underline{\quad} = 10$$

$$2 + \underline{\quad} = 7$$

b. $6 - 3 = \underline{\quad}$

$$8 - 5 = \underline{\quad}$$

$$9 - 2 = \underline{\quad}$$

$$10 - 8 = \underline{\quad}$$

c. $10 - 0 = \underline{\quad}$

$$8 - 2 = \underline{\quad}$$

$$6 - 6 = \underline{\quad}$$

$$7 - 4 = \underline{\quad}$$

Chapter 3: Place Value Within 0-100

Introduction

This chapter covers place value with two-digit numbers, which means students learn to think of two-digit numbers in terms of their tens and ones. They also compare two-digit numbers and learn to count up to 120.

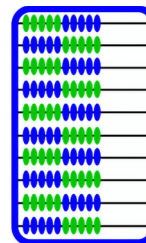
In kindergarten, children have written numbers from 11 to 19 as a sum of $10 +$ some other number, preparing to learn place value. They have also learned to count to 100 by ones and by tens. However, learning to count does not equal understanding place value. In this chapter your child or students learn that the two digits of a two-digit number represent the amounts of *tens* and *ones*. In second grade, they learn that the three digits of a three-digit number represent the amounts of hundreds, tens, and ones. This knowledge is then extended to yet larger numbers in later grade levels.

Understanding place value is crucial to all further learning of mathematics. A student that does not grasp how 72 really means 7 tens and 2 will have trouble with most math topics from this point on, such as adding and subtracting two-digit numbers, whether with mental math or with paper-and-pencil methods.

As children learn to count, they basically perceive numbers as some kind of continuum that goes on without end. With simple counting, the child might not catch on to the inherent structure of the number system. Our number system is based on the idea that the efficient way to count lots of objects *and* denote them is with *groups* of tens, hundreds, and thousands — not individually.

The crucial point in understanding the concept of place value is therefore that a **certain position represents a group of a specific size**. The digit in each position tells us how many groups of that size there are. For example, in the number 2,381, the 8 represents eight tens, and not just “8” and the 3 represents three hundreds, and not just “3”. The place of the digit tells us the size of the group, and the digit itself tells how many of that group.

The initial lessons in the chapter that introduce tens and ones use a **100-bead abacus** extensively. This 100-bead abacus or school abacus simply contains ten beads on ten rods, for a total of 100 beads. It is not the special abacus used by the Chinese or the Japanese. Each bead simply represents one. The 100-bead abacus lets children both “see” the numbers and use their touch while making them.



You will need to purchase this school abacus separately, such as on Amazon, or make your own. You can browse Amazon’s abacus collection at this link:

https://www.amazon.com/s?k=abacus+100+beads&ref=nb_sb_noss_1&tag=mathmammoth-20

Here is a link to an online virtual abacus: <https://apps.mathlearningcenter.org/number-rack/>

Or, you can make one on your own. This is a fairly easy craft project and you can easily find instructions for it on the Internet (search for example for “DIY abacus”).

Besides the abacus, we also use a visual model of blocks where ten of them “snap” together to form a stick. If you already have these base-ten blocks, you can use them along with the visual exercises, if you prefer.

Moreover, we also use number lines and a 100-chart. Number lines help visualize how numbers continue indefinitely and also relate to the concept of measuring. The 100-chart helps the child to be familiar with the numbers and find patterns.

While most of the lessons here focus on place value, students also learn some mental math. They learn to add 10 to any two-digit number (e.g. $78 + 10$), subtract 10 from any two-digit number (e.g. $45 - 10$), and add and subtract multiples of ten (e.g. $20 + 70$ or $50 - 20$). We base these mental calculations on our understanding of place value. We add 10 to, say, 78, by adding 1 to the tens digit, and similarly for subtracting 10. The sum $20 + 70$ is easily found by thinking of 2 tens + 7 tens = 9 tens.

Skip-counting is presented a few times in the context of the 100-chart, but is an optional topic. It will be revisited in 2nd and 3rd grades.

Children have encountered ordinal numbers in kindergarten and in daily life, and thus it is probably an easy topic. It is presented here for completeness' sake. If time is an issue, it can be omitted.

Good Mathematical Practices

- Mathematics is a science of *patterns* and *structures*. Learning about the place value system means studying the basic structure of numbers themselves. Like already mentioned, the topics of this chapter are extremely foundational and important for all future mathematics work.
- Children will use the 100-bead abacus as a basic tool in this chapter. They also encounter the 100-chart and number lines. Another common way to illustrate numbers is base-ten blocks, and the lessons use visual illustrations of them. You are welcome to use physical base-ten blocks with the lessons also, if you like. All of these are useful tools, and becoming familiar with them will allow students in later grades to choose the best tool(s) for the problem at hand.

Pacing Suggestion for Chapter 3

Please add one day to the pacing for the test if you will use it.

The Lessons in Chapter 3	page	span	suggested pacing	your pacing
Counting in Groups of 10	124	2 pages	1 day	
Naming and Writing Numbers 1	126	2 pages	1 day	
Naming and Writing Numbers 2	128	3 pages	1 day	
The “Teen” Numbers	131	3 pages	2 days	
The 100-Chart, Part 1	134	2 pages	1 day	
The 100-Chart, Part 2	136	2 pages	1 day	
Add and Subtract with Tens	138	2 pages	1 day	
Add or Subtract 10, Part 1	140	2 pages	1 day	
Add or Subtract 10, Part 2	142	2 pages	1 day	
Comparisons, Part 1	144	2 pages	1 day	
Comparisons, Part 2	146	2 pages	1 day	
Numbers Past 100	148	2 pages	1 day	
More Practice	150	2 pages	1 day	
Skip-Counting Practice (optional).....	152	3 pages	1 day	
Ordinal Numbers	155	2 pages	1 day	
Chapter 3 Mixed Review	157	2 pages	1 day	
Chapter 3 Review	159	2 pages	1 day	
Chapter 3 Test (optional)				
TOTALS		34 pages	17 days	
with optional content		(37 pages)	(18 days)	

Games and Activities

Big and Bigger!

You need: A standard deck of playing cards from which you remove the face cards and tens, leaving only numbers from 1 through 9.

Game Play: In each round, each player is dealt *three* cards, and they make a two-digit number using *two* of the cards, however they want. The player with the largest two-digit number then wins all the cards used in that round. If there is a tie, those players draw two additional cards each, and resolve the tie with those. After enough rounds have been played to use all of the cards, the player with the most cards wins.

Variations:

- * The player with the smallest number wins each round.
- * The goal is to make a number that is as close as possible to 50.
- * Each player is dealt 4 cards, and they choose two of them to make their two-digit number.
- * Each player is dealt 4 cards, they form two 2-digit numbers out of them, and add those. The player with the largest (and correct) sum wins the cards. This may be too challenging, but you can make it a bit easier by removing 9s, 8s, 7s, and 6s from the deck.

Abacus

The lessons in the chapter include several activities to be done with the 100-bead abacus. Please follow the instructions in the lessons.

Add 10 (version 1; with a board game)

You need: A large deck of number cards from 1-9 (two standard decks of playing cards from which you remove the face cards and tens will work). Any board game where you advance a marker on each turn.

Game Play: Place the deck of cards face down in a pile in the middle. Follow the rules of your chosen board game, except that at the beginning of your turn, *first* draw two cards from the deck. Then, form a 2-digit number with the cards either way you wish. Next, add 10 to your number and say the answer aloud. If your answer is correct, you get to advance your marker on the board according to the rules of the board game. (Such as, perhaps you roll the dice and advance that number of steps.)

The winner is determined according to the rules of the board game.

Add 10 (version 2)

You need: Two standard decks of playing cards from which you remove the face cards and tens, leaving only numbers from 1 through 9. Other number cards work as well.

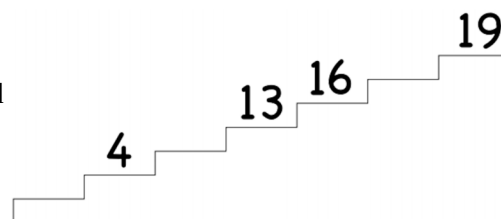
Game Play: Place the deck of cards face down in a pile in the middle. At your turn, draw two cards from the deck. Then, form a 2-digit number with the cards either way you wish. Next, add 10 to your number and say the answer aloud. If your answer is correct, you get to keep the cards and put them in your personal pile.

After enough rounds have been played to use all of the cards, the player with the most cards wins.

Going Up Some More

You need: A deck of number cards from 0 to 9. Another deck that only has numbers 0 and 1 - preferably at least three copies of zero and three copies of 1. Place both decks face down in the middle of the table. Paper and pencil for each player. Each player draws a staircase on their paper, with anywhere from 4 to 10 steps (more steps for older players).

Game play: On your turn, take a number card from both decks, and create a number. The number you get from the deck containing only 0s and 1s will be the tens digit, and the other card provides the ones digit. If possible, you must place this number on your steps. The first player to fill their steps in increasing order from bottom to top wins.



Variation: You can make the game easier by allowing neighboring steps to have the same value.

This game is from <https://www.earlyfamilymath.org> and published here with permission.

Order Us!

You need: Two standard decks of playing cards from which you remove the face cards and tens, leaving only numbers from 1 through 9. Other number cards work as well. Paper and pencil to keep track of points.

Game Play: Place the deck of cards face down in a pile in the middle. On your turn, draw six cards. You will then form three 2-digit numbers using the cards any way you wish, and lastly order those from smallest to largest. If they are ordered correctly, you keep the cards and place them into your personal pile.

After enough rounds have been played to use all of the cards, the player with the most cards wins.

Variations:

* Each player draws 8 cards instead of 6, and makes four 2-digit numbers.

Mammoths & Ladders

The traditional snakes & ladders game with a mammoth theme! This is a good game to practice the counting sequence, and is technically kindergarten level but I know some children will still benefit and enjoy it in 1st grade also. We have two versions, one goes from 1-50 and the other from 51 to 100.

<https://www.mathmammoth.com/download/Mammoths-Ladders-50.pdf>

<https://www.mathmammoth.com/download/Mammoths-Ladders-100.pdf>

Games and Activities at Math Mammoth Practice Zone

Place Value

Choose Max Digits 2 to get 2-digit numbers.

<https://www.mathmammoth.com/practice/place-value>

Give the Cat a Mouse

In this game, the child is asked to find a specific number on a 100-chart. The options allow you to change the starting numbers and the step, allowing you to create, not only a chart from 1-100, but others such as a chart from 0 to 990 by tens, or a chart from 1 to 199 by twos, etc. You can also choose how many numbers the child will find, and the percentage of numbers that are already pre-filled in the chart.

<https://www.mathmammoth.com/practice/hundred-chart>

Skip-count in a 100-chart

In this activity, the child fills in five numbers on a 100-chart in a specific skip-counting pattern. This helps the child to understand what skip-counting is, and to see different patterns that skip-counting forms on a 100-chart. The options allow you to change the starting and ending number of the grid. You can also choose the percentage of numbers that are already pre-filled in the chart.

<https://www.mathmammoth.com/practice/skip-count-hundred-chart>

Ordering Numbers — Online Game

In this activity, children practice ordering numbers by dragging them into the given boxes in the correct order. Choose two-digit numbers.

<https://www.mathmammoth.com/practice/order-numbers>

Plot numbers on the number line

In this activity, the child drags a dot to position the given number on a number line, practicing place value and estimation. In the “normal” mode, you will see plenty of tick marks on the number line. In the “Estimation” mode, there are only a few tick marks, and you have to estimate where to place the dot on the number line. Choose the 2-digit numbers.

<https://www.mathmammoth.com/practice/number-line>

Fruity Math

Add a multiple of ten and a single-digit number (such as $2 + 70$). Click the fruit with the correct answer and try to get as many points as you can within two minutes.

https://www.mathmammoth.com/practice/fruity-math#o=a&d=120&m=m&c=1,9x1__1,9x10&s=120

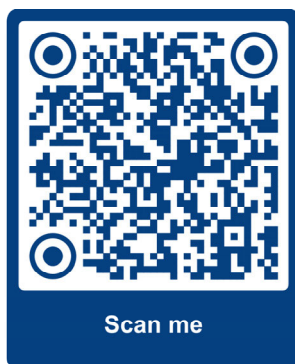
Further Resources on the Internet

We have compiled a list of Internet resources that match the topics in this chapter, including pages that offer:

- **online practice** for concepts;
- online **games**, or occasionally, printable games;
- **animations** and interactive **illustrations** of math concepts;
- **articles** that teach a math concept.

We heartily recommend you take a look! Many of our customers love using these resources to supplement the bookwork. You can use these resources as you see fit for extra practice, to illustrate a concept better and even just for some fun. Enjoy!

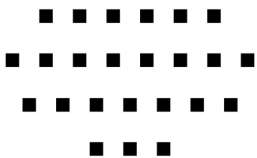
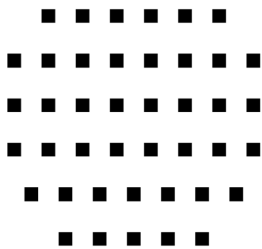
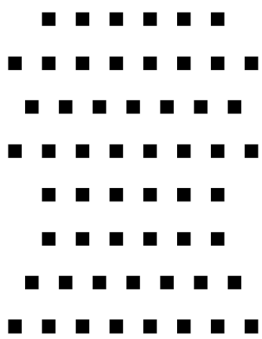
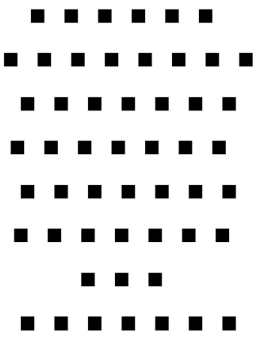
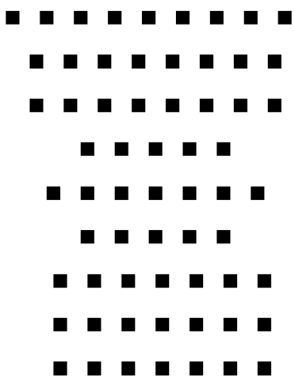
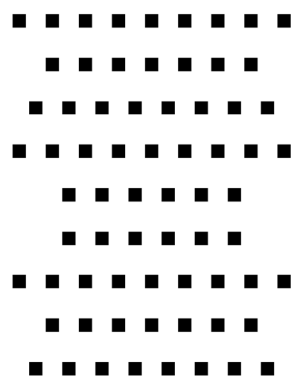
<https://l.mathmammoth.com/2026/gr1ch3>



Counting in Groups of 10

When counting many items, it is much quicker to count them if we first group them into **groups of ten**.

- Counting Activity 1.** (Optional) Put some small items on a table. Ask the child to make groups of ten. Then ask the child to count the groups of ten and the individual items left over. Do not yet use the words eleven, twelve, twenty, thirty, forty, *etc.*, but instead use “*four tens and six*”, “*seven tens and one*”, *etc.* Repeat with different amounts of items, taking turns.
- Count in groups of TEN. Count ten dots, and circle them. Write how many groups of ten you get. Write how many ones are left over.

 groups of ten ones a.	 groups of ten ones b.	 groups of ten ones c.
 groups of ten ones d.	 groups of ten ones e.	 groups of ten ones f.

3. **Counting Activity 2.** You need: Small items to count, such as sticks, beads, or beans. Containers, such as small bags or bowls. (If using sticks, rubber bands work well to group them.)

Before the game: Place 10 of the items in the middle and the rest in a pile on the side.

Play: On their turn, each player adds one more item to the middle pile on the table, and names the number that is formed. Whenever a whole ten is completed, those ten items are grouped together with a rubber band or by placing them in a small bag or bowl.

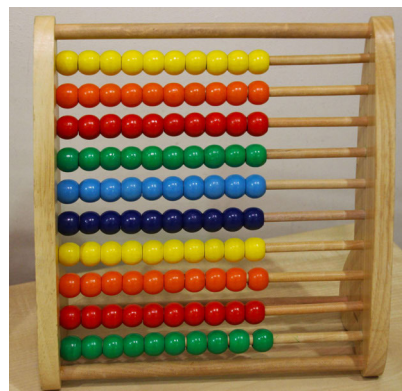
In this lesson, use only the number words from one to ten when counting in this game. Words like eleven, thirteen, fifty, sixty, *etc.* are not allowed. For example, eleven is said as “ten and one”, twelve is “ten and two”, twenty is “two tens”, twenty-five is “two tens and five”, *etc.*

Variation: Each player adds *two* (or more) items to the pile instead of just one.

This abacus has ten rows of 10 beads. Each bead counts as *one*.

These ten rows of 10 make a total of **one hundred** beads. That is why we call this abacus a **100-bead abacus**.

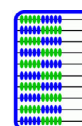
With it, we can illustrate numbers from 1 to 100. For example, to show 6 tens and 2 ones on the abacus, move six entire rows, plus two beads from the seventh row, to one side of the abacus.



4. Make these amounts with the 100-bead abacus.

- | | | |
|-------------------|-------------------|-------------------|
| a. 6 tens, 5 ones | e. 2 tens, 1 one | i. 4 tens, 6 ones |
| b. 2 tens, 7 ones | f. 8 tens, 9 ones | j. 6 tens |
| c. 7 tens | g. 9 tens, 3 ones | k. 7 tens, 1 one |
| d. 1 ten, 5 ones | h. 1 ten, 1 one | l. 1 ten, 8 ones |

5. **Activity 3.** This is an activity for two people. Take turns telling each other what number to make on the abacus, such as “7 tens and 9” or “1 ten and 7”.



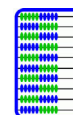
(Do not proceed farther until the student has mastered this! This is the foundation for understanding place value, and thus is a crucial concept.)

Naming and Writing Numbers 1

Learn the names of the numbers that consist of entire tens:

ten =	ten	four tens =	forty	seven tens =	seventy
two tens =	twenty	five tens =	fifty	eight tens =	eighty
three tens =	thirty	six tens =	sixty	nine tens =	ninety
ten tens = one hundred					

1. Say the number names above, from ten to a hundred, aloud a few times while also making those numbers on the 100-bead abacus. It almost sounds like a rhyme! You can also use bags with 10 items in each.

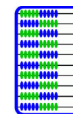


We use the words twenty, thirty, forty, fifty, *etc.* when naming numbers. We write the number of tens and the number of ones next to each other with numerals.

Examples: “6 tens and 7 ones” is sixty-seven, or 67.
 “4 tens and 5 ones” is forty-five, or 45.
 “2 tens and 2 ones” is twenty-two, or 22.

We will look at the numbers between 10 and 20 in the next lesson.

2. Make these numbers with the 100-bead abacus and **name** them (orally). Lastly, write them with numerals.



- | | | | |
|------------------|-----------|------------------|-------|
| a. 5 tens 6 ones | <u>56</u> | f. 3 tens 5 ones | _____ |
| b. 7 tens 2 ones | _____ | g. 2 tens 3 ones | _____ |
| c. 2 tens 1 one | _____ | h. 8 tens 7 ones | _____ |
| d. 3 tens 1 one | _____ | i. 9 tens 4 ones | _____ |
| e. 4 tens 8 ones | _____ | j. 6 tens 6 ones | _____ |

3. **Counting Activity** again — this time with number names from 20 on.

You need: Counting items, such as sticks, beads, or beans.

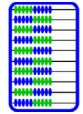
Containers, such as small bags or bowls. (If using sticks, then use rubber bands to group them, rather than containers.)

Preparation: Place 20 of the items in the middle and the rest in a pile to one side.

Play: During their turn, each player adds one more item to the middle pile on the table, and names the number that is formed. Whenever a whole ten is completed, those ten items are grouped together with a rubber band or by placing them in a small bag or bowl.

Variation: Each player adds *two* (or more) items to the pile instead of just one.

4. **Activity** for two people. Take turns telling each other what number to make on the abacus, such as “fifty-two”. Do not use the numbers between 10 and 20 yet.



5. Fill in the missing parts.

a. twenty-four 2 tens 4 ones = 24

b. fifty-three tens ones =

c. _____ 3 tens 9 ones =

d. sixty-two tens ones =

e. eighty-eight tens ones =

f. _____ tens ones = 31

g. _____ tens ones = 79

h. ninety-nine tens ones =

i. forty-seven tens ones =

Naming and Writing Numbers 2

Places:

tens ones

3	5
---	---

In the number 35, the three and the five are **digits**. The digit “3” means three *tens*, not simply “three”. And the digit “5” means five ones.

The digit “3” is located **in the tens place**. Think of the **place** as the address for the digit. It is as if the digit “3” “lives” in the tens place.

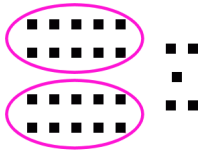
Similarly, the digit “5” is located or “lives” **in the ones place**.

tens ones

7	0
---	---

In the number 70, there are just seven entire tens, and no ones, so we write “7” in the tens place and a **zero** in the ones place.

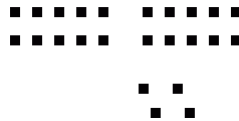
1. Circle each group of ten dots. Then count the groups of ten and the ones left over. Write the digits for the tens and the ones. Also, tell or write the name of the number. Notice how quick it is to count the dots this way!



tens ones

2	5
---	---

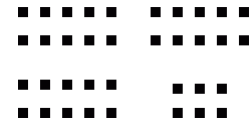
a. _____



tens ones

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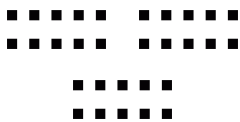
b. _____



tens ones

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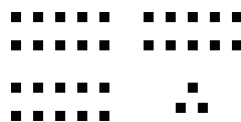
c. _____



tens ones

--	--

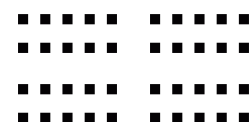
d. _____



tens ones

--	--

e. _____

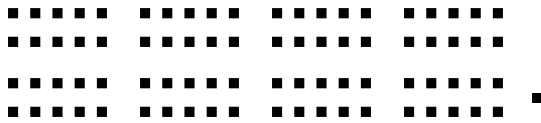


tens ones

--	--

f. _____

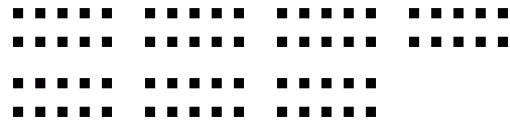
2. Circle each group of ten dots. Then count the groups of ten and the ones left over. Write the number with a numeral. Also, tell *or* write the name of the number.



tens ones

--	--

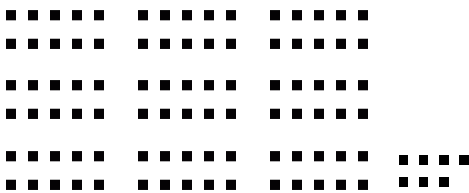
a. _____



tens ones

--	--

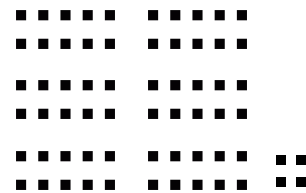
b. _____



tens ones

--	--

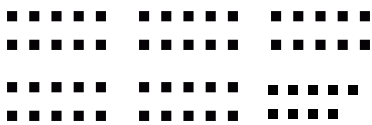
c. _____



tens ones

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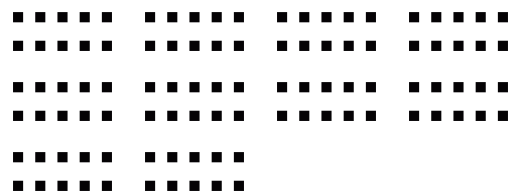
d. _____



tens ones

--	--

e. _____

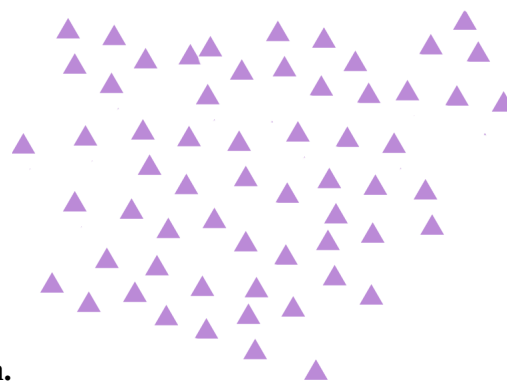
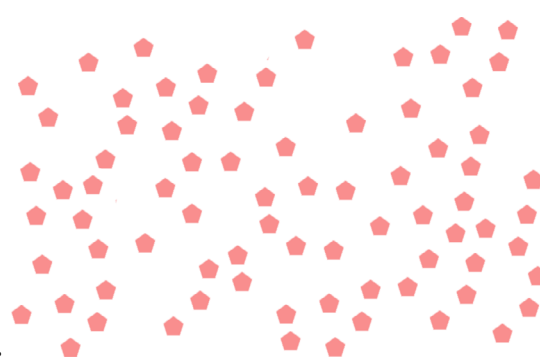
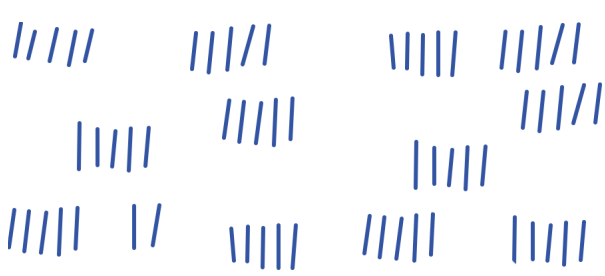


hundreds tens ones

1	0	0
---	---	---

f. one hundred

3. Write the number of items. (First make ten-groups.)

a.  _____	b.  _____
c.  _____	d.  _____

4. Now draw dots yourself. Make groups of ten.

a. 23	b. 50	c. 39
--------------	--------------	--------------

The “Teen” Numbers

The names of the numbers between 10 and 20 are:

1 ten 1 = 11 = eleven

1 ten 6 = 16 = sixteen

1 ten 2 = 12 = twelve

1 ten 7 = 17 = seventeen

1 ten 3 = 13 = thirteen

1 ten 8 = 18 = eighteen

1 ten 4 = 14 = fourteen

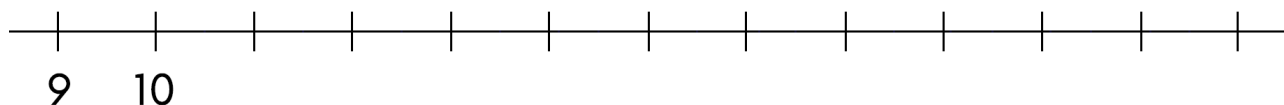
1 ten 9 = 19 = nineteen

1 ten 5 = 15 = fifteen

2 tens = 20 = twenty

The word “teen” actually comes from “ten”. So, seventeen is actually “seven-ten”, or one ten and seven.

1. Write the numbers on this number line under the tick marks. Say the numbers aloud.



2. Make these numbers with the 100-bead abacus and **name** them (orally).
Lastly, write them with numerals.



a. 1 ten 2 ones _____

d. 5 tens _____

b. 1 ten 7 ones _____

e. 1 ten 5 ones _____

c. 7 tens _____

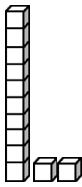
f. 1 ten 3 ones _____

3. Write as numbers.

a. eleven _____	b. sixteen _____	c. sixty _____
d. thirty _____	e. thirteen _____	f. twelve _____
g. nineteen _____	h. fifty _____	i. fifteen _____

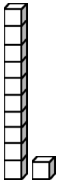
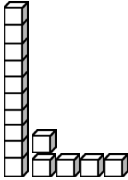
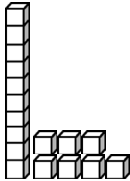
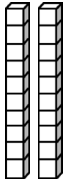
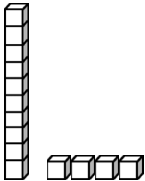
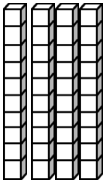
Ten blocks form a pillar, plus there are two little blocks.

In total, there are twelve blocks.



tens	ones
1	2
10 + 2	

4. Fill in the missing parts.

a. <u>Eleven</u>  <u>10</u> + <u>1</u> <table><tr><th>tens</th><th>ones</th></tr><tr><td></td><td></td></tr></table>	tens	ones			b. _____  _____ + _____ <table><tr><th>tens</th><th>ones</th></tr><tr><td></td><td></td></tr></table>	tens	ones		
tens	ones								
tens	ones								
c. _____  _____ + _____ <table><tr><th>tens</th><th>ones</th></tr><tr><td></td><td></td></tr></table>	tens	ones			d. _____  20 + <u>0</u> <table><tr><th>tens</th><th>ones</th></tr><tr><td></td><td></td></tr></table>	tens	ones		
tens	ones								
tens	ones								
e. _____  _____ + _____ <table><tr><th>tens</th><th>ones</th></tr><tr><td></td><td></td></tr></table>	tens	ones			f. _____  _____ + _____ <table><tr><th>tens</th><th>ones</th></tr><tr><td></td><td></td></tr></table>	tens	ones		
tens	ones								
tens	ones								

5. Match the names and the numbers.

17	70	27	72
seventy-two	twenty-seven	seventeen	seventy

6. Fill in the missing parts.

a. $18 = 10 + \underline{\quad}$	b. $19 = \underline{\quad} + 9$	c. $21 = 20 + \underline{\quad}$
d. $81 = \underline{\quad} + 1$	e. $12 = \underline{\quad} + 2$	f. $\underline{\quad} = 10 + 5$

7. Count backwards from 20 to 10 (orally). Do it several times.

8. (optional) Choose one (or both) of the following games to practice the teen numbers some more.

Practice “Teens” board game

You need: A board game where you move a piece along a path. Number cards from 1 to 9 plus one number card with 10, from a standard deck of playing cards or UNO cards, etc.

Before the game: Place the card with 10 right-side-up in the middle. Place the deck next to it, face down.

Play: To start a turn, a player draws a card from the deck, placing it next to the 10-card. Then they have to *name* the number formed by the 10-card and the card they drew. For example, if 7 is drawn, the player has to say “seventeen”. If they name the number correctly, the player can move their piece the number of steps that was on the card they drew (not including the ten). Otherwise, follow the rules of the board game you are using.

Teens out!

You need: A standard deck of playing cards. Each face card (jack, queen, king) counts as a 10.

Before the game: Deal each person 7 cards. Place the remaining cards (the deck) face down in the middle.

Play: During your turn, take one card from the deck. Then, discard from your hand any two cards that when added together are more than 10 but less than 20. You *also* have to name the number when putting the cards down. For example, 10 and 5 make fifteen and can be discarded. Eight and five make thirteen, and can be discarded. Remember, any face card counts as a 10.

A player may discard several pairs of cards in one turn. If a player fails to name the number correctly when discarding, they have to keep the cards in their hand. The game ends when someone is able to discard all of their cards from their hand.

Variation: Instead of discarding the cards, the player places a “teen” pair of cards face up in front of them as a “book”. The game is played till all the cards are used from the deck. If a person does not have any cards left in their hand, then at their turn, they simply draw one card from the deck and the turn passes to the next person. In the end, whoever has the most “books” wins.

The 100-Chart, Part 1

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1. This is a 100-chart (a “hundred chart”).
 - a. The “sixties” row is colored, ending in 70. Find the “forties” row. Color it if you like.
 - b. Say aloud the numbers from 11 to 20. Color the “teen” numbers (from thirteen to nineteen).
 - c. What interesting things can you find in the chart?
 - d. Find the column that has the numbers that end in “5”, such as 5, 15, 25, and so on. Color them some color you like.
 - e. Find the column with the multiples of ten (ten, twenty, thirty, and so on). What is similar about those numbers?
 - f. Learn to skip-count by fives, starting with 5. It goes like this: 5, 10, 15, 20, ... You add five every time, or in other words, you list every 5th number.

How can you use the chart to skip-count by fives?

2. You can use the 100-chart if you like. What number is...

a. two more than 65?

c. two more than 57?

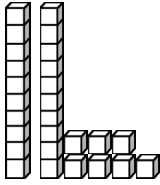
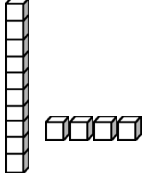
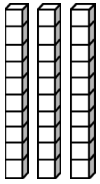
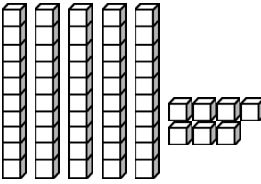
e. three less than 42?

b. three more than 88?

d. four less than 85?

f. four more than 96?

3. Fill in the missing parts.

a. _____  _____ + _____ tens ones	b. _____  _____ + _____ tens ones
c. _____  <u>30</u> + <u>0</u> tens ones	d. _____  _____ + _____ tens ones

4. Break each number into tens and ones.

a. $73 = \underline{70} + \underline{3}$	c. $45 = \underline{\quad} + \underline{\quad}$	e. $98 = \underline{\quad} + \underline{\quad}$
b. $91 = \underline{\quad} + \underline{\quad}$	d. $83 = \underline{\quad} + \underline{\quad}$	f. $64 = \underline{\quad} + \underline{\quad}$

5. Match each number to its name.

46	forty-two
64	forty-six
55	fifty-five
70	fifty-nine
69	sixty-four
59	seventy
42	sixty-nine

99	seventy-six
81	eighty-one
90	ninety-one
91	ninety
79	seventy-nine
76	ninety-nine
100	one hundred

The 100-Chart, Part 2

1. Fill in the numbers missing from this 100-chart.

1		3		5				9	10
11	12							19	
21		23			26				
							38		
41			44						50
		53			56			59	
61			64			67			
		73				77		79	
81			84						
						97	98		100

2. Find these numbers on the chart: 26, 58, 34, 91, 70, and 45. You can color them.

3. Practice finding numbers on the chart with your friend or teacher. Take turns giving each other a number to find. Or, play this online game:

Give the Cat a Mouse —Find the hidden mouse on a 100-chart.

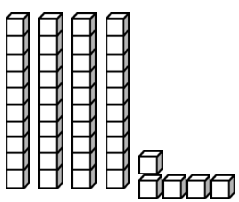
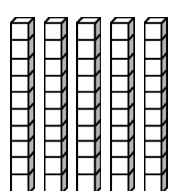
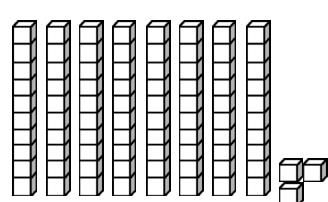
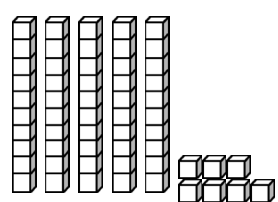
<https://www.mathmammoth.com/practice/hundred-chart>

4. Write the number that is... (You can also make a game out of this.)

a.	b.	c.
three more than 98 _____	two more than 79 _____	two less than 86 _____
three more than 27 _____	two more than 38 _____	three less than 70 _____

5. Count backwards from 90 to 70. (Orally.)

6. Fill in the missing parts.

a. _____  _____ + _____ tens ones 	b. _____  _____ + _____ tens ones
c. _____  _____ + _____ tens ones 	d. _____  _____ + _____ tens ones

7. Fill in the missing parts.

a. $60 + 7 = \underline{\hspace{2cm}}$ $80 + 3 = \underline{\hspace{2cm}}$	b. $\underline{\hspace{2cm}} = 4 + 50$ $\underline{\hspace{2cm}} = 8 + 80$	c. $86 = 6 + \underline{\hspace{2cm}}$ $47 = \underline{\hspace{2cm}} + 40$
--	--	---

8. Fill in the missing numbers on these two (different) number lines.

a.

$\leftarrow$

|

|

|

|

|

|

$\rightarrow$

46

47

b.

$\leftarrow$

|

|

|

|

|

|

$\rightarrow$

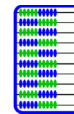
70

71

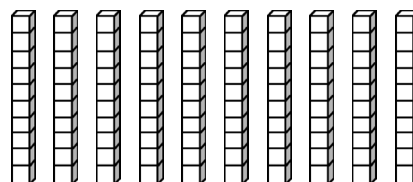
137

Add and Subtract with Tens

1. Add and subtract with tens. Try to find a **shortcut** for these types of problems.
Use the abacus and/or base ten blocks for help.



a. $30 + 20 = \underline{\hspace{2cm}}$	b. $40 + 40 = \underline{\hspace{2cm}}$
c. $40 + 30 = \underline{\hspace{2cm}}$	d. $20 + 70 = \underline{\hspace{2cm}}$
e. $30 - 20 = \underline{\hspace{2cm}}$	f. $60 - 30 = \underline{\hspace{2cm}}$
g. $50 - 30 = \underline{\hspace{2cm}}$	h. $90 - 60 = \underline{\hspace{2cm}}$



Shortcut

Adding **60 + 30** means **6 tens + 3 tens = 9 tens**.

So, you can simply add $6 + 3 = 9$, and just give the answer as 9 tens, or 90.

Subtracting **80 - 70** means **8 tens - 7 tens = 1 ten**.

So I can simply subtract $8 - 7 = 1$, and just give the answer as 1 ten, or 10.

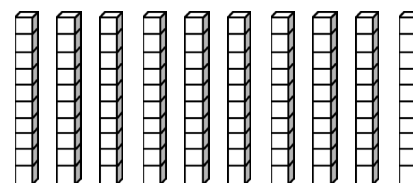
2. Solve.

a. $10 + 50 = \underline{\hspace{2cm}}$	b. $60 + 30 = \underline{\hspace{2cm}}$	c. $70 - 30 = \underline{\hspace{2cm}}$
$60 + 20 = \underline{\hspace{2cm}}$	$70 - 10 = \underline{\hspace{2cm}}$	$90 - 40 = \underline{\hspace{2cm}}$

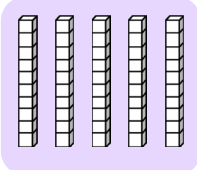
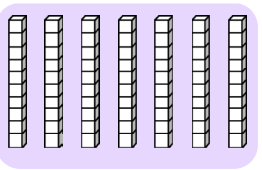
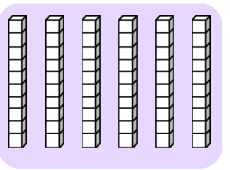
3. Subtract from 100. Use the abacus if you like.



a. $100 - 50 = \underline{\hspace{2cm}}$	b. $100 - 30 = \underline{\hspace{2cm}}$
c. $100 - 20 = \underline{\hspace{2cm}}$	d. $100 - 10 = \underline{\hspace{2cm}}$
e. $100 - 90 = \underline{\hspace{2cm}}$	f. $100 - 70 = \underline{\hspace{2cm}}$



4. Subtract.

a.  $50 - 20 - 10 = \underline{\hspace{2cm}}$	b.  $70 - 10 - 30 = \underline{\hspace{2cm}}$	c.  $60 - 20 - 20 - 20 = \underline{\hspace{2cm}}$
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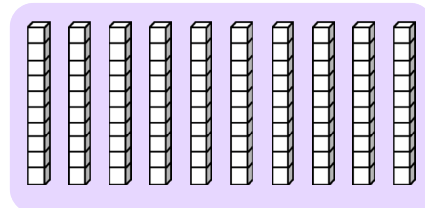
5. Add.

a. $10 + 10 + 10 + 10 = \underline{\hspace{2cm}}$

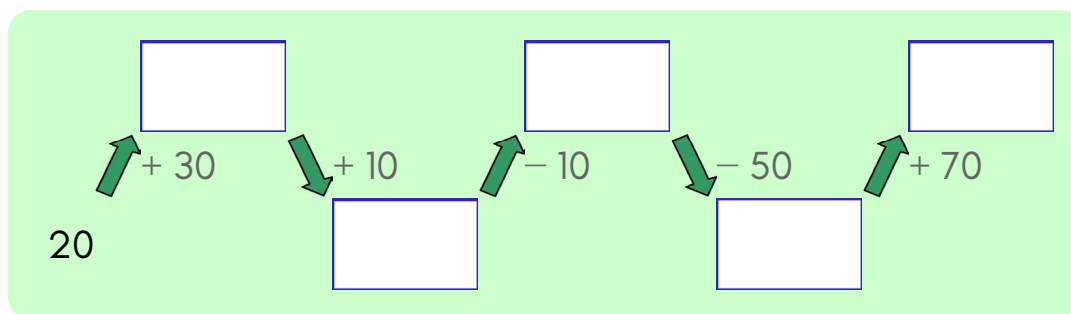
b. $30 + 30 + 30 = \underline{\hspace{2cm}}$

c. $10 + 40 + 20 + 20 = \underline{\hspace{2cm}}$

d. $20 + 20 + 20 + 20 + 20 = \underline{\hspace{2cm}}$



6. Add and subtract.



7. Fill in the missing numbers.

a. $60 + \boxed{\hspace{2cm}} = 90$ $20 + \boxed{\hspace{2cm}} = 100$	b. $80 - \boxed{\hspace{2cm}} = 30$ $80 - \boxed{\hspace{2cm}} = 40$	c. $\boxed{\hspace{2cm}} - 10 = 10$ $\boxed{\hspace{2cm}} - 20 = 20$
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Puzzle Corner

Quick additions! Find a way to add these numbers quickly.

Hint: first add any numbers that make $\underline{\hspace{2cm}}$.

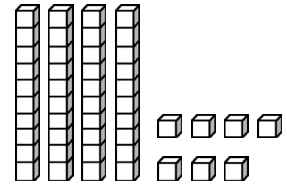
$10 + 2 + 20 + 4 + 5 + 1 + 8 + 9 + 9 + 20 + 5 + 6 = \underline{\hspace{2cm}}$

$8 + 6 + 7 + 9 + 5 + 30 + 4 + 3 + 5 + 2 + 1 + 20 + 1 = \underline{\hspace{2cm}}$

Add or Subtract 10, Part 1

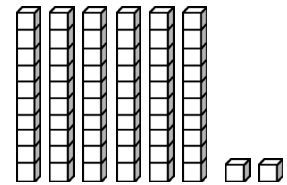
What number is 10 more than 47?
In other words, what is $47 + 10$?

Draw another ten-pillar in the picture to find out.

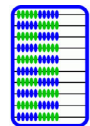


What number is 10 less than 62?
In other words, what is $62 - 10$?

Cross out one ten-pillar to find out.



1. **a.** Make 38 on the 100-bead abacus. Then add 10 to it. What do you get?
(Think: what is the quickest way to add 10 on the abacus?)



Use the abacus also to find what is $33 + 10$ and other sums your teacher asks.

- b.** Make 84 on the abacus. What number is 10 less than that?
(Think: what is the quickest way to subtract 10 from a number on the abacus?)

Use the abacus also to find what is $37 - 10$ and $89 - 10$.

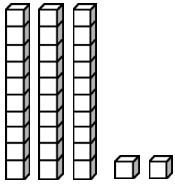
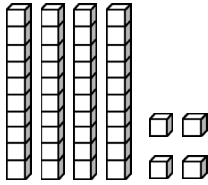
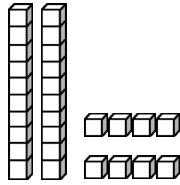
2. For each highlighted number, find the number that is exactly ten more, and color it.

3. For each underlined number, find the number that is exactly ten less, and underline it.

4. (optional) Count by fives, starting at 5. Can you do that without looking at the chart?

1	2	3	4	5	6	7	8	9	10
11	12	<u>13</u>	14	15	16	17	18	19	20
21	22	23	24	25	26	<u>27</u>	28	29	30
31	32	33	34	35	36	37	38	39	<u>40</u>
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	<u>57</u>	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	<u>84</u>	85	86	87	88	89	90
91	92	93	94	95	96	97	98	<u>99</u>	100

5. Draw another ten-pillar to add 10, or cross out a ten-pillar to subtract 10.

a.  $32 + 10 = \underline{\hspace{2cm}}$	b.  $44 + 10 = \underline{\hspace{2cm}}$	c.  $28 - 10 = \underline{\hspace{2cm}}$
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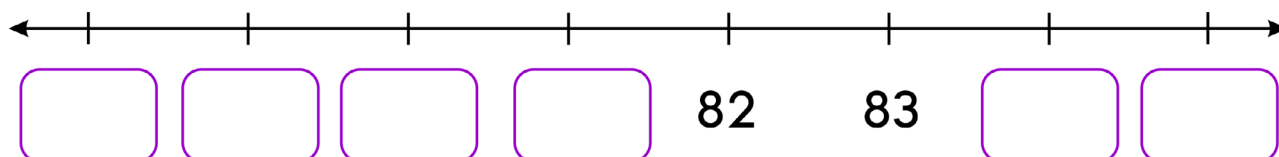
6. Tell or write the number that is... (You can also use the 100-bead abacus to help.)

a. ten more than 18 $\underline{\hspace{2cm}}$ ten more than 35 $\underline{\hspace{2cm}}$	b. ten more than 60 $\underline{\hspace{2cm}}$ ten more than 59 $\underline{\hspace{2cm}}$	c. ten less than 24 $\underline{\hspace{2cm}}$ ten less than 37 $\underline{\hspace{2cm}}$
---	---	---

7. Add or subtract 10.

a. $51 + 10 =$	b. $16 + 10 =$	c. $67 - 10 =$
d. $74 - 10 =$	e. $65 - 10 =$	f. $18 - 10 =$
g. $74 + 10 =$	h. $90 + 10 =$	i. $39 - 10 =$

8. Fill in the missing numbers.



Add or subtract.

a. $43 + 20 = \underline{\hspace{2cm}}$

b. $66 + 30 = \underline{\hspace{2cm}}$

c. $92 - 40 = \underline{\hspace{2cm}}$

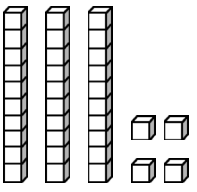
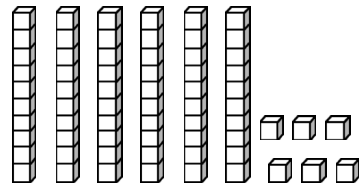
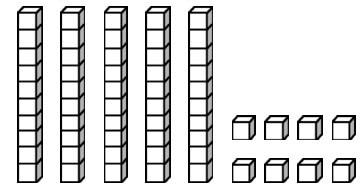
d. $99 - 50 = \underline{\hspace{2cm}}$

e. $37 + 50 = \underline{\hspace{2cm}}$

Puzzle Corner

Add or Subtract 10, Part 2

1. Find the number that is 10 more or 10 less than the number shown by the blocks.

a.  10 more: _____	b.  10 less: _____	c.  10 more: _____
--	--	--

2. See Jodie's work on the right.

Explain to Jodie what she has done wrong, and how to do the problem correctly.

Jodie's work:

$$44 + 10 = \underline{45}$$

3. Add and subtract 10.

a. $61 + 10 =$ _____ $61 - 10 =$ _____	b. $37 + 10 =$ _____ $37 - 10 =$ _____	c. $89 + 10 =$ _____ $89 - 10 =$ _____
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4. **a.** Count aloud from 45 to 62.

b. Count aloud backwards from 83 to 68.

c. Count aloud backwards from 100 to 85.

5. Add and subtract.

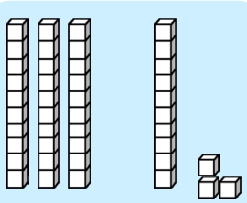
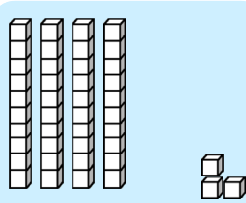
a. $50 + 20 =$ _____ $50 - 20 =$ _____	b. $90 + 10 =$ _____ $40 - 30 =$ _____	c. $20 + 40 =$ _____ $70 - 20 =$ _____
---	---	---

6. Write each number as (tens) + (ones).

a. $91 = \underline{\quad\quad} + \underline{\quad\quad}$	b. $79 = \underline{\quad\quad} + \underline{\quad\quad}$	c. $58 = \underline{\quad\quad} + \underline{\quad\quad}$
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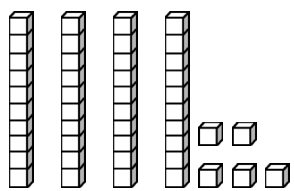
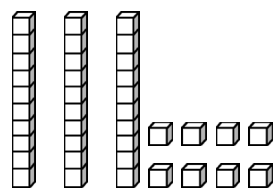
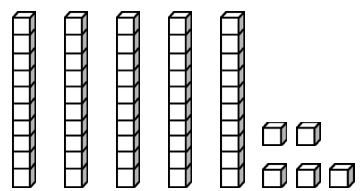
7. Fill in the missing numbers.

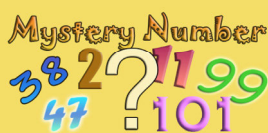
a. $20 + 6 = \underline{\quad\quad}$ $80 + 2 = \underline{\quad\quad}$	b. $60 + \underline{\quad\quad} = 67$ $6 + \underline{\quad\quad} = 56$	c. $74 = 70 + \underline{\quad\quad}$ $93 = 3 + \underline{\quad\quad}$
---	--	--

 30 + 13	 40 + 3	(optional) We can write 43 as $40 + 3$, adding its tens and ones. But we can also write it as $30 + 13$. (What other ways are there?)
--	---	---

8. Make 49 on the abacus. Now, 49 is $40 + 9$.
But it is also $30 +$ (some number). What number?

9. Write these numbers as sums in different ways. You can work with a friend or on your own. (You can separate the ten-pillars and the one-blocks into two groups in different ways.)

a.  $\underline{\quad\quad} + \underline{\quad\quad}$ $\underline{\quad\quad} + \underline{\quad\quad}$	b.  $\underline{\quad\quad} + \underline{\quad\quad}$ $\underline{\quad\quad} + \underline{\quad\quad}$	c.  $\underline{\quad\quad} + \underline{\quad\quad}$ $\underline{\quad\quad} + \underline{\quad\quad}$
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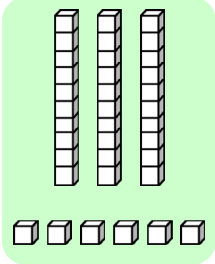


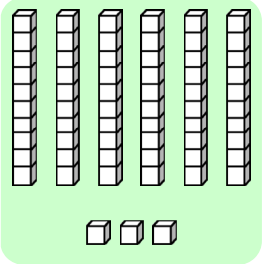
*I'm between 84 and 92. I don't have any ones.
Who am I?*

Comparisons, Part 1

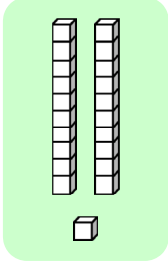
1. Write each number shown by the blocks. Then circle the number that is more.

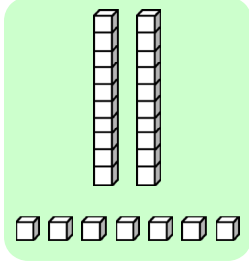
a.





b.





2. How can you tell which is more, 29 or 90? Or, which is more, 66 or 47?
You can make the numbers on the abacus or with base ten blocks to help.

3. Circle the amount or number that is more. You can use the abacus to help.

a. 5 tens 2 ones	3 tens 9 ones	b. 1 ten 7 ones	1 ten 9 ones
c. 76	72	d. 3 + 70	40 + 9
e. 51	25	f. six tens	16

4. Based on your work so far, fill in.

To tell which two-digit number is greater, first check how many _____
the numbers have.

If the numbers have the same amount of _____, then _____

81 has more *tens* than 38, so 81 is greater than 38.

$$81 > 38$$

(We don't need to compare the amount of ones, because the tens count so much more. Even *one* ten is more than the maximum amount of ones [which is 9] a number can have.)

“81 is greater than 38.”

62 has the same amount of tens as 66, but it has less *ones* than 66.

$$62 < 66$$

“62 is less than 66.”

Think of each of the symbols $>$ and $<$ as a hungry alligator's mouth! The open end, or the open “mouth”, of the symbol ALWAYS points to the **bigger** number—the alligator wants to eat as much as it can!

5. Write $<$ or $>$ between the numbers. (To help you, you can use the abacus or base-ten blocks.)

a. 45 54	b. 34 24	c. 50 54
d. 16 61	e. 70 17	f. 35 64

6. Compare the numbers. Write $<$, $>$, or $=$ between them.

a. 15 56	b. 29 64	c. 81 90
d. 77 47	e. 34 34	f. 80 68

7. Find the comparisons that are true.

$$14 > 22$$

$$68 < 68$$

$$94 > 27$$

$$55 > 65$$

$$15 > 92$$

$$52 < 59$$

$$49 > 51$$

8. Which of the given numbers fits the comparison below? Write it on the empty line.

a. 76 89 39 _____ $<$ 73	b. 29 36 50 _____ $>$ 36	c. 53 55 35 _____ $<$ 41
-----------------------------------	-----------------------------------	-----------------------------------

Comparisons, Part 2

Here we compare two sums (two additions).

First, find out what number each sum makes.
Then compare them.

$20 + 5$		$70 + 1$
↓		↓
25	$<$	71

1. Which side “wins”, so to speak? Or, which side has the greater number?

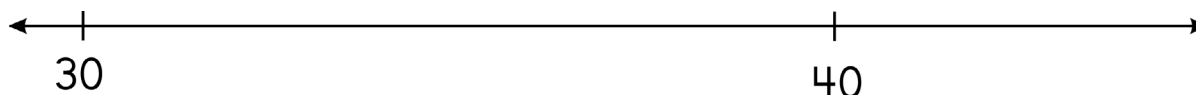
Write $<$, $>$ or $=$. (Some of these may be equal!)

a. $80 + 2$ $80 + 7$ ↓ ↓ 	b. $10 + 7$ $70 + 5$ ↓ ↓
c. $50 + 7$ $70 + 3$ ↓ ↓ 	d. $6 + 60$ $6 + 80$ ↓ ↓
e. $60 + 5$ $40 + 5$	f. $5 + 40$ $40 + 5$
g. $40 + 5$ $4 + 40$	h. $70 + 5$ $7 + 50$

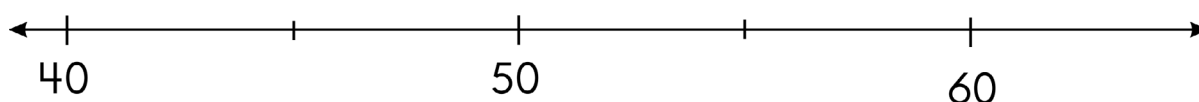
2. Which of the given numbers fits the comparison below? Write it on the empty line.

a. 15 51 45 _____ $<$ 44	b. 28 81 33 72 $<$ _____	c. 70 48 96 69 $>$ _____
--	--	--

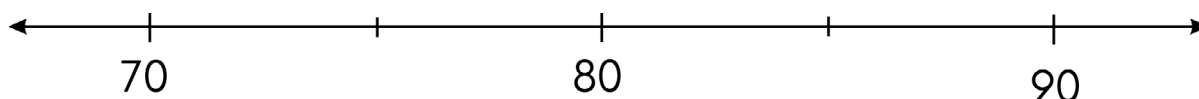
3. Mark these numbers (approximately) on the number line: 37, 41, 33.



4. Mark these numbers (approximately) on the number line: 59, 45, 62.



5. If a number is greater than 75, yet less than 85, where could it be on the number line?

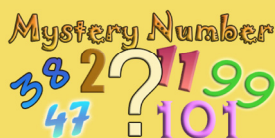


6. Compare the numbers. Write $<$, $>$, or $=$ between them.

a. 81 88	b. 95 59	c. 16 60
d. 91 91	e. 90 88	f. 49 94
g. 69 83	h. 50 55	i. 22 12

7. (challenge) Write the numbers in order from the smallest to the largest.

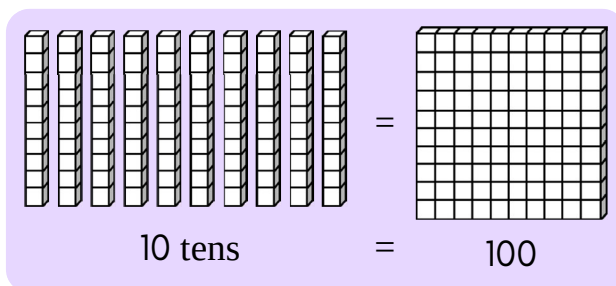
a. 68, 67, 46 ____ < ____ < ____	b. 33, 53, 37 ____ < ____ < ____
c. 48, 46, 18 ____ < ____ < ____	d. 87, 78, 80 ____ < ____ < ____
e. 48, 84, 46, 50 ____ < ____ < ____ < ____	f. 98, 87, 67, 76 ____ < ____ < ____ < ____



Who am I? I have two more tens than 40, and the same amount of ones as 17.

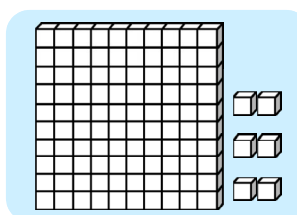
Numbers Past 100

Imagine 10 ten-pillars joined together: they become a hundred-flat, so to speak.



This image shows us **one hundred six**.
We write it as 106.

The digit “1” is in its own place:
in the hundreds place.



hundreds	tens	ones
1	0	6

Counting past 100 is easy. You just start over, but instead of 1, 2, 3, 4, and so on, you count, “one hundred one”, “one hundred two”, “one hundred three”, and so on.

Count aloud with your teacher from one hundred to one hundred twenty.
Look at how these numbers are written:

100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110,
111, 112, 113, 114, 115, 116, 117, 118, 119, 120.

1. **a.** Count aloud backwards from 110 to 95.

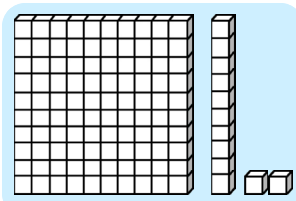
b. Count aloud from 97 to 114.

c. Count aloud backwards from 120 to 99.

2. Fill in the missing numbers.

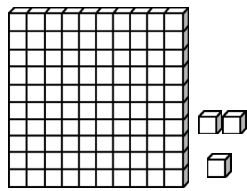
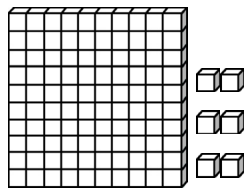
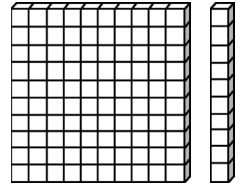
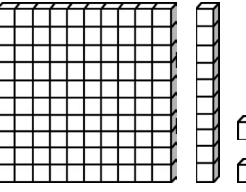
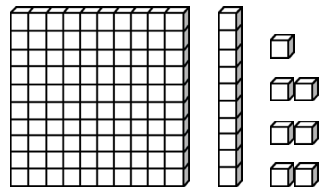
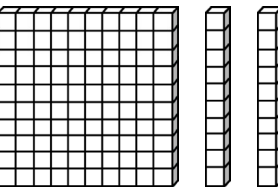
91			94			97			
		103			106				110
111									

Here we have one hundred, one ten, and one one (or one unit). In total, it is 112 (*one hundred twelve*) blocks.



hundreds	tens	ones
1	1	2

3. Fill in the missing parts. Also, say each number aloud.

a.  hundreds tens ones <table border="1"> <tr> <td></td> <td></td> <td></td> </tr> </table>				b.  hundreds tens ones <table border="1"> <tr> <td></td> <td></td> <td></td> </tr> </table>			
c.  hundreds tens ones <table border="1"> <tr> <td></td> <td></td> <td></td> </tr> </table>				d.  hundreds tens ones <table border="1"> <tr> <td></td> <td></td> <td></td> </tr> </table>			
e.  hundreds tens ones <table border="1"> <tr> <td></td> <td></td> <td></td> </tr> </table>				f.  hundreds tens ones <table border="1"> <tr> <td></td> <td></td> <td></td> </tr> </table>			

4. Write the number that is...

a. one more than 109 _____ one more than 113 _____	b. two more than 116 _____ two more than 99 _____	c. three more than 101 _____ three more than 116 _____
d. one less than 108 _____ one less than 120 _____	e. two less than 110 _____ two less than 117 _____	f. three less than 120 _____ three less than 111 _____

More Practice

1. **a.** Count aloud from 94 to 113.

b. Count aloud backwards from 114 to 88.

2. Fill in the missing numbers on the chart.

		83							
	92	93	94	95	96				
101				105					110
					116	117			120

3. Write the number that is...

a.	b.	c.
two more than 67 _____	one less than 54 _____	three less than 36 _____
one more than 57 _____	three less than 81 _____	three less than 70 _____
three more than 48 _____	two less than 60 _____	four less than 66 _____

4. Compare the numbers, writing $<$, $>$, or $=$ between them.

a. 42 47	b. 60 58	c. 91 19
----------------------------------	----------------------------------	----------------------------------

5. Add and subtract 10.

a. $59 + 10 =$ _____ $59 - 10 =$ _____	b. $88 + 10 =$ _____ $88 - 10 =$ _____	c. $90 + 10 =$ _____ $90 - 10 =$ _____
--	--	--

6. Which side “wins”, so to speak? Write $<$, $>$ or $=$.

a. $70 + 7$ $40 + 8$	b. $40 + 3$ $3 + 40$
c. $2 + 80$ $80 + 7$	d. $50 + 5$ $6 + 40$

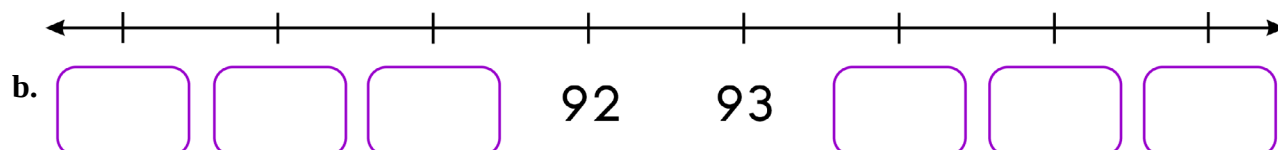
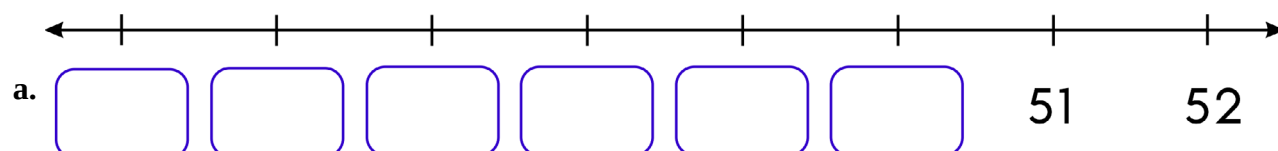
7. Count by ones to fill in the missing numbers.

_____, _____, _____, _____, _____, 110, 111

8. Add and subtract.

a. $40 + 50 =$ _____ $100 - 20 =$ _____	b. $50 + 30 =$ _____ $50 - 20 =$ _____	c. $60 - 50 =$ _____ $80 - 50 =$ _____
--	---	---

9. Fill in the missing numbers on these two (different) number lines.



10. (optional) Find the sums that equal 75.

$60 + 17$

$5 + 70$

$50 + 15$

$17 + 15$

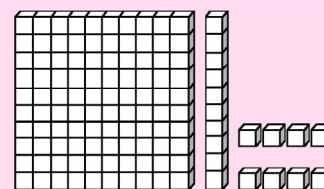
$30 + 25$

$15 + 60$

$7 + 50$

Puzzle Corner

Find the number that is 32 more than the number in the illustration.



Skip-Counting Practice

(This lesson is optional.)

1. Fill in the chart.

Then color every fourth number on the number chart starting at 4. It will create a pretty pattern — **the skip-counting pattern by 4**.

1	2	3	4	5	6	7	8	9	10
									20
	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38		
41							48	49	50
51	52	53				57			
61			64	65			68		
				75	76	77			
	82	83				87	88	89	
91	92	93			96	97	98	99	100

2. Count by twos.

(This can be done orally.)

Use the chart to help.

Notice the pattern!

2, 4, _____, _____, _____,
 _____, _____, _____, _____, _____,
 _____, _____, _____, _____, _____,
 _____, _____, _____, _____, _____

3. Count by twos again, starting at 1 this time.

(This can be done orally.)

Use the chart to help.

Notice the pattern!

1, 3, _____, _____, _____,
 _____, _____, _____, _____, _____,
 _____, _____, _____, _____, _____,
 _____, _____, _____, _____, _____

4. Count by fives. (This can be done orally.) Use the chart to help.

0, 5, _____, _____, _____, _____, _____, _____, _____,
 _____, _____, _____, _____, _____, _____, _____, _____

5. Color every third number on the number chart starting at 3. It should create a pretty pattern — **the skip-counting pattern by 3.**

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

6. Count by tens. Use the chart to help. Notice the patterns.

a. 0, 10, _____, _____, _____

b. 21, 31, _____, _____, _____

c. _____, 12, _____, 32, _____

d. _____, _____, 65, 75, _____

e. _____, 26, _____, 46, _____

f. _____, _____, _____, 89, 99

7. Add and subtract.

a. $60 + 20 =$ _____

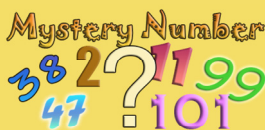
$15 - 10 =$ _____

b. $34 - 10 =$ _____

$100 - 30 =$ _____

c. $30 + 50 =$ _____

$55 - 10 =$ _____



*I have as many tens as 15 and
as many ones as 99. Who am I?*

Try your own coloring patterns on this page!

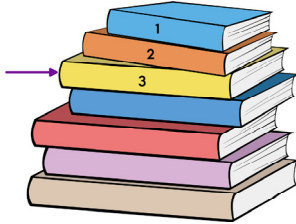
1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Ordinal Numbers

Ordinal numbers are used when talking about the *order* of things.

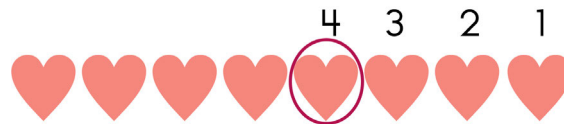
The **third** book from the top.



The **fifth** letter from the left:

1 2 3 4 5
U M B R E L L A

The **fourth** heart from the right:



Most of the time, the ordinal number is formed by adding “th” to the regular number word but there are some exceptions (in purple).

Ordinal Number

1st

first

2nd

second

3rd

third

4th

fourth

5th

fifth

6th

sixth

7th

seventh

8th

eighth

Ordinal Number

9th

ninth

10th

tenth

11th

eleventh

12th

twelfth

13th

thirteenth

14th

fourteenth

15th

fifteenth

16th

sixteenth

1. Circle.

a. The second puppy from the left.



b. The fourth puppy from the right.

c. The sixth tomato from the left.

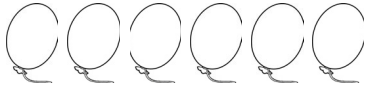


d. The third tomato from the right.

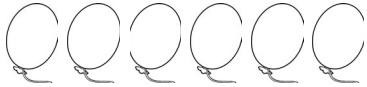
e. The twelfth drop from the right.



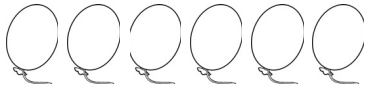
2. Color.



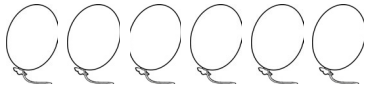
a. The third balloon from the left.



b. The first three balloons on the left.



c. The fifth balloon from the right.



d. The first five balloons on the right.

3. Circle the second, fifth, sixth, ninth, and tenth letters from the top to find Liam's favorite food.

B
T
E
D
A
C
R
O
O
S

M
I
A
J
N
X
C
G
E
O

4. Which letters should you circle to make the word "mango" (Liam's favorite fruit)? Use ordinal numbers.

5. Find the letters in the grid, and find out what is Emily's favorite day.

The second row from the top,
the second letter from the left.

The fourth row from the top,
the third letter from the left.

The fifth row from the top,
the first letter from the right.

The fifth row from the bottom,
the third letter from the right.

The 1st row from the bottom,
the 2nd letter from the left.

The sixth row from the bottom,
the third letter from the left.

The 6th row from the top,
the 4th letter from the right.

The 4th row from the top,
the 1st letter from the left.

H	E	D	I	V	S
N	B	P	T	L	F
D	A	U	D	W	M
Y	Z	I	Y	I	C
D	T	S	O	Q	R
T	H	A	V	E	L

Chapter 3 Mixed Review

1. Solve.

a. Lucas has three more marbles than Anita. *Anita* has five marbles. Draw the children's marbles.

Lucas:

Anita:

b. Lucas has three more marbles than Anita. *Lucas* has five marbles. Draw the children's marbles.

Lucas:

Anita:

2. Add in your head. Then compare, writing $<$, $>$, or $=$.
Compare the problems in each pink box.

a. $3 + 3$ 7

$4 + 3$ 7

$5 + 3$ 7

b. $4 + 1$ 5

$4 + 3$ 5

$4 + 4$ 5

c. 8 $6 + 4$

8 $4 + 4$

8 $5 + 4$

3. Write two subtractions to match the given addition. You can draw sticks to help.

a. $3 + 7 = \underline{\quad}$

$\underline{\quad} - \underline{\quad} = \underline{\quad}$

$\underline{\quad} - \underline{\quad} = \underline{\quad}$

b. $6 + 2 = \underline{\quad}$

$\underline{\quad} - \underline{\quad} = \underline{\quad}$

$\underline{\quad} - \underline{\quad} = \underline{\quad}$

4. Solve. Write both an addition *and* a subtraction for each problem. You can draw something to help you.

- a. Emilio has nine toy cars. Three of them don't work and the rest are good.
How many are good?

_____ are good.

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$\underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

- b. An apple tree has ten apples, some big and some small. Two of the apples are small.
How many are big?

_____ are big.

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$\underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

5. Write an addition for each illustration.



a. $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$



b. $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$



c. $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$



d. $\underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

6. Write different sums where the sum (the answer) is 7 and 8.

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 7$$

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 7$$

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 7$$

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 7$$

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 7$$

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 8$$

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 8$$

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 8$$

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 8$$

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}} = 8$$

Chapter 3 Review

1. Fill in the missing parts.

a. 1 ten 5 ones

15

b. tens ones

sixty-seven

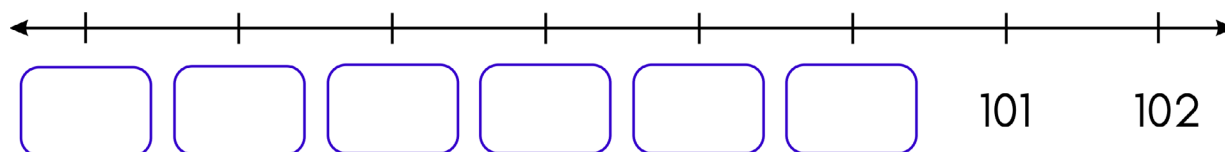
c. 4 tens 0 ones

d. 10 tens 0 ones

e. tens one

fifty-one

2. Fill in the missing numbers.



3. Write < or > between the numbers.

a.	b.	c.	d.	e.
78 87	22 25	56 57	68 80	101 11

4. Count. You can also say this aloud with your teacher.

97, 98, _____, _____, _____, _____, _____,
 _____, _____, _____, _____, _____, _____.

5. Break the numbers into their tens and ones.

a. $68 = \underline{\quad} + \underline{\quad}$	b. $25 = \underline{\quad} + \underline{\quad}$	c. $91 = \underline{\quad} + \underline{\quad}$
---	---	---

6. Add.

a. $50 + 7 = \underline{\hspace{2cm}}$ $20 + 40 = \underline{\hspace{2cm}}$	b. $8 + 10 = \underline{\hspace{2cm}}$ $49 + 10 = \underline{\hspace{2cm}}$	c. $60 + 40 = \underline{\hspace{2cm}}$ $9 + 60 = \underline{\hspace{2cm}}$
--	--	--

7. Subtract.

a. $50 - 10 = \underline{\hspace{2cm}}$ $90 - 40 = \underline{\hspace{2cm}}$	b. $38 - 10 = \underline{\hspace{2cm}}$ $63 - 10 = \underline{\hspace{2cm}}$	c. $100 - 60 = \underline{\hspace{2cm}}$ $80 - 50 = \underline{\hspace{2cm}}$
---	---	--

8. Compare, and write $<$, $>$ or $=$ in the box.

a. $56 \square 5 + 60$

b. $20 + 8 \square 33$

c. $60 + 5 \square 50 + 6$

d. $34 \square 30 + 6$

e. $4 + 90 \square 49$

f. $80 + 2 \square 70 + 9$

9. Find the statements that are true.

$67 > 66$

$80 > 18$

$94 = 9 + 40$

$25 > 52$

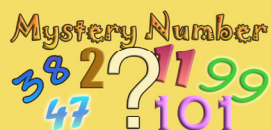
$17 < 40$

$58 = 55$

$93 > 49$

10. (optional) Put the numbers in order.

a. 57, 17, 75 $\underline{\hspace{1cm}} < \underline{\hspace{1cm}} < \underline{\hspace{1cm}}$	b. 18, 48, 44, 41 $\underline{\hspace{1cm}} < \underline{\hspace{1cm}} < \underline{\hspace{1cm}} < \underline{\hspace{1cm}}$
---	--



*I have five fewer ones than 39,
and one more ten than 47.*

Chapter 4: Time

Introduction

In this chapter we cover reading analog and digital clocks to whole and half hours, and some basics of the calendar.

The first lesson uses an analog clock that only has the hour hand. We omit the minute hand for a reason: this way the child can concentrate on the hour hand only, and learning to tell whole and half hours becomes much easier. We also practice telling what time it is one hour or a half-hour later than a given time.

The next lesson focuses on minutes. The aim of this lesson is to learn that one hour is 60 minutes, that a half-hour is 30 minutes, and how the phrases “o’clock” and “half past” relate to the hours and minutes. For example, the child is to learn that “half past eight” is written 8:30, and the “30” part is the number of minutes, so half an hour is just 30 minutes.

This lesson also includes a few exercises about reading the clock to five-minute intervals using a special clock that includes the numbers for the minute hand; however, these can be skipped if desired, because in second grade, the student will get a lot of practice reading the clock to the nearest five minutes.

The next lesson deals with morning and afternoon hours: AM and PM. The goal is for the student to understand that the clock starts at 12 midnight, and goes through all the A.M. hours from 1 to 12 until it is 12 noon, and then goes through all of the P.M. hours from 1 to 12 until it is 12 midnight again.

We will also briefly look at the calendar and practice the names of the months.

In second grade, students tell and write time from analog and digital clocks to the nearest five minutes, using a.m. and p.m. Then in third grade, they tell and write time to the nearest minute, measure time intervals in minutes, and solve word problems involving time intervals.

Reading the clock is a skill that can and should be practiced in everyday situations from now on so that children can learn by experience, and not just by filling in pages in their math book.

Pacing Suggestion for Chapter 4

Please add one day to the pacing for the test if you will use it.

The Lessons in Chapter 4	page	span	suggested pacing	your pacing
Whole and Half Hours 1	164	3 pages	1 day	
Whole and Half Hours 2	167	3 pages	1 day	
Practicing with Time	170	2 pages	1 day	
AM and PM	172	3 pages	1 day	
The Calendar (optional)	175	2 pages	1 day	
Chapter 4 Mixed Review	177	2 pages	1 day	
Chapter 4 Review	179	1 page	1 day	
Chapter 4 Test (optional)				
TOTALS		14 pages	6 days	
with optional content		(16 pages)	(7 days)	

Games and Activities

Tell the Time!

You need: An analog clock that allows you to turn the clock hands, or an app that allows you to do so.

In this activity, ask your child or student to turn the clock hands to a specific time (whole hours or half hours). Once they do so, then it is their turn to give you a time that you will set the clock to. You can use random times, and also important, specific times, such as, “Set the clock to the time when we eat supper (6 o’clock).”

Find the Date!

You need: A wall calendar

In this simple activity, ask your child or student to find a specific date on the calendar. Once they do so, then it is their turn to tell you a date to find. You can find random dates, and also important, specific dates, such as, “Find my birthday, September 12.”

Months Match

This is a simple activity to practice matching the names of the months to their numbers.

You need: A set of 12 number cards with numbers from 1 to 12 on them. You can use cards from a standard deck if your child understands Jack as 11 and Queen as 12.

Shuffle the cards. Ask the child to turn the cards from the deck one by one, and at each card, say the name of the month that corresponds to that number. For example, if the child gets 7, they should say “July”.

Once the child can go through all 12 cards without any mistakes, give them a small reward.

Games and Activities at Math Mammoth Practice Zone

Telling Time

Practice telling time on an analog clock with this interactive online exercise. Choose “Whole hours” or “Whole hours and half hours” for this grade level.

<https://www.mathmammoth.com/practice/tell-time>

Further Resources on the Internet

We have compiled a list of Internet resources that match the topics in this chapter, including pages that offer:

- **online practice** for concepts;
- online **games**, or occasionally, printable games;
- **animations** and interactive **illustrations** of math concepts;
- **articles** that teach a math concept.

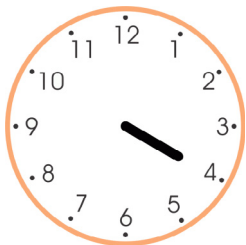
We heartily recommend you take a look! Many of our customers love using these resources to supplement the bookwork. You can use these resources as you see fit for extra practice, to illustrate a concept better and even just for some fun. Enjoy!

<https://l.mathmammoth.com/2026/gr1ch4>

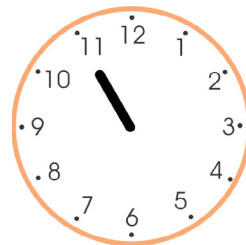


Whole and Half Hours 1

In this lesson, the clock has only one hand – the HOUR hand.



The hour hand points to four – it is four hours, or “four o’clock.”

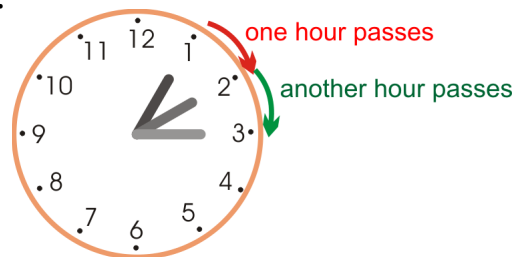


The hour hand points to eleven – it is eleven hours, or “eleven o’clock”.

The hour hand moves slowly around the clock face: from 1 to 2 to 3, and so on.

When the hour hand moves from 1 to 2, exactly one hour of time has passed.

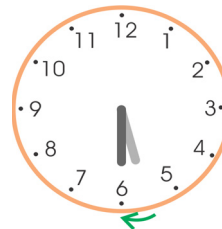
The same is true when the hour hand moves from 2 to 3. It takes the hour-hand one hour to do that.



On this clock, the hour hand pointed to 5 when it was five o’clock. Then it moved to **halfway** between 5 and 6. We say it is **half past five**.

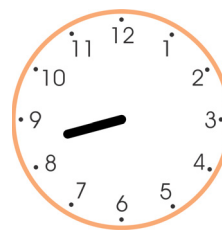
It takes the hour hand one-half hour to move from five to halfway between five and six.

The hour hand will *also* take a half hour to move from half past five to six o’clock.

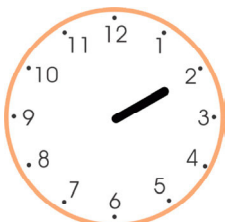
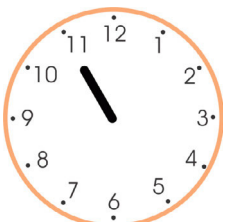
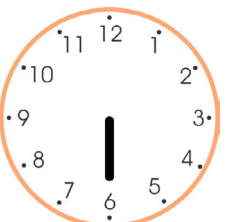
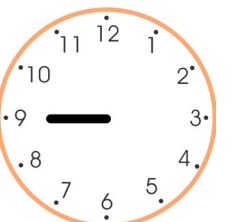


Here the hour hand has moved past eight o’clock, and is halfway between 8 and 9. We say it is half past eight.

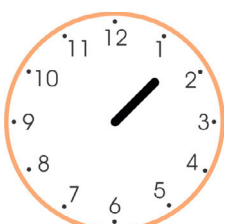
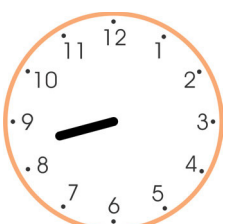
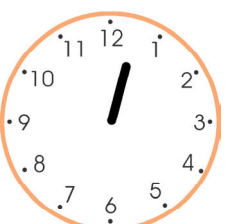
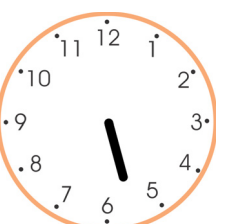
In half an hour from now, it will be nine o’clock.
In an hour from now, it will be half past nine.



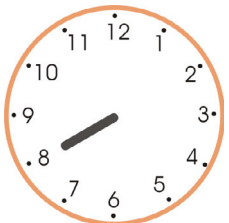
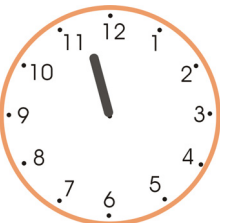
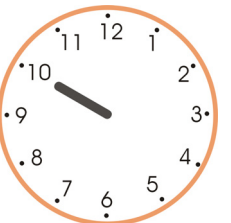
1. Tell or write the time.

 a. ____ o'clock	 b. ____ o'clock	 c. ____ o'clock	 d. ____ o'clock
--	--	---	--

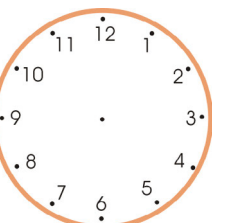
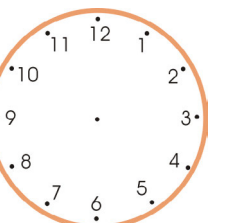
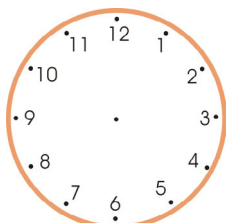
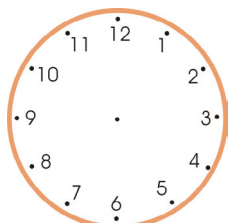
2. Tell or write the time.

 a. Half past ____	 b. Half past ____	 c. Half past ____	 d. Half past ____
--	--	---	--

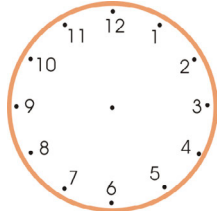
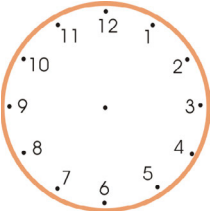
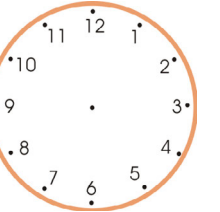
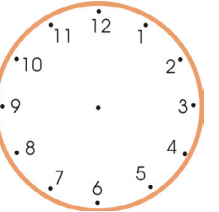
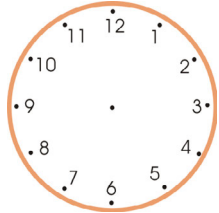
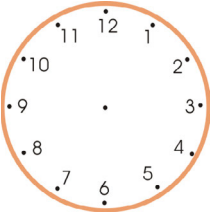
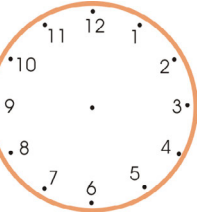
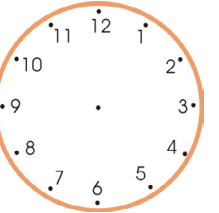
3. Tell or write the time.

 a. _____ _____	 b. _____ _____	 c. _____ _____	 d. _____ _____
---	---	--	---

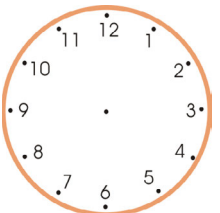
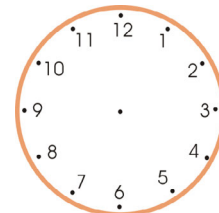
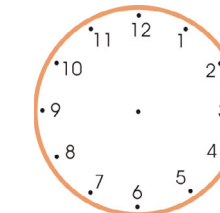
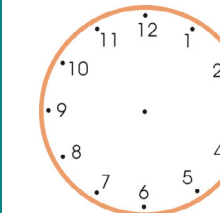
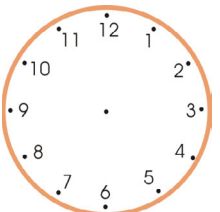
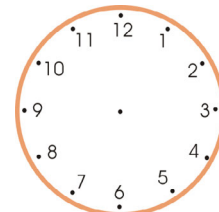
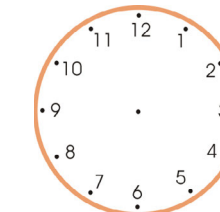
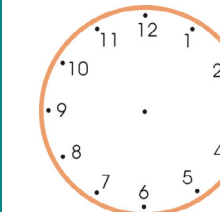
4. Draw the hour hand.

 a. Five o'clock	 b. Half past three	 c. Twelve o'clock	 d. Half past six
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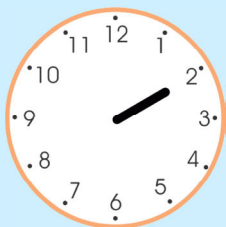
5. Draw an hour hand on each clock. In the second row, show the time a half-hour later.

Draw the hour hand.				
	a. Five o'clock	b. One o'clock	c. Half past six	d. Half past three
A half-hour later →				

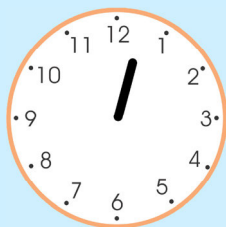
6. Draw an hour hand on each clock. In the second row, show the time an hour later.

Draw the hour hand.				
	a. Half past two	b. Twelve o'clock	c. Half past nine	d. Eight o'clock
An hour later →				

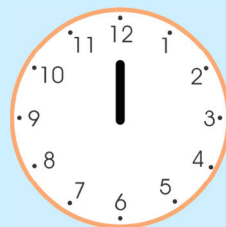
7. Match the clocks and the times.



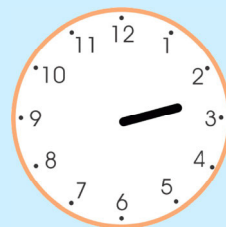
Twelve o'clock



Half past twelve



Half past two



Two o'clock

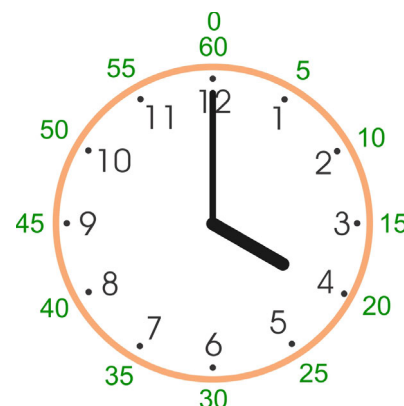
Whole and Half Hours 2

The minute hand on the clock is the thinner and longer hand.

It shows us the minutes — but the numbers 1 through 12 on the clock face do NOT tell us the minutes.

The green, smaller numbers are for the minute hand. They are not normally written on the clock face at all.

The time on the clock is 4 o'clock, or 4:00, which is four hours and zero minutes.



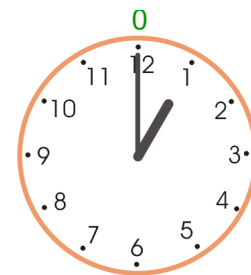
Find a clock that has a knob you can turn to move the hour and minute hands. Set the hour hand pointing to one, and the minute hand pointing straight “up”.

Move the hour hand from 1 to 2, and observe the minute hand!

Now move the hour hand from 2 to 3.
What does the minute hand do?

If you make the hour hand travel from 7 to 10,
how many “rounds” does the minute-hand make? _____

Make the minute hand travel backwards as well.



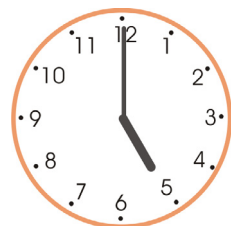
It is 1 o'clock
or **1:00**
(1 hour and 0 minutes)

Use your clock. How much time passes when the hour hand travels from 1 to 2? _____ hour

At the same time, the minute hand travels **from 0 to 60 minutes**, or once around the clock.

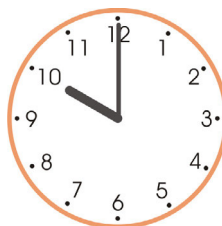
1 HOUR = 60 MINUTES

1. The minute hand is added to the clocks. Tell or write the time in two ways: using the expression *o'clock* or using numbers.



a. _____

_____ : _____



b. _____

_____ : _____

1 HOUR = 60 MINUTES**HALF HOUR = 30 MINUTES**

Use your practice clock again. Now let the hour hand travel from 2 to half-past 2 (only half an hour). The minute hand traveled from 0 minutes to 30 minutes.

Thirty minutes is half an hour.

We write the time as 2:30. It means 2 hours and 30 minutes -- or two and a half hours. That is why we can also say it has "half past two".

Now let the hour hand travel from half-past 2 to 3 — another half an hour.

The minute hand traveled from 30 minutes to _____ (or 0) minutes.



It is 2:30.

The numbers outside the clock face are for the minutes.

One hour = 60 minutes

Half hour = 30 minutes

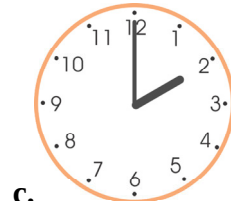
2. Tell or write the time in two ways: (1) using the expressions *o'clock* or *half past* and (2) with numbers.



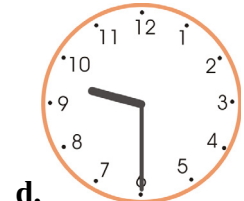
_____ : _____



_____ : _____



_____ : _____



_____ : _____



_____ : _____



_____ : _____



_____ : _____



_____ : _____

This is a digital clock. It shows time using numbers only.

This clock shows the time two-thirty, or half past 2.

In text, we normally write it with a colon (:), like this: 2:30.



3. Tell or write the time.

a.



b.



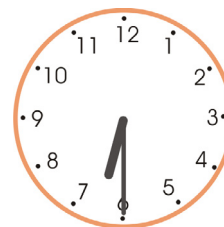
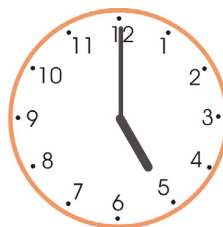
c.



d.



4. Match the clocks that show the same time.

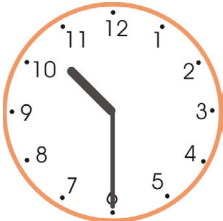


5. Write the time a half-hour later. Use numbers.

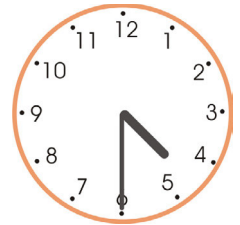
Now:	3:00	6:30	9:00	2:30	11:30
Half hour later:					

Practicing with Time

1. Tell or write the time each clock shows, using numbers. Then tell or write the time it will be a half-hour later.

	a.  _____ : _____	b.  _____ : _____	c.  _____ : _____	d.  _____ : _____
half hour later →	_____ : _____	_____ : _____	_____ : _____	_____ : _____

2. Anna is learning how to tell time. She says this clock shows “half five.” Is that correct? If not, explain to Anna how to tell the time correctly.



3. Tell the time shown by each clock, using “o’clock” and “half past.”



4. Order the clocks above in time-order, from the earliest time to the latest.
(You can say or write here the order using a, b, and c.)

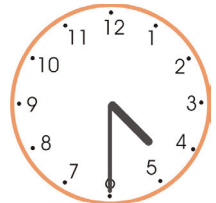
5. Write the time an hour later. Use numbers.

Now:	7:30	12:00	11:30	9:30	11:00
One hour later:					

6. Match the clocks that show the same time.

The image shows four analog clocks and four digital watches arranged in two rows. The top row contains four analog clocks with the following times: 11:30, 1:00, 3:30, and 12:30. The bottom row contains four digital watches with the following times: 12:30, 2:00, 11:30, and 2:30.

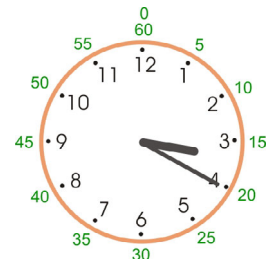
7. It is 3 o'clock, and Mom tells Daniel, "You can play half an hour more." Daniel stopped playing when the clock showed this:



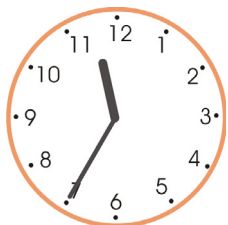
Did he play for half an hour only? Explain.

Puzzle Corner

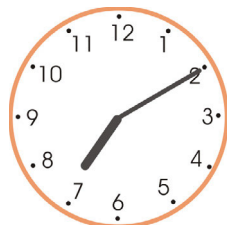
Here, the hour hand is past 3, and the minute hand points to 20. It is 3:20. Tell the time shown by each clock below.



a. _____ : _____



b. _____ : _____



c. _____ : _____



d. _____ : _____

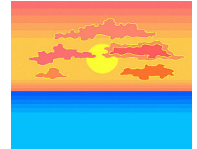
AM and PM

Night is when it is dark and most people sleep. After night comes morning and the sun rises. All through the morning, the sun keeps climbing higher.



Noon is **12:00**. It is when the sun is at its highest point in the sky.

After that, we call it **afternoon**. Afternoon ends at 6 o'clock. That's when evening starts. The sun sets and the darkness starts (usually) some time during the evening.



The 12 clock hours before noon are marked with AM.

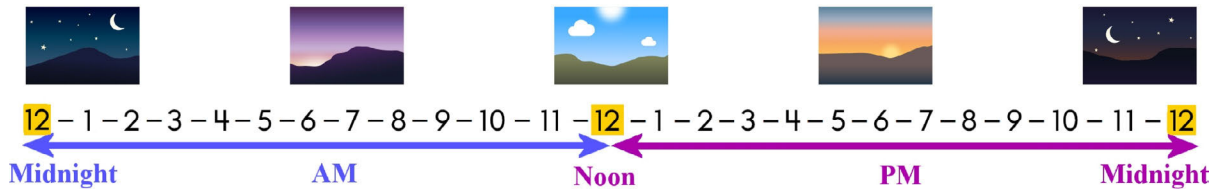
The 12 clock hours after noon are marked with PM.

For example, you might eat breakfast at 8 o'clock in the morning, or 8 AM.

You might go to bed at 8 or 9 o'clock in the evening — 8 PM or 9 PM.

AM is short for Latin *Ante Meridiem*, which means “before noon.”

PM comes from Latin *Post Meridiem*, which means “after noon.”



So, from midnight, for the rest of the night, at sunrise, and all through the morning until NOON (when the sun is at its highest), the hours are AM.

After noon, in the afternoon, when the sun is going down, when the sun sets, and all through the evening until midnight, the hours are PM.

The clock has only 12 hours, but our day and night have 24 hours all total.

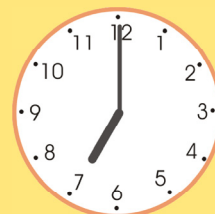
That is why the hour hand travels through those 12 hours *two* times in one day-night period: $12 + 12 = 24$.

So every clock hour (such as 7:00) happens two times during one day-night period.



7:00 AM

This is before noon.

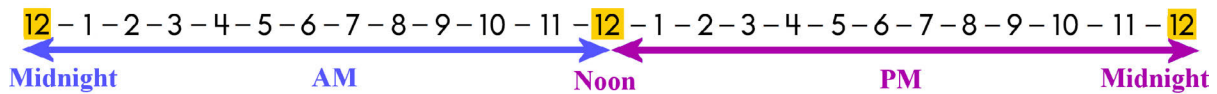


7:00 PM

This is after noon, during the evening.

Because of the meaning of AM and PM, noon is technically neither. The same is true of midnight. However, most people consider noon to be 12 PM (the beginning of the PM) and midnight to be 12 AM (the beginning of the AM). You can say, “12 o'clock noon” or “12 o'clock midnight” to avoid confusion.

1. **a.** Mark on the hour line when you usually wake up and when you usually go to bed.
Mark your sleeping hours with black.
- b.** Mark on the hour line when you eat some of your main meals.



2. What are you usually doing at these times?

- a.** 5:00 AM _____
- b.** 8:00 AM _____
- c.** 12:00 noon _____
- d.** 3:00 PM _____
- e.** 6:00:PM _____
- f.** 12:00 midnight _____

3. Is it before noon, or not? Tell the time, and include either AM or PM.

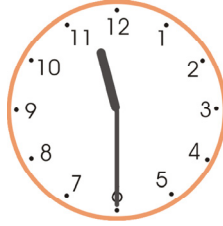
a. The birds are awake and singing, but most people are still in bed.		b. Mom is watching the evening news on TV.	
c. It is time to start schoolwork!		d. The sun is shining brightly.	
e. We are eating lunch.		f. Jack went to bed two hours ago.	

4. Tell the time, using numbers. Include AM or PM.



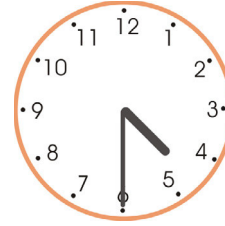
a. People are going to work.

_____ : _____ _____



b. Angi is watching the stars.

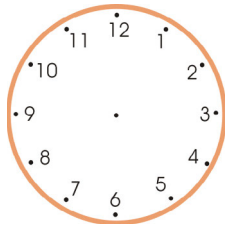
_____ : _____ _____



c. Mom is buying groceries.

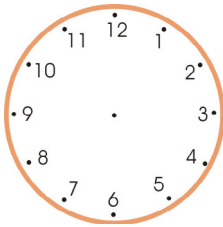
_____ : _____ _____

5. Draw the hands on the clock to match the situation. Then write the time. Include AM or PM.



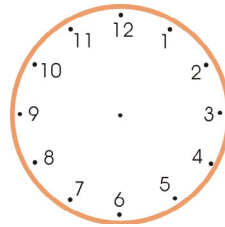
a. I wake up.

_____ : _____ _____



b. I eat lunch.

_____ : _____ _____



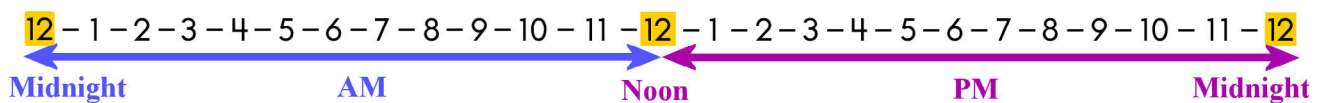
c. I eat supper.

_____ : _____ _____

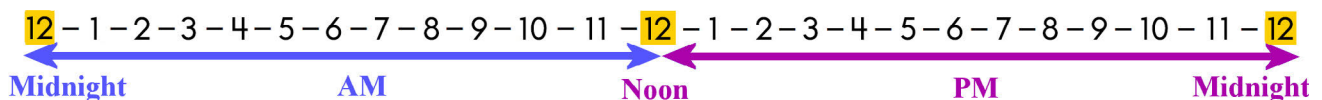
6. (optional) Mark on the hour line when the SUN usually rises and when it usually sets.

Color the nighttime hours black. (Ask your teacher to help you make these for different seasons.)

a. the middle of winter



b. spring



c. summer



The Calendar

(This lesson is optional.)

January Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31	February Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28	March Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
April Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30	May Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31	June Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
July Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31	August Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31	September Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
October Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31	November Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30	December Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

1. Circle these dates on the calendar above.

a. March 15

b. November 18

c. December 29

d. February 8

e. November 11

f. October 30

g. today

h. your birthday

i. any other important dates

2. What do the letters “Su Mo Tu We Th Fr Sa” mean?

3. Try not to look at the calendar. Write the name of the month that goes in between.

- | | | |
|------------|-------|----------|
| a. August | _____ | October |
| b. March | _____ | May |
| c. June | _____ | August |
| d. January | _____ | March |
| e. October | _____ | December |

4. (Use the calendar above.) What day of the week is...

- | | | | |
|--------------|-------|------------------|-------|
| a. January 4 | _____ | b. May 14 | _____ |
| c. July 23 | _____ | d. March 10 | _____ |
| e. today | _____ | f. your birthday | _____ |

5. Find all of the MONDAYS in June. Write their dates below:

June _____	June _____	June _____	June _____	June _____
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6. Find all of the THURSDAYS in April. Write their dates below:

April _____	April _____	April _____	April _____	April _____
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7. Can you name the months in order from memory?

Practice memorizing them in groups of four.

- January, February, March, April;
- May, June, July, August;
- September, October, November, December.
(These are all the months ending in “ber”!)

Chapter 4 Mixed Review

1. Write the four facts (two additions, two subtractions) for the given numbers.

a. Numbers: 7, 2, 5

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

b. Numbers: 10, 8, 2

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

$$\underline{\quad} - \underline{\quad} = \underline{\quad}$$

2. Draw a line to the correct answer.

a.

$$8 - 6$$

$$8 - 4$$

$$6 - 2$$

$$7 - 3$$

$$10 - 8$$

$$5 - 3$$

2

4

b.

$$9 - 6$$

$$9 - 4$$

$$7 - 4$$

$$8 - 3$$

$$10 - 5$$

$$10 - 7$$

5

3

3. Fill in the missing parts.

a.

$$50 + 7 = \underline{\quad}$$

$$20 + 4 = \underline{\quad}$$

b.

$$\underline{\quad} = 9 + 80$$

$$\underline{\quad} = 3 + 30$$

c.

$$75 = 5 + \underline{\quad}$$

$$36 = \underline{\quad} + 30$$

4. Compare the numbers. Write $<$, $>$, or $=$ between them.

a. 98

89

b. 60

16

c. 37

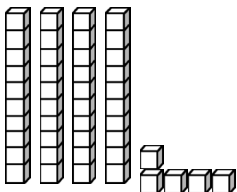
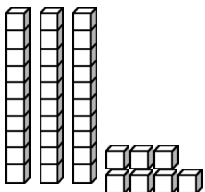
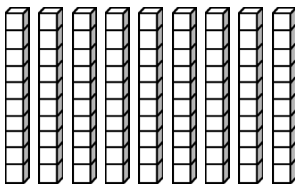
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5. Count by ones to fill in the missing numbers.

a. _____, _____, _____, _____, _____, 119, 120

b. _____, _____, 97, 98, _____, _____, _____

6. Write the number that is **10 more** than the number shown in the illustration.

 a. _____	 b. _____	 c. _____
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7. Are these equations true or false? Circle the correct word.

a. $2 = 7 - 6$ true *or* false

b. $1 + 1 = 5 - 3$ true *or* false

c. $0 = 0 + 0$ true *or* false

d. $10 = 30 - 10$ true *or* false

8. Choose two “math notes” or stickers that are equal in value. Then write an equation, using them and an equals sign.

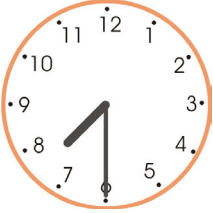
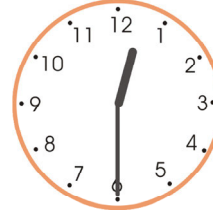
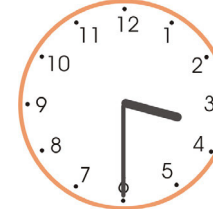
a. 9 $3 + 5$ $6 + 3$ Equation: _____ = _____	b. 4 $2 + 3$ $7 - 2$ 7 Equation: _____ = _____
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9. Add and subtract.

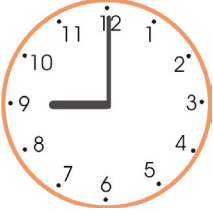
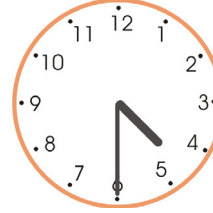
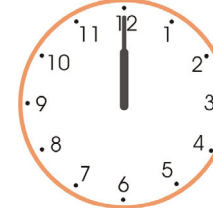
a. $30 + 70 =$ _____ $100 - 50 =$ _____ $60 - 30 =$ _____	b. $40 + 30 =$ _____ $100 - 20 =$ _____ $80 - 70 =$ _____	c. $82 - 10 =$ _____ $65 + 10 =$ _____ $27 - 10 =$ _____
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Chapter 4 Review

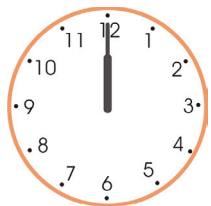
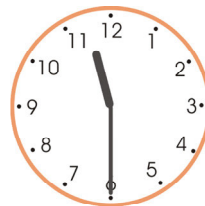
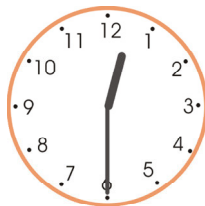
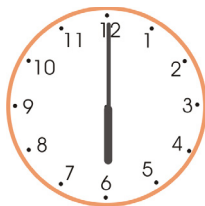
1. Tell or write the time using the expressions *o'clock* or *half past*.

 a. _____ _____	 b. _____ _____	 c. _____ _____	 d. _____ _____
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2. Write the time using numbers.

 a. _____ : _____	 b. _____ : _____	 c. _____ : _____	 d. _____ : _____
---	---	--	---

3. Match the digital clock to the correct analog clock.



4. Write the time at a half-hour later. Use numbers.

Now:	a. 5:00	b. 10:30	c. 12:00	d. 1:30	e. 5:30
A half-hour later:					

5. Fill in either “AM” or “PM.”

a. Jack wakes up. It is 8 _____.	b. Jack plays in the afternoon. It is 3 _____.	c. Jack is sleeping. It is dark. It is 2 _____.
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